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Benchmarking Holistic vs Sentence-Level Reward-Model Preferences for Factual Summarization via Granular Credit Assignment (GCA)

Master's Thesis

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the degree of Master of Science (M.Sc.)
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submitted by
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Abstract

Preference-based alignment of abstractive summarization models increasingly relies on automatically generated preference labels rather than human judgments. It remains unclear whether a single holistic judgment over a multi-sentence summary uses the available supervision as effectively as a judgment assembled from sentence-level evidence. This thesis studies that question as a controlled comparison of two preference-construction procedures: a holistic condition, where a complete candidate summary is scored against the source article, and a Granular Credit Assignment (GCA) condition, where each summary is decomposed into sentences, each is scored separately, and the sentence scores are aggregated into a summary-level preference. The candidate summaries, the factuality evaluator (AlignScore), the Bradley–Terry reward-model architecture, and the training procedure are held fixed across four research questions, so feedback granularity is isolated as the primary experimental variable.

RQ1 asks whether GCA produces a more learnable preference signal, measured as reward-model pairwise validation accuracy. Across twenty independently seeded runs at $n = 1{,}000$, GCA led in every run, a mean advantage of 3.95 percentage points (bootstrap 95% CI $[+3.19, +4.70]$, run-level Wilcoxon $p < 0.001$, Cohen's $d = 2.22$).

RQ2 asks how evaluator configuration and aggregation strategy affect this comparison. The aggregation rule matters substantially: an early weakest-link penalty formula measurably hurt GCA relative to a simple mean. A single-run sweep across four AlignScore modes suggested the evaluator mode matters too, but the confirmatory campaigns do not bear that out, so it is reported as exploratory rather than established.

RQ3 asks whether the RQ1 advantage is stable across dataset scale; it is not. Twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$, at the same resolution, give mean gaps of -0.22 and -0.11 percentage points, both statistically indistinguishable from zero, with equivalence tests excluding any effect larger than ± 0.61 and ± 0.36 points: the advantage is absent, not merely undetected. Absolute accuracy improves for both conditions as data grows even as their difference vanishes.

RQ4 asks whether the RQ1 advantage also holds against an independent, held-out ground truth: 500 articles disjoint from every training set, scored by the same judge but never used in training. It does, more sharply: at $n = 1{,}000$, GCA is significantly more precise than holistic on comparisons with a genuine quality gap (up to $+9.25$ percentage points, $p < 0.001$). It does not merely fail to persist, as RQ1 does; it reverses to a significant advantage for holistic at $n = 10{,}000$ on three of four ground-truth constructions ($p = 0.010$–0.036). Two candidate explanations, near-zero-margin training pairs and a length/sentence-count shortcut, were tested and both rejected; the mechanism behind the reversal remains open.

These results characterize when granular feedback changes a preference signal's structure, not that GCA is generally preferable: it is not, once data are abundant or the comparison is ambiguous. Higher accuracy shows a preference relation is more recoverable by this architecture, not that GCA labels better match human judgments or that trained models are more factually consistent, both future work.

Acknowledgments

I would like to thank my first supervisor, Prof. Dr. Frank Hopfgartner of the Institute for Web Science and Technologies, for his guidance throughout this project and in particular for the feedback that led me to narrow the study to a single, cleanly controlled comparison rather than a broader but less conclusive pipeline. I am equally grateful to my second supervisor, Dr. Zeyd Boukhers of the Fraunhofer Institute for Applied Information Technology FIT, for his technical input on reward modeling and factuality evaluation. I owe particular thanks to my mentor, Lingxiao Kong of the Fraunhofer Institute for Applied Information Technology FIT, who guided me through this project from its early design decisions to its final submission. His feedback shaped the direction of the experiments, including the design of the independent ground-truth evaluation in Chapter 6, and his close reading of multiple drafts of this thesis directly produced the reorganization of the results around the research questions rather than the chronology of the project, and much of the precision in how the limitations are stated.

The experiments in this thesis were carried out on the MOGON NHR cluster at Johannes Gutenberg University Mainz. I gratefully acknowledge the compute time made available there; the repeated-run and multi-scale campaigns reported in Chapter 6 would not have been feasible without access to that infrastructure.

Finally, I thank the University of Koblenz for providing the research environment in which this work was completed, and my family and friends for their support over the course of it.

Declaration on the Use of AI Assistance

I acknowledge the use of large language models to support the writing of this thesis, specifically for improving the clarity, structure, and grammatical correctness of the text, and as a coding assistant during the implementation of the experimental pipeline. The research design, the experiments, the analysis of the results, and the conclusions drawn from them are my own. All generated text and code were reviewed, verified, and revised by me, and I take full responsibility for the content of this thesis.

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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
CI	Confidence Interval
CV	Cross-Validation
DPO	Direct Preference Optimization
GCA	Granular Credit Assignment
GPU	Graphics Processing Unit
LLM	Large Language Model
MDE	Minimum Detectable Effect
MoDPO	Multi-Objective Direct Preference Optimization
NHR	Nationales Hochleistungsrechnen (National High-Performance Computing)
NLI	Natural Language Inference
RLAIF	Reinforcement Learning from AI Feedback
RLHF	Reinforcement Learning from Human Feedback
RM	Reward Model
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SD	Standard Deviation
TOST	Two One-Sided Tests (equivalence testing procedure)
`_sp`	Suffix marking the AlignScore evaluation modes that split the input internally (`nli_sp`, `bin_sp`)

1. Introduction

1.1. Motivation

Large language models (LLMs) are increasingly used for multi-sentence generation tasks such as news summarization. In this setting, the perceived quality of an output is often dominated not by its overall fluency but by a small number of high-impact factual errors: a wrong entity, a flipped relation, or a fabricated claim can outweigh otherwise well-written text. The quality dimension this thesis is concerned with throughout is *factual consistency* with the source document, not stylistic polish or fluency.

Preference-based fine-tuning, most prominently reinforcement learning from human feedback (RLHF), updates a model using comparative judgments between candidate outputs rather than token-level supervision [4, 31, 38]. Collecting high-quality human preferences at scale is expensive and slow, so reinforcement learning from AI feedback (RLAIF) replaces human annotators with an automatic judge [1, 24]. This trades annotation cost for a new source of label noise and bias.

A further question arises for long, multi-sentence outputs such as news summaries, one that is largely independent of whether the feedback comes from a human or an AI judge: how granular should the preference signal be? A single holistic judgment ("summary A is better than summary B") collapses all of the evidence in a multi-sentence output into one binary decision, and cannot express *where* in the output the deciding evidence lies. Sentence-level judging can in principle localize exactly which claims are, and are not, supported by the source, and that localized evidence can then be aggregated into a summary-level decision. This thesis asks whether that extra granularity actually produces a better training signal, and under what conditions.

The intuition that originally motivated the design was more specific: that a summary which is mostly accurate but contains one fabricated sentence ought to be penalized relative to one that is uniformly mediocre but fabricates nothing, and that a single holistic label cannot make that distinction. It is worth stating that intuition explicitly, because the results reported here do not bear it out. The aggregation rule built to act on it, a weakest-link penalty, performed measurably *worse* than a simple mean of the same sentence scores (Chapter 6, Section 6.5.1). The simple mean is therefore the rule used in this thesis's final configuration, and a mean has the same limitation as a holistic score: it can dilute a single fabricated sentence rather than penalize it (Chapter 8, Section 8.4).

What survives of the granularity question is narrower and more conditional than the motivating intuition suggests.

1.2. Problem Statement

When constructing AI-generated preference data for training a factuality-oriented reward model, it is unclear whether coarse, holistic preferences are sufficient, or whether they systematically underuse the supervision signal available in long, multi-sentence outputs. The problem addressed here, under controlled conditions, is whether sentence-level factuality judgments, aggregated into a summary-level preference through a mechanism referred to as *Granular Credit Assignment* (GCA), yield a more learnable preference signal for reward-model training than purely holistic judgments of the same candidate summaries.

Learnability, as used throughout this thesis

The pairwise validation accuracy of a Bradley–Terry reward model trained on the resulting preference pairs (Chapter 3, Section 3.1): given a held-out preference pair, how often does the trained reward model rank the summary the judge preferred above the alternative?

1.3. Thesis Objective

The objective of this thesis is to determine, through a controlled empirical comparison, whether Granular Credit Assignment produces a more learnable factuality-preference signal than holistic scoring. The two conditions differ only in how factuality scores are obtained and converted into a pairwise label:

> **Holistic:** article + complete candidate summary → factuality score → pairwise preference.
>
> **GCA:** article + individual sentences → sentence factuality scores → aggregation → pairwise preference.

Both preference sets are then used to train the same Bradley–Terry reward-model architecture under the same procedure. Holding the source articles, the candidate summaries and their generation settings, the evaluator family, the reward-model architecture, and the training procedure fixed allows feedback granularity to be studied as the primary experimental variable, rather than as one difference among several.

The thesis does not propose a new reward-model architecture, a new factuality metric, or a new training objective. Its subject is the construction of the preference data itself.

1.4. Research Questions

Four research questions follow from this objective. They are stated here in summary form; Chapter 4, Section 4.1 gives them together with the working hypotheses and the scope conditions under which each can be answered, and Chapter 8, Section 8.2 answers them against the evidence collected.

- **RQ1.** Does sentence-level factuality feedback aggregated via GCA produce a more learnable pairwise preference signal than holistic summary-level feedback, as measured by reward-model pairwise validation accuracy?
- **RQ2.** How do evaluator configuration and sentence-score aggregation strategy affect the relative learnability of holistic and GCA-derived preferences?
- **RQ3.** How stable are the observed differences across repeated reward-model training runs and dataset sizes?
- **RQ4.** Does the learnability advantage measured in RQ1 also hold when the trained reward models are evaluated against an independent, held-out ground truth, rather than against their own training-style labels, and does that hold as the training-set size grows?

RQ1 is the confirmatory question and is addressed by a repeated-run comparison. RQ2 is addressed by an exploratory sweep, and therefore supports statements about association between configuration and outcome rather than about causal mechanism. RQ3 motivates both the repeated runs at a single dataset size and the multi-scale check at $n = 5{,}000$ and $n = 10{,}000$. RQ4 was added after that multi-scale check, to test the learnability finding against data the trained models never saw and never influenced their selection (Chapter 4, Section 4.8).

1.5. Contributions

This thesis makes the following scientific contributions:

1. **A controlled framework for studying feedback granularity in automatic preference construction.** The thesis formulates holistic and sentence-level feedback as two alternative procedures for constructing factuality-oriented preference data. The source articles, candidate summaries, factuality evaluator, reward-model architecture, and training procedure are held fixed across both conditions, allowing the effect of preference-construction granularity to be studied without simultaneously modifying the downstream learning algorithm (Chapter 4).

2. **A Granular Credit Assignment procedure for constructing pairwise preferences from sentence-level factuality evidence.** The thesis introduces a decompose–score–aggregate procedure in which each candidate summary is segmented into sentences, the sentences are evaluated individually against the source article, and their factuality scores are aggregated into a summary-level score. The resulting scores are then converted into standard chosen–rejected preference pairs that can be consumed by an unmodified Bradley–Terry reward model. The aggregation formulation includes a parameter controlling the influence of low-scoring sentences (Chapter 4, Section 4.3).

3. **A systematic methodology for analyzing the design choices involved in granular preference construction.** The thesis treats sentence-score aggregation and factuality-evaluator configuration as explicit methodological variables rather than fixed implementation details. Multiple aggregation strategies and evaluator modes are examined to determine how these choices influence the preference data subsequently used for reward-model learning (Chapter 6, Section 6.5).

4. **A repeated-run and multi-scale evaluation protocol for comparing preference-construction methods.** The thesis develops an evaluation procedure that compares holistic and GCA-derived preferences across independently seeded reward-model runs and multiple training-set sizes. The run, rather than the individual cross-validation fold, is treated as the unit of independent replication, and the analysis combines bootstrap confidence intervals, paired non-parametric testing, effect-size estimation, power analysis, and equivalence testing (Chapter 4, Section 4.5).

5. **A held-out evaluation procedure for analyzing learned preference rankings beyond training-style validation labels.** In addition to cross-validation on preference pairs, the thesis constructs a set of articles disjoint from the reward-model training pools and evaluates retained reward-model checkpoints on previously unseen candidate summaries. This provides an additional way to examine how reward models trained from different preference-construction procedures behave outside the preference pairs used during training. The asymmetry introduced by constructing this evaluation criterion from sentence-level AlignScore scores is explicitly documented as a limitation (Chapter 4, Section 4.8).

The contributions of this thesis therefore concern the construction, analysis, and evaluation of automatically generated factuality preferences. The thesis does not introduce a new factuality evaluator, reward-model architecture, or preference-learning objective, and it does not claim that GCA universally produces more accurate factuality labels or more factual generated summaries.

1.6. Scope

- **Task:** single-document abstractive news summarization, using the CNN/DailyMail dataset [16, 30].
- **Candidate generation:** a single instruction-tuned base model, Mistral-7B-Instruct-v0.3, sampled at two temperatures ($T = 0.7$ and $T = 1.0$) to produce an A/B candidate pair per source article.
- **Judge:** a single fixed, deterministic factuality judge, `yzha/AlignScore` [41], intended throughout in its `nli` evaluation mode, though the confirmatory campaigns were, due to an implementation oversight found during final review, actually run under its `nli_sp` mode (Chapter 5, Section 5.4). No generative LLM-as-judge and no external API dependency is used in the final pipeline.
- **Optimization target:** the learnability of the resulting preference data for Bradley–Terry reward-model training, measured by pairwise validation accuracy under 5-fold cross-validation. A downstream fine-tuned generation policy is not evaluated in the final comparison; an early Direct Preference Optimization (DPO) prototype was explored but dropped from the final scope (Chapter 4).

1.7. Thesis Structure

Chapter 2 situates this work within the literature on preference-based alignment, LLM-as-a-judge reliability, fine-grained feedback for long-form generation, and automatic factual-consistency evaluation. Chapter 3 introduces the technical mechanisms this thesis builds on: the Bradley–Terry preference model, Direct Preference Optimization, the AlignScore factuality metric, and the statistical tools used to validate the results. Chapter 4 presents the experimental design, including the GCA aggregation formula itself, which is this thesis's own contribution rather than prior work, and an explicit account of how the study that was actually carried out diverges from the original proposal, and why. Chapter 5 describes the pipeline, codebase, and compute infrastructure. Chapter 6 reports the results, organized by research question and leading with the confirmatory findings: the learnability comparison, its behavior across dataset scale, and the independent ground-truth evaluation that answers RQ4, followed by the two configuration decisions and the evidence justifying them. Chapter 7 interprets these results, including why the ground-truth result reverses rather than merely fading where the learnability result only fades, and Chapter 8 answers the research questions, states the limitations of this work, and outlines directions for future work. Appendix D retains the record of the exploratory and superseded experiments that preceded the final design, which are referenced from the main text where they justify a decision but are not themselves findings of this thesis.

2. Related Work

This chapter reviews four bodies of work that this thesis draws on: preference-based alignment and how preference data becomes a training signal (Section 2.1); the reliability of AI judges when they replace human annotators (Section 2.2); the granularity of feedback in long-form generation, which is the variable this thesis manipulates (Section 2.3); and automatic factual-consistency evaluation, which supplies the judge (Section 2.4). Section 2.5 draws these together into the specific gap this thesis addresses, and Section 2.6 states what separates the present work from the studies closest to it.

2.1. Preference-Based Alignment: RLHF, RLAIF, and Offline Optimization

Preference-based alignment collects pairwise comparisons between candidate model outputs and converts those comparisons into a training signal. The approach originates in preference learning for reinforcement learning, where human annotators compare short trajectory segments and a reward model is fitted to reproduce their comparisons, allowing an agent to be trained on objectives that are easy to judge but hard to specify numerically [4]. The choice to elicit comparisons rather than absolute scores is deliberate: relative judgments between two concrete outputs are more consistent between annotators than absolute ratings on an abstract scale, and the Bradley–Terry model (Chapter 3, Section 3.1) provides a principled way to recover a scalar reward from them.

Stiennon et al. [38] adapted this recipe to summarization, and their pipeline is the direct ancestor of the one used in this thesis. They collect human comparisons between candidate summaries, train a reward model on those comparisons, and then fine-tune a policy against the learned reward with reinforcement learning. Two of their findings matter here. First, the reward model's own held-out agreement with human preferences is reported as a diagnostic in its own right, which establishes the precedent for treating reward-model accuracy as a meaningful measurement rather than merely an intermediate artifact, as this thesis does. Second, they observe that optimizing too hard against a learned reward degrades true quality, an early instance of the reward-overoptimization problem that motivates careful attention to the quality of the preference signal itself. RLHF was subsequently scaled to general instruction-following [31], establishing the reward-model-plus-policy-optimization pattern as the default alignment recipe.

Because human comparison data is expensive, reinforcement learning from AI feedback (RLAIF) substitutes a model for the human annotator. Constitutional AI generates critiques and revisions from a written set of principles and derives preference labels from them, removing human annotation from the harmlessness training loop almost entirely [1]. Lee et al. [24] compare RLAIF and RLHF directly on summarization and dialogue tasks and report broadly comparable downstream performance, which is the empirical result that makes AI-generated preference data a credible substitute rather than a compromise. They also emphasize that the outcome depends substantially on how the judge is prompted and calibrated, a dependency that recurs throughout Section 2.2 and, in a different form, in the evaluator-configuration effects observed here (Chapter 6, Section 6.5.2).

On the optimization side, RLHF pipelines have typically used on-policy algorithms such as Proximal Policy Optimization [35], which require sampling from the policy during training and maintaining a separate reward model. Direct Preference Optimization (DPO) removes both requirements by reparameterizing the RLHF objective so that the optimal policy can be recovered in closed form directly from preference pairs [33]. DPO was the objective originally proposed for this thesis, chosen because it isolates the effect of the preference data from the dynamics of reinforcement learning. As Chapter 4 describes, DPO was implemented and produced early results before being dropped from the final comparison in favor of evaluating Bradley–Terry reward models directly. That change narrows the comparison further than DPO alone would: it removes both the optimization dynamics that motivated adopting DPO and the decoding variance introduced by generating and scoring a fine-tuned policy's outputs, leaving a more targeted question about whether the preference data itself is more learnable.

2.2. Reliability Risks in AI Supervision and LLM-as-a-Judge

Substituting a language model for a human annotator introduces a distinct class of error. Benchmark studies of LLM-as-a-judge document sensitivity to prompt wording, position bias favoring whichever candidate appears first, verbosity bias favoring longer answers regardless of quality, inconsistent scoring across repeated calls, and self-enhancement effects in which a judge prefers outputs from models similar to itself [43]. Liu et al. [27] document related calibration problems in LLM-based evaluation of generated text, and a recent survey synthesizes the accumulated evidence on these failure modes and the mitigations proposed for them, including rubric enforcement, pairwise rather than absolute evaluation, and position swapping [13].

These issues are more consequential in RLAIF than in evaluation. An unreliable evaluation metric produces a noisy estimate of a model's quality; an unreliable judge used to construct training data produces systematically mislabeled supervision, and the model is then optimized to fit that mislabeling. Casper et al. [3] catalog failure modes of this kind across the RLHF pipeline, including reward misspecification and the divergence between training loss and intended behavior, and argue that problems in the preference data propagate into the trained policy in ways that are difficult to detect after the fact.

One response to this literature is to treat the judge as a fallible instrument and wrap it in a reliability protocol: confidence gating, order randomization, evidence-grounded rationale requirements, and prompt-robustness checks. The design adopted here removes the exposure at its source instead, by using a fixed, deterministic factuality model rather than a generative judge (Section 2.4). A deterministic model cannot exhibit prompt sensitivity, position bias, or run-to-run inconsistency, because it is never prompted and returns no natural-language output. This eliminates the category of risk the protocol above is designed to manage, at two costs: it does not make the evaluator correct, and it removes the rationale text that a human audit would examine. Chapter 4, Section 4.4 states what this implies for the claims the thesis can support.

2.3. Feedback Granularity and Credit Assignment in Long-Form Generation

The variable this thesis manipulates is how localized the supervision signal is. A single preference over a multi-sentence output provides one bit of information about a text that may contain a dozen independently checkable claims, and it does not indicate which claim caused the preference. This is a credit assignment problem, and it is present even in the original preference-learning formulation: Christiano et al. [4] have annotators compare short trajectory segments rather than whole episodes precisely to make the resulting supervision denser and easier to attribute.

The closest prior work to this thesis is Wu et al. [40], who study fine-grained human feedback for long-form question answering. Their approach differs from holistic RLHF along two dimensions simultaneously. First, feedback is collected at the density of individual sentences or sub-sentence spans rather than whole responses, so an annotator marks the specific span that is irrelevant, untruthful, or incomplete. Second, they train several distinct reward models, one per error category, and combine their outputs into a dense reward signal during reinforcement learning, so the policy receives a reward at each segment rather than a single scalar at the end of the sequence. They report that this improves both the resulting policy and the reward models themselves relative to holistic feedback, and that the multiple reward models allow the relative weight of different objectives to be adjusted after training.

Three differences separate their setting from this thesis. Their fine-grained labels come from trained human annotators, whereas this thesis asks whether the same localization strategy is still useful when the localized judgments come from an automatic judge, which is cheaper by orders of magnitude but noisier per label. Their fine-grained signal is consumed as a dense per-segment reward during reinforcement learning, whereas GCA aggregates local scores back into a single summary-level preference pair so that the resulting dataset remains compatible with standard pairwise preference optimization, requiring no change to the training objective. And they vary feedback density and objective decomposition together, whereas this thesis holds the objective fixed at factual consistency alone and varies only granularity, which isolates the granularity question at the cost of addressing a narrower one.

Three more recent lines of work place sentence- or step-level signals at other points in the alignment pipeline, and each should be distinguished carefully from the present study, since none leaves the question examined here open in general.

Gu et al. [14] introduce Mask-DPO, which incorporates sentence-level factuality signals directly into the DPO objective. Rather than treating a whole response as preferred or dispreferred, Mask-DPO uses sentence-level factuality information to learn only from the factually correct sentences within a preferred response, and to avoid penalizing factually correct content that appears inside a dispreferred one. The motivation is that response-level factuality supervision is noisy, because preferred and dispreferred responses each mix accurate and inaccurate sentences. The intervention point is the policy-optimization objective: the preference pair remains a response-level object, and the granular signal is applied as a mask during training. The present thesis intervenes earlier and leaves the training objective untouched. It asks how automatic factuality judgments at different granularities should be converted into the pairwise label in the first place, and measures the learnability of the resulting labels under a fixed Bradley–Terry architecture.

Qiu et al. [32] modify the reward model itself, proposing a sentence-level reward model that assigns scores at sentence boundaries and aggregates them into a response-level score, while still learning from response-level human preferences. Their aim is to address reward sparsity by producing an intermediate granularity between response-level and token-level reward modeling. The distinction from this thesis is architectural: they change what the reward model computes, and the preference labels they learn from are conventional response-level human judgments. This thesis does not modify the reward-model architecture at all. The Bradley–Terry model, its backbone, and its training procedure are held fixed across both conditions precisely so that differences in outcome can be attributed to the preference labels rather than to the model consuming them.

Closer still to GCA's decompose-then-aggregate structure is process reward modeling for multi-step reasoning [25]. Rather than training a single model to judge a complete solution as correct or incorrect, Lightman et al. [25] have human annotators label the correctness of each individual reasoning step in a chain-of-thought solution, train a reward model to reproduce those per-step judgments, and show that this process-level supervision outperforms a response-level outcome reward for selecting correct solutions among many sampled candidates. The structural resemblance to GCA is real: both decompose a long output into smaller units, score each unit, and use the per-unit scores rather than a single holistic judgment. Two differences remain. Process reward models keep the per-step scores as the object consumed downstream, typically for reranking or search over sampled solutions, whereas GCA aggregates its per-sentence scores back into one summary-level score so that the output is a conventional pairwise preference, compatible with unmodified Bradley–Terry training. And the per-step labels in Lightman et al. [25] come from trained human annotators judging mathematical correctness, a domain with a comparatively well-defined notion of a correct step, whereas GCA's per-sentence labels come from an automatic factuality judge applied to open-ended news summaries, where what counts as a faithful sentence is judged by a model rather than verified against a solution. The result that step-level supervision outperforms outcome-level supervision in their setting is consistent with the motivation for GCA, but it is evidence from a different domain, a different label source, and a different point of consumption in the pipeline, not a direct precedent for the automatic, aggregated case this thesis studies.

Reward models are also increasingly evaluated in their own right rather than only as components of a larger pipeline. RewardBench assembles prompt-chosen-rejected trios across several domains and evaluates reward models by how often they rank the preferred response above the dispreferred one [23]. That pairwise-ranking accuracy is the same quantity used as the dependent variable in this thesis (Chapter 4, Section 4.4), which places the measurement on established ground; the difference is that RewardBench varies the reward model against fixed data, whereas this thesis varies the data construction against a fixed reward model.

2.4. Factual Consistency in Summarization and Evaluation

News summarization remains a standard benchmark domain, with CNN/DailyMail widely used for both training and evaluation [16, 30]. Abstractive systems have long produced fluent output that nonetheless contradicts or embellishes the source [36], and this persistent gap between fluency and faithfulness is what makes factual consistency a worthwhile optimization target rather than a solved preprocessing concern.

Automatic factuality evaluation has developed along two main lines. Entailment-based methods treat the source as a premise and the summary as a hypothesis: FactCC trains a classifier on synthetically corrupted summaries to detect factual conflicts [20]. Question-answering-based methods instead generate questions from the summary and check whether the source yields the same answers, an approach developed in FEQA [9] and refined in QAFactEval, which systematically compares components of the QA-based pipeline [11].

Decomposition and aggregation in SummaC. The work most directly relevant to GCA's design is SummaC [21], which observes that natural language inference models trained on sentence-level data transfer poorly to document-level consistency checking, partly because the inputs are far longer than anything seen in training. Their solution is to decompose the problem: they compute an entailment matrix between each source sentence and each summary sentence, then aggregate that matrix into a single score. Critically for this thesis, they show that *how* the matrix is aggregated changes performance substantially. Their zero-shot variant takes, for each summary sentence, the maximum entailment score across all source sentences, and then averages these maxima across summary sentences. Their learned variant instead trains a convolutional layer over the distribution of entailment scores per summary sentence, retaining information that the maximum discards, and performs better.

That aggregation strategy is a design decision with measurable consequences is therefore established in the evaluation literature before this thesis takes it up. The contribution here is to ask the same question one step upstream, at the point where scores are converted into training labels rather than into an evaluation metric, and Chapter 6, Section 6.5.1 finds the same lesson holds: the choice between a mean and a penalty-weighted aggregation changes the outcome more than the decision to decompose at all.

AlignScore. The judge used throughout this thesis, AlignScore, was not part of the literature surveyed in the original proposal. Zha et al. [41] recast a broad collection of natural language processing tasks, including natural language inference, question answering, paraphrase detection, and fact verification, into a single "information alignment" format, and train one model on the union. The resulting function takes a context and a claim and returns a scalar in $[0, 1]$ estimating whether the context supports the claim. Because it is trained across many task formats rather than on entailment data alone, it generalizes across factuality benchmarks better than the single-task methods above.

AlignScore also inherits the decomposition question from SummaC, and resolves it internally: in its split evaluation modes, the context is divided into chunks and the claim into sentences, and the final score aggregates the resulting matrix by taking a maximum over chunks and then a mean over claim sentences, which is structurally the same rule as SummaC's zero-shot variant. This internal behavior turns out to matter a great deal for the present work. Chapter 6, Section 6.5.2 finds that whether GCA outperforms holistic scoring depends on which AlignScore mode is used, and Chapter 7, Section 7.2 argues that this is because the split modes already perform, inside the judge, much of the decomposition that GCA performs outside it.

Fine-grained evaluation. More recent work evaluates summaries at sub-summary granularity rather than assigning a single score. FineSurE assesses faithfulness at the sentence level and content coverage against a list of extracted key facts, producing a diagnostic profile rather than one number [37]. Like SummaC, and unlike this thesis, it is positioned as an evaluation framework: it tells a practitioner where a summary went wrong, but does not convert that localized judgment into supervision for training.

FActScore decomposes a generation into individual atomic facts and checks each against a knowledge source, then reports the fraction supported as a single precision score [29]. Its decompose-score-aggregate structure is the closest evaluation-side analogue to GCA's own aggregation step (Chapter 4, Section 4.3): both split a generation into small checkable units, score each independently, and combine the results into one number. The differences are the same ones that separate this thesis from SummaC and FineSurE. FActScore decomposes into atomic facts extracted from the generation itself and verifies them against an external knowledge source, whereas GCA decomposes into sentences and scores each against the source article with a single fixed judge; and FActScore's aggregate score is reported as a diagnostic metric, not converted into a pairwise preference for training a reward model. The aggregation choice it makes, a simple fraction of atomic facts supported, is nonetheless a further data point for the same lesson this thesis draws in Section 6.5.1: how per-unit scores are combined is a design decision with measurable consequences, examined here independently by two unrelated methods on two different granularities of decomposition.

2.5. Synthesis: The Gap This Thesis Addresses

Table 2.1 summarizes the closest lines of work along the dimension that distinguishes the present study: the point in the pipeline at which sentence-level information is introduced, and what each approach holds fixed while doing so.

Table 2.1 organizes this literature by *where* sentence-level information is injected, which is the dimension along which the present study is distinguished. Sentence-level factuality signals are clearly not unexplored: they have been used to mask the policy-optimization objective [14], to restructure the reward model [32], to supply dense rewards during reinforcement learning [40], to train a reward model that outputs per-step judgments directly [25], and to build evaluation metrics [21, 29, 37]. What distinguishes this thesis is the insertion point. Each of the training-oriented works above takes the preference labels as given, in most cases as response-level human judgments, and modifies what happens downstream of them. The step examined here is the construction of those labels: how the output of an automatic factuality evaluator, queried at different granularities, is converted into a pairwise preference, and how learnable the resulting labels are when everything downstream is held constant. This is attractive because automatic labels are cheap, and uncertain because they are noisier per label than human ones, so whether the additional granularity recovers more signal than noise is not obvious in advance.

2.6. Positioning of This Thesis

The contribution of this thesis is not sentence-level reward modeling itself, nor sentence-level factuality alignment in general, both of which are active areas with established

Table 2.1.: Where sentence-level information enters the pipeline in related work.

Work	Sentence-level signal enters at	Held fixed
Stiennon et al. [38]	not used	–
Lee et al. [24]	not used	–
Wu et al. [40]	reward signal during RL (human labels)	response-level objective
Gu et al. [14]	the DPO objective, as a sentence mask	preference pair construction
Qiu et al. [32]	the reward-model architecture	preference labels (human, response-level)
Lightman et al. [25]	the reward model's own output (per-step)	training objective; labels (human, per-step)
Laban et al. [21]	evaluation metric only	–
Song et al. [37]	evaluation metric only	–
Min et al. [29]	evaluation metric only	–
Zha et al. [41]	internal to the evaluator	–
This thesis	**construction of the preference label**	reward-model architecture and training procedure

results. It is a controlled comparison of *preference-construction procedures*: the downstream Bradley–Terry architecture and training procedure are held fixed while the granularity at which automatic factuality evidence is converted into pairwise preference labels is varied. No new factuality scorer is introduced; AlignScore is used off the shelf, and the measurement instrument is the learnability of the resulting labels.

Six adjacent lines of work are distinguished as follows:

- **RLAIF and LLM-as-a-judge research** [1, 13, 24, 43] studies judge reliability at the level of a whole-response verdict, rather than the effect of querying the judge at a finer granularity.
- **Fine-grained human feedback** [40] establishes that localized supervision helps when labels are reliable; the present study asks the cheaper and less certain version, in which each local label is produced automatically.
- **Sentence-level factuality alignment** [14] applies granular factuality information inside the policy-optimization objective while leaving preference construction unchanged; here the objective is unchanged and preference construction varies.
- **Sentence-level reward modeling** [32] changes the reward model's architecture while learning from conventional response-level preferences; here the architecture is fixed and the preferences vary.

- **Process reward modeling** [25] trains a reward model to output human-labeled, per-step correctness judgments directly, consumed downstream for reranking or search; here the reward model outputs a single scalar per summary, unchanged from the holistic condition, and what varies is only how that scalar's training label was constructed upstream.
- **Fine-grained evaluation frameworks** [21, 29, 37] decompose summaries in order to diagnose them rather than to supervise with them.

The comparison is conducted on preference data generated by one evaluator on one corpus, so the findings describe behavior under specific conditions rather than a general property of granular supervision (Chapter 7, Section 7.11).

3. Theoretical Background

3.1. Preference Optimization: The Bradley–Terry Model

The Bradley–Terry model [2] is a standard probabilistic model of pairwise comparison. It assumes that every item i has an underlying, unobserved scalar quality r_i, and that the probability of item i being preferred over item j in a pairwise comparison is

$$P(i \succ j) = \sigma(r_i - r_j), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}. \tag{3.1}$$

Given a dataset of observed pairwise preferences (w, l), where w (the "winner") was judged preferable to l (the "loser"), the model parameters are fit by maximizing the likelihood of the observed comparisons under Equation 3.1. Here the parameters belong to a neural network R_θ that maps an (article, summary) pair to a scalar reward, and maximizing the likelihood is equivalent to minimizing the negative log-likelihood

$$\mathcal{L}(\theta) = -\,\mathbb{E}_{(w,l)\sim\mathcal{D}}\Big[\log \sigma\big(R_\theta(w) - R_\theta(l)\big)\Big], \tag{3.2}$$

where $\mathcal{D}$ is the preference dataset, holistic or GCA in this thesis. Equation 3.2 is exactly the loss implemented in `src/reward_model/train.py` (`bradley_terry_loss`), and it belongs to the same family of reward-modeling loss used inside RLHF pipelines more broadly [31, 38]. The difference here is that the reward model R_θ is not an intermediate component on the way to fine-tuning a generation policy; it is the final object of comparison. The research question this thesis asks, whether GCA produces a more learnable preference signal than holistic scoring, is operationalized directly: does a Bradley–Terry reward model trained on GCA-derived preferences achieve higher held-out pairwise accuracy than one trained on holistic-derived preferences, under an otherwise identical training recipe?

This corresponds to the inverse reinforcement learning perspective: given a set of preference demonstrations, the task is to recover the latent reward function that best explains them. Comparing two reward models trained on two differently constructed preference datasets is then a controlled test of which construction procedure yields the more consistently recoverable latent signal, without requiring a downstream generation policy to be fine-tuned and re-evaluated (Section 3.2).

The architecture used throughout this thesis (`BradleyTerryRewardModel` in `src/reward_model/train.py`) encodes the concatenation of a truncated source article and a candidate summary with a pretrained transformer encoder, mean-pools the encoder's final hidden states over non-padding tokens, and maps the pooled representation to a scalar reward through a single linear layer. Reward-model accuracy is defined as the fraction of held-out pairs for which the model correctly ranks the chosen summary above the rejected one, i.e. $R_\theta(w) > R_\theta(l)$ (`_evaluate` in `src/reward_model/train.py`); this is the metric reported throughout Chapter 6.

3.2. Direct Preference Optimization (DPO)

Direct Preference Optimization (DPO) replaces RLHF's reward-model-plus-reinforcement-learning pipeline with a single closed-form supervised objective on a policy, defined relative to a fixed reference policy [33]. It is included here only as background on the study's earlier design: DPO was the objective originally proposed for this thesis and produced the early results reported in Appendix D, Section D.1, but it is not used in the final experiments (Chapter 4, Section 4.9).

The relevant point is structural. The DPO objective consumes exactly the same pairwise preference pairs used to train the reward models in Section 3.1, but converts them directly into a policy update, so the quality of the preference data and the behavior of the optimizer both influence the resulting summaries. Separating those two influences is what motivated measuring the preference data on its own, which is what the final study does.

3.3. AlignScore: Factual Consistency Scoring

AlignScore is a unified alignment function trained across a mixture of natural language processing tasks recast into a single "information alignment" format [41]. Given a context c, the source article in this thesis, and a claim a, a candidate summary or a single sentence from one, AlignScore produces a scalar score in $[0, 1]$ estimating the degree to which a is supported by c. This thesis uses AlignScore unmodified as the sole factuality judge for both preference-construction regimes (`src/judging/reward_model_judge.py`, class `RewardModelJudge`).

AlignScore exposes several evaluation modes, of which four are used in this thesis (the `alignscore-evaluation-mode` argument of `build_reward_preferences.py`):

- `nli_sp`: the context is first split into sentences before scoring, and the claim is scored against this sentence-split representation via AlignScore's NLI (three-way entailment/neutral/contradiction) output head. This was the default mode inherited from the earliest AlignScore-based experiments in the project.

- `nli`: the same NLI output head, without the sentence-splitting preprocessing step.
- `bin_sp` / `bin`: the same two preprocessing variants, but using AlignScore's binary (aligned / not aligned) output head instead of the three-way NLI head.

Chapter 6, Section 6.5.2 reports an empirical sweep across all four modes. The choice of mode is associated with substantial changes in the outcome, alongside the GCA aggregation formula (Chapter 4, Section 4.3). The `nli` mode produced the largest GCA advantage in that sweep and was selected as the intended final configuration for the validated $n = 1{,}000$ campaign and the scale-robustness checks (Appendix D, Section D.4), though Chapter 5, Section 5.4 reports that the confirmatory twenty-run campaigns were, due to an implementation oversight, actually trained under `nli_sp` instead. This should be read as a judge-*configuration* finding specific to this pipeline and this task, not as a general property of sentence-level aggregation independent of the underlying judge; Chapter 7 discusses what can and cannot be concluded from it.

3.4. Statistical Validation: Bootstrap Confidence Intervals and the Wilcoxon Signed-Rank Test

Two statistical tools are used throughout Chapter 6 to assess whether an observed GCA-versus-holistic accuracy gap is distinguishable from noise.

Bootstrap confidence intervals. The bootstrap [10] estimates the sampling distribution of a statistic by resampling the observed data. Given a set of paired fold-level accuracy differences $\{d_1, \ldots, d_m\}$, where $d_k = \mathrm{acc}_{\mathrm{GCA},k} - \mathrm{acc}_{\mathrm{Holistic},k}$ for cross-validation fold k, pooled across independent training seeds, a bootstrap 95% confidence interval for the mean difference $\bar{d}$ is constructed by resampling $\{d_1, \ldots, d_m\}$ with replacement B times, computing the mean of each resample, and taking the 2.5th and 97.5th percentiles of the resulting distribution of means. This thesis uses $B = 10{,}000$ resamples with a fixed NumPy random seed of 42.

Wilcoxon signed-rank test. As a complementary non-parametric test suited to a small number of paired, non-normally-distributed differences, the two-sided Wilcoxon signed-rank test [39] is reported on the same paired differences. Unlike a paired t-test it does not assume normality, which is appropriate given the small number of observations and the fact that fold accuracies are bounded proportions.

An important qualification: the unit of replication. Both procedures assume that the observations they are applied to are independent, and this determines which level the tests may legitimately be applied at. Fold-level differences are *not* independent: the five folds within a run partition a single dataset and therefore share most of their training data, so applying either procedure to 30 such observations treats correlated measurements as independent replications, which understates uncertainty and produces intervals that are too narrow and p-values that are too small. Run-level differences, by contrast, come from separately seeded training runs and satisfy the assumption.

This thesis therefore reports two kinds of figure, and the distinction should be kept in view when reading Chapter 6. Fold-level intervals and p-values, which appear only for the superseded six-run campaign (Appendix D, Section D.4), are anti-conservative and are presented as exploratory summaries of spread, never as hypothesis tests. All confirmatory statistics (the twenty-run campaigns at each dataset size, their bootstrap intervals, Wilcoxon tests, effect sizes, and equivalence tests) are computed over run-level differences, where the independence assumption holds. Chapter 7, Section 7.3 sets out what each level can support.

Effect size. A p-value indicates whether a difference is distinguishable from zero, not how large it is, so both are reported. For paired differences $\{d_1, \ldots, d_n\}$ the standardized effect size is Cohen's $d = \bar{d}/s_d$, the mean difference expressed in units of its own standard deviation [7]. Alongside it, the matched-pairs rank-biserial correlation is reported as the effect size native to the Wilcoxon test: it is the difference between the sums of positive and negative signed ranks, divided by their total, and therefore ranges from -1 (every run favors one condition) to $+1$ (every run favors the other), capturing consistency of direction independently of magnitude.

Equivalence testing. A non-significant result does not by itself establish that an effect is absent; it may equally reflect insufficient data. Establishing absence requires reversing the burden of proof, which is what the two one-sided tests (TOST) procedure does [22]. Given an equivalence bound $\pm\delta$ representing the smallest effect that would be practically meaningful, TOST tests the two null hypotheses that the true effect is at least $+\delta$ and at most $-\delta$. Rejecting both places the true effect inside $(-\delta, +\delta)$ with the stated confidence, which is positive evidence of a negligible effect rather than mere failure to detect one. This procedure is applied in Chapter 6, Section 6.3 to the $n = 5{,}000$ and $n = 10{,}000$ campaigns, where the substantive claim is precisely that no advantage remains.

Pooling across runs rather than reporting a single run is nonetheless deliberate, and responds to an early observation (Appendix D, Section D.3): a promising single-run result, a $+1.4$ percentage-point advantage from one hyperparameter configuration, did not reproduce when the same configuration was run again. Single runs at this dataset scale are therefore not reliable evidence on their own, whatever statistics are computed from their folds.

4. Methodology

4.1. Research Questions and Hypotheses

- **RQ1.** Does sentence-level factuality feedback aggregated via GCA produce a more learnable pairwise preference signal than holistic summary-level feedback, as measured by reward-model pairwise validation accuracy?
- **RQ2.** How do evaluator configuration and sentence-score aggregation strategy affect the relative learnability of holistic and GCA-derived preferences?
- **RQ3.** How stable are the observed differences across repeated reward-model training runs and dataset sizes?
- **RQ4.** Does the learnability advantage measured in RQ1 also hold when the trained reward models are scored against an independent, held-out ground truth constructed from the same factuality judge, rather than against the models' own training-style labels, and does that hold as the training-set size grows?

RQ1–RQ3 are addressed by the experiments in Chapter 6 and answered in Chapter 8. RQ2 is addressed by an exploratory sweep rather than a controlled ablation, so it supports statements about association between configuration and outcome, not about causal mechanism (Section 4.6). RQ4 was added after the $n = 1{,}000/5{,}000/10{,}000$ campaign, in direct response to a limitation that campaign left open: pairwise validation accuracy is measured against labels supplied by the same evaluator that constructed the preferences, so a reward model could in principle learn to recover those labels' idiosyncrasies rather than genuine factuality signal (Section 4.4). Section 4.8 describes the independent evaluation constructed to address this, and Section 6.4 reports its results.

Two substantive working hypotheses guided the design, alongside one methodological precaution. **H1**, the primary hypothesis, is that sentence-level aggregated preferences are more learnable than holistic preferences, because they localize the factuality signal within a long output. **H2** is that the evaluator's configuration affects whether any such difference is observable at all, so that granularity cannot be assessed independently of the judge it is applied on top of. **H3** is not a prediction about the world but a precaution against generalizing from one setting: that effects observed at one dataset scale need not persist at another, and so must be checked rather than assumed. It is stated alongside the other two because it turned out to determine the shape of the final result. The status of all three against the evidence collected is assessed in Chapter 8, Section 8.3.

4.2. Study Design

Figure 4.1 gives an overview of the full experimental pipeline before the individual stages are described. The comparison is between two procedures for converting factuality scores into pairwise preference labels. For every source article, two candidate summaries are generated once and reused by both conditions:

Holistic. The complete candidate summary is scored against the source article, producing one score per candidate. The two scores are compared to yield a pairwise preference.

GCA. The candidate summary is segmented into sentences; each sentence is scored separately against the source article; the sentence scores are aggregated into one summary-level score. The two aggregated scores are compared to yield a pairwise preference.

The following are held identical across conditions: the source articles; the two candidate summaries and the decoding settings that produced them; the evaluator family and checkpoint; the reward-model architecture; and the reward-model training and evaluation procedure. The manipulated variable is the granularity at which the evaluator is queried, together with the aggregation step that GCA requires and holistic scoring does not. Table 4.1 lists the final configuration.

4.3. Granular Credit Assignment (GCA)

Granular Credit Assignment (GCA) is the aggregation layer that converts per-sentence AlignScore scores into a single summary-level score, which is then compared against the corresponding score for the other candidate summary to construct a preference label. A candidate summary is first split into sentences with a regex-based sentence splitter (`src/judging/sentence_level.py`, `split_into_sentences`). Each sentence is then scored independently against the full source article using AlignScore (`RewardModelJudge.score_sentences`), yielding a list of per-sentence scores

[1] This is the configuration selected in Chapter 6, Section 6.5.2 and intended as the locked design. The confirmatory campaigns that produced the results in Chapter 6 were, due to an implementation oversight found during final review, actually trained under evaluation mode `nli_sp` throughout, and, at $n = 1{,}000$ specifically, under $\alpha = 0.5$ rather than $\alpha = 0.0$. This discrepancy is reported in full in Chapter 5, Section 5.4, and its consequences are discussed in Chapter 7, Section 7.2.

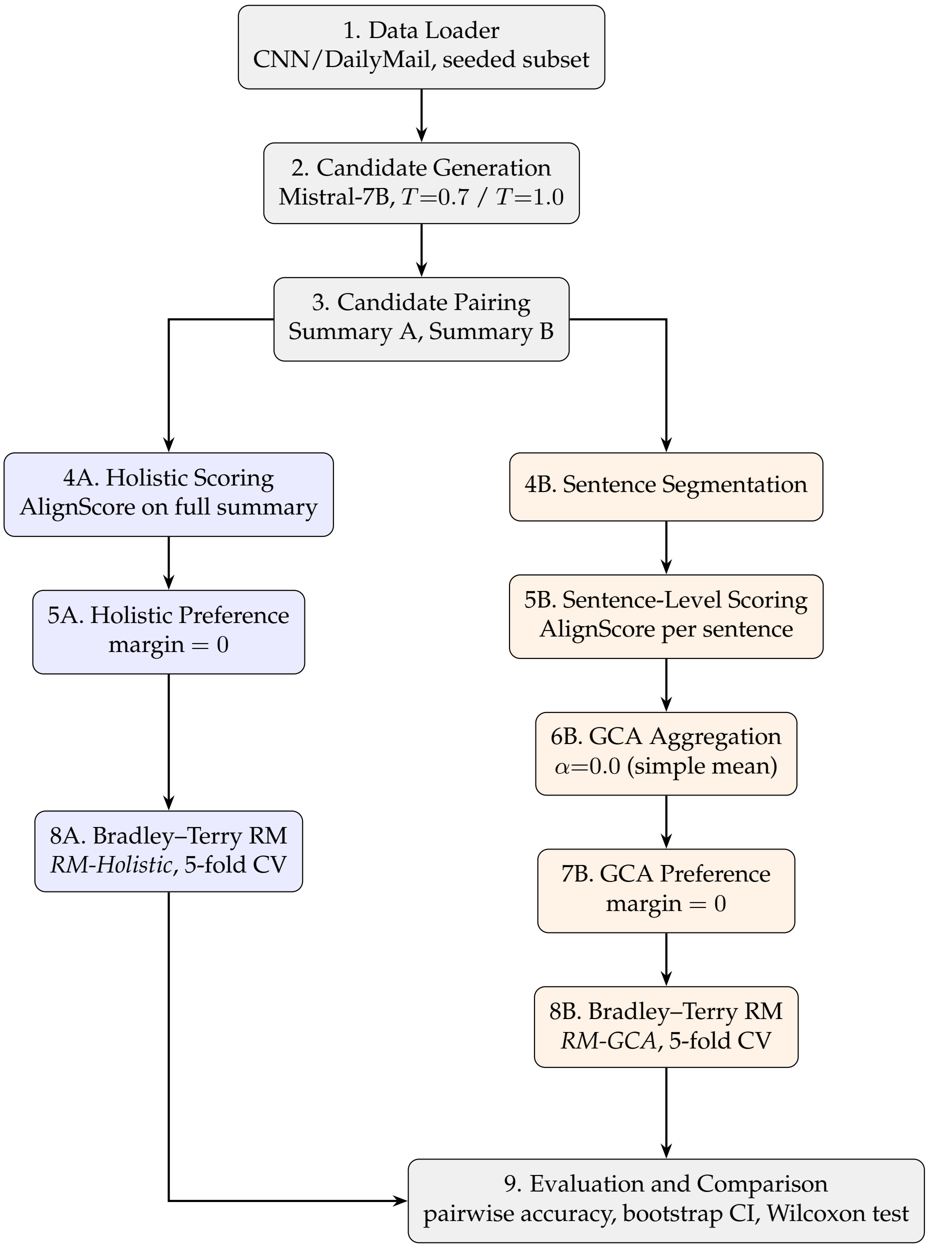


Figure 4.1.: The final pipeline. Shared setup (gray) produces one candidate pool from which two independent preference-construction branches run: holistic (blue) and GCA (orange). Each branch trains its own Bradley–Terry reward model under an identical recipe, and the two are compared in the final evaluation step. Compare to the pipeline described in the original proposal (Section 4.9), which additionally included a DPO fine-tuning stage and a cross-summary sentence-alignment step, neither of which appears here. Chapter 5, Section 5.1 walks through every stage's concrete input and output.

Table 4.1.: Final experimental configuration.

Factor	Value
Task	Single-document abstractive summarization (news)
Dataset	CNN/DailyMail; runs at $n = 1{,}000$, $5{,}000$, and $10{,}000$ (subset seed 200)
Candidate generation	Mistral-7B-Instruct-v0.3, two-temperature sampling ($T = 0.7$ / $T = 1.0$)
Evaluator	`yzha/AlignScore`, mode `nli`[1]
Manipulated variable	Feedback granularity: holistic vs. sentence-level + GCA
GCA aggregation	Simple mean ($\alpha = 0.0$)[1]
Margin	0 (all pairs used)
Reward model	Bradley–Terry, `FacebookAI/roberta-base` backbone, mean-pool + linear scalar head
RM training	epochs = 5, lr = 2×10^{-5}, batch size = 8, max length = 512, 5-fold cross-validation
Dependent variable	Reward-model pairwise validation accuracy

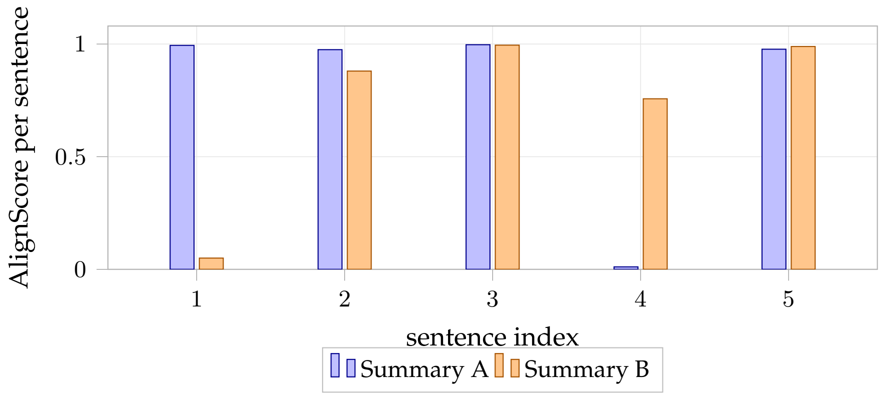
$s = (s_1, \ldots, s_n)$. These per-sentence scores are aggregated into a single summary-level score via (`src/judging/gca.py`, `aggregate_sentence_scores`):

$$\mathrm{gca}(s) = \bar{s} \cdot \left(\frac{\min(s)}{\bar{s}}\right)^{\alpha}, \qquad \bar{s} = \frac{1}{n}\sum_{k=1}^{n} s_k, \tag{4.1}$$

where $\alpha \geq 0$ is a penalty exponent. When $\alpha = 0$, Equation 4.1 reduces to the simple mean, $\mathrm{gca}(s) = \bar{s}$; larger α increasingly penalizes summaries containing at least one low-scoring, "weakest-link" sentence, even when the average sentence score is otherwise high. This aggregation rule, GCA itself, is the contribution of this thesis rather than an established method it builds on, which is why it is defined here alongside the rest of the study design rather than in Chapter 3.

Two distinct configurations of Equation 4.1 were used at different points in the project. The original, proposal-era configuration used $\alpha = 0.5$, a moderate weakest-link penalty intended to make GCA sensitive to exactly the failure mode motivating this thesis: a summary that is mostly accurate but contains one fabricated or unsupported sentence. Empirically, this formula achieved only 82.2% agreement with the corresponding holistic decision on the same 1,000 candidate pairs, and when used to construct reward-model training data it caused the GCA reward model to *underperform* the holistic reward model (56.0% vs. 57.2% mean pairwise accuracy; Appendix D, Section D.2). A subsequent optimization campaign (Chapter 6, Section 6.5.1) tested nine alternative aggregation strategies and found that the simple mean, $\alpha = 0.0$, achieved 97.6% agreement with holistic decisions, the highest of all strategies tested, and, combined with the AlignScore `nli` mode (Chapter 3, Section 3.3), was intended as the configuration behind the repeated-run GCA advantage (Appendix D, Section D.4). $\alpha = 0.0$ was accordingly locked as the final configuration; whether the `nli` mode was too is qualified in Chapter 5, Section 5.4 and Chapter 7, Section 7.2.

Figure 4.2 illustrates the difference between the two settings of α on a real candidate pair drawn from the $n = 200$ preference set. Summary A contains four well-supported sentences and one that the evaluator scores at 0.011; summary B is more uniformly mediocre. Under the simple mean the single weak sentence is diluted by the other four and A is preferred, matching the holistic decision. Under the penalty formula the minimum term dominates and the decision reverses. This is the behavior the penalty was designed to produce, and also the behavior that made it noisier in practice (Chapter 6, Section 6.5.1).



Aggregation	Summary A	Summary B	Preferred
Holistic score (full summary)	0.7910	0.7345	A
GCA, $\alpha = 0.0$ (simple mean)	0.7910	0.7345	A
GCA, $\alpha = 0.5$ (penalty)	0.0938	0.1924	B

Figure 4.2.: Worked example on pair `cnn_00233` from the $n = 200$ preference set (evaluator mode `nli_sp`). Summary A has one sentence scored 0.011 among four strong ones. The simple mean preserves the holistic ordering, while the $\alpha = 0.5$ penalty inverts it. Values are computed from the stored per-sentence scores in `data/preferences/gca_reward_preferences_200.jsonl`.

No cross-summary sentence alignment. One design point is worth stating explicitly, since sentence-level methods in this area sometimes align the sentences of one candidate to those of another before comparing them. GCA as defined here performs no such cross-summary alignment. Each candidate summary's sentences are scored independently against the source article, and the two aggregated scores are compared only afterwards. The method therefore contains no alignment step that could introduce matching errors, and correspondingly offers no opportunity to study alignment quality as a variable.

4.4. Scope of the Evaluation

The dependent variable throughout this thesis is the pairwise validation accuracy of a Bradley–Terry reward model trained on a given preference set. Within this thesis, *learnability* denotes exactly this quantity: the extent to which the preference relation encoded in a dataset can be recovered by the chosen reward-model architecture under a fixed training procedure.

Because this term is easily over-read, it is worth stating precisely what a higher accuracy does and does not establish. A higher value indicates that the generated preference relation is more consistently recoverable by the selected reward-model architecture. It does not by itself establish any of the following:

- that the preference labels agree better with human factuality judgments;
- that the labels are objectively more accurate as factuality annotations;
- that a summarization model trained on those preferences produces more factually consistent summaries;
- that GCA is generally preferable to holistic scoring.

The first two are inaccessible here because no independent human annotation was collected, so the labels produced by AlignScore [41] function as the reference standard rather than as verified ground truth. Substituting an automatic judge for human annotators is the defining move of RLAIF [1, 24], and it carries this limitation with it: the resulting labels are cheap and consistent, but unaudited. The third is inaccessible because the final study does not include a downstream policy-optimization stage (Section 4.9). The fourth is addressed empirically and answered in the negative: Chapter 6 reports configurations in which GCA does not lead. Claims made in later chapters are confined to learnability in the sense defined here.

4.5. Reward-Model Evaluation Protocol

Reward-model accuracy is evaluated by 5-fold cross-validation [19] rather than a single fixed held-out split. At the dataset sizes used here, a fixed split would leave either too little data for training or too little for a stable accuracy estimate, whereas cross-validation uses the full dataset for both and yields five accuracy readings per run.

Cross-validation alone does not account for variance introduced by the reward model's initialization and by which pairs fall into which fold, and at this dataset size that variance is large enough to reverse a single run's conclusion (Appendix D, Section D.3). Every reported comparison is therefore a repeated-run campaign, and the *run* is treated as the unit of independent replication throughout: each confirmatory result pools *twenty independently seeded runs* (seeds 1–20, fixed before any was executed) under a training procedure in which PyTorch and CUDA are seeded per fold, so that a run is fully determined by its seed (Chapter 5, Section 5.5). The same twenty-run resolution is used at all three dataset sizes, so no scale carries more evidential weight than another.

Two consequences of this design are carried through the rest of the thesis. The first concerns which level the statistics may be applied at. Fold-level differences are not independent observations: the five folds within a run partition a single dataset, so their training sets overlap heavily and their errors are correlated, and procedures assuming independence overstate the effective sample size. All confirmatory statistics reported in this thesis (the bootstrap intervals, the Wilcoxon tests, the effect sizes, and the equivalence tests of Section 6.3) are therefore computed over run-level differences, for which the assumption does hold; fold-level figures appear only as descriptive summaries of spread (Chapter 7, Section 7.3).

The second concerns campaign size. With n paired same-signed observations the smallest attainable two-sided Wilcoxon p-value is $2/2^n$, which at $n = 6$ is 0.031 and leaves no room for a single discordant run, while at $n = 20$ it is 1.9×10^{-6}. Twenty runs is thus the smallest campaign size that permits a result below the conventional threshold with a realistic margin, which is why the earlier six-run campaign could not have established run-level significance at any true effect size (Appendix D, Section D.4).

4.6. Exploratory and Confirmatory Experiments

The experiments in Chapter 6 do not all have the same evidential status, and the distinction is material to how the results should be read.

The aggregation-formula comparison (Section 6.5.1), the evaluator-mode sweep (Section 6.5.2), and the hyperparameter search (Appendix D, Section D.3) were all carried out on the $n = 1{,}000$ setting, and the final configuration was selected after observing their outcomes. These experiments are exploratory, and are reported as such: the first two in condensed form because they justify a design decision, the rest in the appendix. The configuration they produced was not specified in advance, and reporting it as though it had been pre-registered would overstate the evidence: with enough configurations examined, some separation between conditions is expected by chance alone.

The twenty-run campaign at $n = 1{,}000$ (Section 6.1) tests whether the selected configuration produces a stable effect under repetition, but it reuses the dataset on which that configuration was chosen and therefore cannot establish that it generalizes beyond that setting.

The $n = 5{,}000$ and $n = 10{,}000$ experiments provide a weaker check than would be ideal, for a reason that must be stated explicitly. All three subsets are drawn from the same CNN/DailyMail test split by the same deterministic procedure at the same seed, differing only in the number of samples requested, and they are consequently nested rather than disjoint (Section 4.7). The larger-scale runs therefore re-use almost all of the data on which the configuration was selected, and are best understood as measurements of how the effect behaves as the dataset grows, not as out-of-sample replication on independent data. No experiment in this thesis evaluates the selected configuration on data disjoint from the set used to choose it.

4.7. Relationship Between the Dataset Sizes

The three dataset sizes used in the final experiments are not independent samples, and this constrains how the larger-scale runs can be interpreted.

All three subsets are produced by the same deterministic procedure (`src/data/subset.py`, `select_subset`) from the same CNN/DailyMail test split, under the same length filters and the same subset seed (200). Only the number of requested samples differs. Because the selection draws from one eligible pool of 10,896 articles with a fixed random seed, the resulting subsets overlap heavily. Reproducing the selection with the recorded parameters gives the intersections in Table 4.2, shown schematically in Figure 4.3.[2]

Table 4.2.: Overlap between the subsets used in the final experiments. All three use subset seed 200 and differ only in sample count.

Pair	Shared sample IDs	Share of the smaller set
$n = 1{,}000$ and $n = 5{,}000$	980	98.0%
$n = 1{,}000$ and $n = 10{,}000$	999	99.9%
$n = 5{,}000$ and $n = 10{,}000$	5,000	100%

The $n = 5{,}000$ subset is contained entirely within the $n = 10{,}000$ subset, and the $n = 1{,}000$ subset is almost entirely contained within both. The $n = 10{,}000$ set additionally covers 91.8% of the whole eligible pool, so at that size little of the test split remains unused in any case.

[2] Computed by re-running `select_subset` with the parameters recorded in `configs/subset_1000.yaml` and `slurm/generate_candidates_scale.sh` (`seed` = 200, test split, article limit 8,000 characters, summary limit 2,000 characters) against the local copy of the test split in `data/cnn_dailymail_test_full`, and comparing the resulting SHA-256-derived sample IDs. The candidate and preference files for these runs are not retained in the repository, so the comparison reproduces the selection rather than reading the stored artifacts; the selection is deterministic given these parameters.

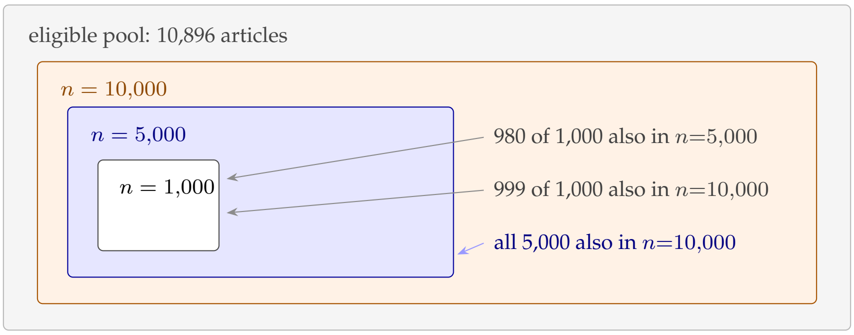


Figure 4.3.: The three subsets are nested rather than independent. Box sizes are schematic; the counts are the measured intersections reported in Table 4.2.

Two consequences follow, and both are carried through the rest of the thesis. The larger-scale experiments do not constitute out-of-sample replication: they re-use almost all of the data on which the configuration was selected, and therefore measure how the effect behaves as further data is added to the same pool rather than whether it transfers to new data. And because the samples are nested, the three results are not statistically independent of one another, so the differences between them should not be read as three separate estimates of the same quantity. Drawing the larger subsets with different seeds would have made them independent, at the cost of the nesting property the generation script was written to provide; this is recorded as a limitation in Chapter 8.

4.8. Independent Ground-Truth Construction

The evaluation described above has two properties that limit what it can establish, both noted in Section 4.6 and Section 8.4: the labels a trained reward model is scored against are supplied by the same evaluator that constructed its training preferences, and the $n = 5{,}000/n = 10{,}000$ checks reuse almost all of the data the configuration was selected on rather than evaluating it on disjoint data. RQ4 was formulated to address both at once, by scoring the already-trained reward models against a held-out set built specifically to be disjoint from every training pool used in this thesis, under an evaluation criterion that does not depend on how any individual training pair happened to be labeled.

Held-out article selection. 500 articles were drawn from the same CNN/DailyMail test split, using `select_disjoint_subset` (`src/data/subset.py`): the function reconstructs the exact `rng.sample()` call that produced the $n = 10{,}000$, seed-200 draw used throughout this thesis, removes those indices from the eligible pool, and draws 500 samples from what remains under a new seed (999). Because the $n = 5{,}000$ subset is fully contained in the $n = 10{,}000$ draw and the $n = 1{,}000$ subset is 99.9% contained in it (Table 4.2), excluding the $n = 10{,}000$ draw is sufficient to make the held-out set disjoint from all three training pools used in the main campaign; two further training pools introduced later in this work, at $n = 2{,}000$ and $n = 3{,}000$ (Section 6.4.3), were verified by direct index comparison to be exact subsets of the $n = 10{,}000$ draw and are therefore disjoint from the held-out set for the same reason.

Candidate generation and scoring. Two candidate summaries were generated per held-out article with the same Mistral-7B-Instruct-v0.3 pipeline and the same two decoding temperatures used throughout this thesis (Section 4.2), giving 1,000 candidate summaries in total. Every candidate was scored with AlignScore under the locked configuration used for the main campaign: sentence-level scoring, GCA aggregation with $\alpha = 0.0$ (Section 6.5.1). This GCA-aggregated score is used only to construct the evaluation criterion below; it is never used as a training signal for the held-out set, and no reward model is trained on these 500 articles.

Evaluation criterion. Two ground-truth constructions were used. The first pairs each article's own two candidates directly, mirroring the structure of the training preferences; the two candidates are generated by the same model at similar temperatures, so their quality gap is typically small (mean GCA score gap approximately 0.04). The second pools all 1,000 scored candidates across all 500 articles, sorts them by GCA score, and compares the top $X\%$ against the bottom $X\%$ for $X \in \{25, 10, 5\}$, which produces progressively larger, more unambiguous quality gaps between the two sides at the cost of fewer summaries per comparison (Section 6.4). At $X = 5$, every high-scoring summary is compared against every low-scoring summary (2,500 pairs from 50 summaries per side) rather than one random pairing, to remove pairing-order noise from the smallest, highest-signal condition. A trained reward model's *success rate* on a given construction is the fraction of comparisons in which it scores the higher-GCA-score summary above the lower-GCA-score one; this is the dependent variable throughout Section 6.4, and it is computed identically for the holistic and GCA reward models, using the same fixed ranking for both.

That last point requires a sharp qualification. The ranking itself, the ground-truth criterion both models are scored against, is constructed using GCA's own aggregation rule (sentence-level AlignScore, mean-pooled at $\alpha = 0.0$), the same rule the GCA reward model is trained to predict. The holistic reward model is trained to predict a different rule (whole-summary AlignScore) and is then scored against a criterion built from the rule it never saw. The *data* are genuinely held out for both conditions, no reward model trains on these 500 articles, but the *criterion* is in-distribution for GCA and out-of-distribution for holistic in a way that plain data held-out-ness does not fix. This does not mean the RQ4 result is void: a higher success rate for GCA under a GCA-defined ranking is still evidence that GCA's aggregated score is a more decisive signal by its own standard, and the two same-model comparisons (GCA vs. its own criterion, holistic vs. the same criterion) are still informative relative to each other. But it is not a symmetric test that treats the two conditions identically, and the possibility that part of GCA's RQ4 advantage reflects this asymmetry rather than a general precision advantage cannot be ruled out with the data collected here. Scoring the held-out pool under a holistic ranking as well, and reporting GCA's and holistic's success against both criteria, would settle this and is recorded as a limitation (Chapter 8, Limitation 4); it was not attempted here because it requires rescoring 1,000 held-out summaries with AlignScore under a second configuration, which is new computation this thesis does not undertake at this stage.

Checkpoint retention. The training procedure used for RQ1–RQ3 (Section 4.5) discards each fold's trained weights after recording its accuracy, since only the accuracy is needed there. RQ4 requires scoring the same trained model against a second, independent criterion, so the training procedure was changed for this evaluation to save the full reward-model checkpoint (encoder weights and reward head) for every run, rather than only its cross-validation accuracy.

4.9. Evolution of the Study Design

The study reported here is narrower in scope than the one outlined in the original research plan, and the reasons are methodological.

The initial design used the generated preferences directly for DPO-based policy optimization, evaluating factual consistency on the fine-tuned model's own generations. Early downstream experiments made it difficult to attribute observed differences to any single stage: a change in generated-summary factuality could originate in the preference-construction procedure, in the quality of the reward signal, or in the policy-optimization step, and the design provided no way to separate these. Increasing the number of stages also increased the number of variance sources standing between the manipulated variable and the measured outcome.

The final study therefore isolates an earlier and more controlled diagnostic question: whether the preference data produced by each construction procedure is recoverable by a fixed reward-model architecture. Removing the policy-optimization stage removes both reinforcement-learning dynamics and decoding variance from the measurement path, leaving a comparison in which granularity is the principal difference between conditions.

The claims of this thesis are correspondingly narrower. The work evaluates the learnability of the preference signal rather than demonstrating improvements in generated-summary factuality (Section 4.4). Two further consequences follow from the same change. Margin-based filtering of near-tie pairs was removed, since AlignScore produces continuous scores for which exact ties are effectively impossible, and the threshold discarded roughly one fifth of the data without a demonstrated benefit. The reliability protocol designed for a generative LLM judge, comprising confidence gating, order randomization, and rationale requirements, no longer applies once the evaluator is a deterministic model that is never prompted and returns no rationale; a planned human audit of judge rationales was not carried out and is recorded as a limitation in Chapter 8.

4.10. Validity Considerations

Ethical and practical considerations. All experiments use CNN/DailyMail [16, 30], a publicly available corpus of news articles and human-written summaries; no personally sensitive data is collected. Because both the evaluator's judgments and the generated summaries can be incorrect, results are reported as measurements of preference-data learnability rather than as statements about factual correctness, and model outputs are not presented as verified fact. Compute use was bounded by working with fixed-size subsets.

Candidate generation and decoding temperature. The two candidates in each pair are produced by the same model at different decoding temperatures ($T = 0.7$ and $T = 1.0$). This introduces a potential confound: preference structure may correlate with systematic stylistic differences induced by temperature rather than with factuality alone. The preference data stored in the repository allow the asymmetry to be quantified directly, and Table 4.3 reports it.[3]

Table 4.3.: Proportion of pairs in which the lower-temperature candidate ($T = 0.7$) is preferred, on the $n = 200$ preference set (evaluator mode `nli_sp`, $\alpha = 0.5$).

Condition	Threshold	T=0.7 preferred	T=1.0 preferred	Share
Holistic	margin = 0	137	63	68.5%
GCA	margin = 0	126	74	63.0%
Holistic	margin = 0.05	109	45	70.8%
GCA	margin = 0.05	97	60	61.8%

The asymmetry is substantial under both conditions and consistently larger under holistic scoring, and it is not an artifact of the margin threshold. Two implications follow. Features correlated with decoding temperature, such as summary length or lexical

[3] Computed from `data/preferences/holistic_reward_preferences_200.jsonl` and `data/preferences/gca_reward_preferences_200.jsonl`. The margin-0 rows recompute the decision from the stored scores at the threshold used in the final configuration; the margin-0.05 rows reproduce the stored decisions, whose aggregate counts also appear in `reports/reward_model_judging_summary.csv`. Preference files for the final $n = 1{,}000$ configuration are not retained in the repository, so these proportions could not be recomputed under the final evaluator mode.

conservativeness, may carry a non-trivial share of the label information, so a reward model could recover part of the preference relation without representing factuality as such. And because the asymmetry differs between conditions by roughly five percentage points, the two preference sets are not equivalent in this respect either, which means the comparison between them is not perfectly clean with respect to temperature.

These figures come from the $n = 200$ set scored under the earlier evaluator configuration and need not hold exactly at the final one. They are reported as evidence that the confound is real and measurable, not as a quantification of its effect on the reported results. Controlling candidate generation more explicitly, by sampling both candidates at one temperature, would remove the confound at source and is recorded as a limitation (Chapter 8, Limitation 13).

How much label information do surface features actually carry? The concern above is only as serious as the amount of label information that temperature-correlated surface features actually carry, and that is measurable directly on the same stored data. Table 4.4 reports the accuracy of the cheapest possible classifiers, ones that ignore the article entirely and decide from a single surface property of the two candidate summaries, against the same decisions the evaluator made.[4]

Table 4.4.: Accuracy of surface-feature classifiers against the evaluator's own decisions, $n = 200$ preference set (mode `nli_sp`, $\alpha = 0.5$). Each row states the accuracy of the better of the two opposing rules (prefer longer/shorter, prefer more/fewer sentences). Mean lengths are in characters.

Predictor	Holistic	GCA
Prefer the longer summary	0.494	0.465
Prefer the shorter summary	0.506	0.535
Prefer more sentences	0.519	0.475
Prefer fewer sentences	0.481	0.525
Mean length, chosen	865.6	840.5
Mean length, rejected	863.5	846.2

None of these predictors performs meaningfully above chance. The mean length of the chosen summary differs from that of the rejected summary by 2 characters under holistic scoring and by -6 characters under GCA, on summaries averaging roughly 850 characters, and sentence counts differ by 0.01 and -0.28 respectively. Length and sentence count, the two most obvious surface proxies for decoding temperature, therefore do not carry the preference signal, and a reward model cannot be recovering its accuracy from them.

[4]Computed by `analysis/surface_feature_baselines.py` from the same two preference files as Table 4.3, restricted to the pairs carrying an A/B decision (154 holistic, 157 GCA).

This narrows the confound without eliminating it, and the distinction matters. The 68.5% and 63.0% asymmetries in Table 4.3 are not themselves a shortcut available to the reward model: the model is trained on (article, chosen) and (article, rejected) text pairs and never observes which candidate came from which temperature, or which position it occupied (Chapter 5, Section 5.5). For the asymmetry to inflate reward-model accuracy, some *textual* property correlated with temperature would have to be learnable from the summary itself. Table 4.4 rules out the two most likely candidates for such a property. Subtler lexical or stylistic correlates are not ruled out, which is recorded as a limitation (Chapter 8, Limitation 13), but the simple version of the objection does not hold.

Threats to validity. Table 4.5 summarizes the threats considered and how each is handled.

Table 4.5.: Threats to validity and how each is addressed.

Threat	Status
Single evaluator (AlignScore)	Not mitigated; all results are conditional on this evaluator's judgments, which are not independent ground truth.
Evaluator and aggregation sensitivity	Measured directly via the formula comparison and evaluator-mode sweep (Chapter 6).
Fold- and run-level noise	Addressed by twenty independently seeded runs at each of the three dataset sizes, under fully deterministic per-fold seeding (Section 4.5).
Non-independence of pooled folds	Mitigated by testing at the run level, where the independence assumption holds; fold-level figures are reported only as descriptive summaries of spread (Section 4.5).
Configuration selected on observed results	Acknowledged, not remedied; the larger-scale subsets are nested supersets of the selection data (Section 4.7), so no run evaluates the configuration on independent data.
Scale sensitivity	Examined explicitly rather than assumed away (Section 6.2).
Decoding-temperature confound	Quantified where data permit; not controlled for in the design. Length and sentence count are excluded as carriers of the signal (Table 4.4); subtler lexical correlates remain untested (see above).
Source-document truncation	Documented in Chapter 5, Section 5.5; discussed as a limitation.
Weak absolute accuracy	Both conditions remain near chance (Chapter 7, Section 7.4).

5. Implementation

5.1. Pipeline Overview

Figure 4.1 (Chapter 4, Section 4.2) shows the final pipeline. It differs from the pipeline described in the original proposal in exactly the ways described in Section 4.9: there is no DPO fine-tuning stage, no MoDPO [44] branch, and no cross-summary sentence-alignment step. The pipeline has four stages. First, a fixed number of CNN/DailyMail articles are selected deterministically (Section 5.2). Second, two candidate summaries are generated per article at different sampling temperatures (Section 5.3). Third, the same candidate pool is scored twice, once holistically and once at the sentence level with GCA aggregation, to produce two separate preference datasets (Section 5.4). Fourth, a Bradley–Terry reward model is trained independently on each preference dataset and the two are compared (Section 5.5).

Table 5.1 states the input and output of every stage in Figure 4.1 explicitly. Stages 4A/5A form the holistic branch, stages 4B/5B/6B/7B form the GCA branch, and stage 9 is the only point at which the two branches are compared against each other.

5.2. Data

All experiments use CNN/DailyMail [16], version `3.0.0` as distributed through the Hugging Face `datasets` library. Subset selection (`src/data/subset.py`, function `select_subset`) filters the chosen split to articles under 8,000 characters and reference summaries under 2,000 characters, then draws a fixed-size sample from the eligible pool using Python's `random.Random` seeded deterministically. Each sample is assigned an ID of the form `cnn_{index}_{hash}`, where `hash` is the first 12 hexadecimal characters of the SHA-256 digest of the article text (`make_sample_id`). This ties every sample's identifier to its content rather than to its position in the dataset, so the same article always receives the same ID regardless of which subset size or seed selected it.

Table 5.2 lists every subset used across the project. The $n = 1{,}000$, $n = 5{,}000$, and $n = 10{,}000$ subsets all use subset seed 200 and the same filtering parameters, differing only in the number of samples requested. They are therefore *nested rather than disjoint*: the smaller sets are almost entirely contained in the larger ones. Chapter 4, Section 4.7

Table 5.1.: Input and output of each stage of the final pipeline shown in Figure 4.1.

Stage	Input	Output
1. Data Loader	CNN/DailyMail dataset	N seeded article samples with content-derived IDs
2. Candidate Generation	Article text	2 candidate summaries per article (T=0.7, T=1.0)
3. Candidate Pairing	2 candidate summaries	Summary A / Summary B pair
4A. Holistic Scoring	Article + full summary	1 AlignScore factuality score per summary
4B. Sentence Segmentation	Full summary	List of sentences
5A. Holistic Preference	Score(A), Score(B)	chosen/rejected label (or `no_preference`)
5B. Sentence-Level Scoring	Article + each sentence	1 AlignScore score per sentence
6B. GCA Aggregation	All sentence scores for one summary	1 aggregated score per summary (α=0.0)
7B. GCA Preference	Aggregated score(A), score(B)	chosen/rejected label (or `no_preference`)
8A. Train RM-Holistic	Holistic preference pairs	Trained Bradley–Terry RM, 5-fold CV accuracy
8B. Train RM-GCA	GCA preference pairs	Trained Bradley–Terry RM, 5-fold CV accuracy
9. Evaluation & Comparison	Both conditions' accuracies, 20 seeds	Mean gap, bootstrap 95% CI, Wilcoxon p-value

quantifies the overlap and states its methodological consequence. Earlier subsets at seeds 42 and 100 were drawn with different seeds and are not part of the final comparison.

Table 5.2.: CNN/DailyMail subsets used across the project.

n	Subset seed	Purpose	Notes
200	42	Early DPO-based prototype	Superseded; see Appendix D, Section D.1
495	100	First Bradley–Terry RM training set	Drawn with a different seed from the $n = 200$ set
1,000	200	Final repeated-run campaign	6 RM training runs; seeds 42, 42, 7, 100, 314, 2026; later extended to 20 (Section 6.1)
5,000	200	Scale-robustness check	Single training run; later extended to 20 (Section 6.2)
10,000	200	Scale-robustness check	Single training run; later extended to 20 (Section 6.2)
2,000	200	RQ4 crossover check	5 training runs; exact subset of the $n = 10{,}000$ draw (Section 6.4.3)
3,000	200	RQ4 crossover check	5 training runs; exact subset of the $n = 10{,}000$ draw (Section 6.4.3)
500	999	RQ4 held-out ground truth	Disjoint from the $n = 10{,}000$ draw by construction; never used to train any reward model (Section 4.8)

5.3. Candidate Generation

Candidate summaries are generated with Mistral-7B-Instruct-v0.3 [18], loaded in `bfloat16` without quantization (`src/generation/candidates.py`). On the A100-SXM4-40GB GPUs used for this project, the model occupies roughly 14 GB in `bfloat16`, comfortably within budget, so 4-bit quantization, though supported in the code via `BitsAndBytesConfig`, is disabled in the configuration actually used (`load_in_4bit: false` in `configs/generation.yaml`). The tokenizer uses left-padding, required for batched generation with a decoder-only model, and articles are truncated to 1,024 input tokens.

Each article is summarized twice, using the fixed prompt template `"Summarize the following news article in a concise paragraph. ..."`, with two different sampling configurations: a low-temperature setting ($T = 0.7$, top-$p = 0.9$) and a high-temperature setting ($T = 1.0$, top-$p = 0.95$), generating up to 256 new tokens each. These two outputs form the candidate pair ("summary A" and "summary B") that both the holistic and GCA preference construction stages score. Generation is resumable: the script checks which sample IDs already appear in the output file and only generates candidates for the remainder, which matters in practice on a shared cluster where a job may need to be resubmitted after a queue timeout.

5.4. Preference Construction

Both preference-construction regimes are implemented in `src/judging/build_reward_preferences.py` and share a single judge object, `RewardModelJudge` (`src/judging/reward_model_judge.py`), configured to use the AlignScore backend.

In the holistic regime (`judge_holistic_pair`), each full candidate summary is scored once against the article with `judge.score(article, summary)`, and the two resulting scores are compared with a margin guard (`make_decision`): summary A is preferred if its score exceeds summary B's by more than the margin, summary B is preferred in the opposite case, and the pair is marked `no_preference` otherwise. With the final margin of 0 (Chapter 4, Section 4.9), every pair with a nonzero score difference produces a decision.

In the GCA regime (`judge_gca_pair`), each candidate summary is first split into sentences with a regex-based splitter, then every sentence is scored independently against the article with `judge.score_sentences(article, sentences)`. The resulting list of per-sentence scores is passed to `aggregate_sentence_scores(scores, alpha)` (Chapter 4, Equation 4.1) to produce a single GCA score per summary, and the same margin-guarded comparison determines the preference. Every output record retains the full sentence-level detail, including each sentence's individual AlignScore score and a count of "low-faithfulness" sentences (score below 0.5), which supports the qualitative disagreement analysis in Chapter 6.

The locked final configuration is invoked as:

```
python src/judging/build_reward_preferences.py
  -mode both -alpha 0.0 -margin 0
  -alignscore-evaluation-mode nli
  -alignscore-backbone FacebookAI/roberta-base
```

A discrepancy between this intended configuration and what the confirmatory campaigns actually ran, found during final review. This is reported here in full because it materially affects how several results in this thesis should be read, and because finding it and reporting it precisely is more useful to the reader than leaving it undiscovered.

The command above is what *should* produce the preference data behind every confirmatory campaign, and it is what a small number of exploratory runs earlier in the project, including the judge-mode sweep itself (Chapter 6, Section 6.5.2) and the six-run campaign that followed it (Appendix D, Section D.4), were in fact built from. It is not, however, what the twenty-run extended campaigns at $n = 1{,}000$, $n = 5{,}000$, and $n = 10{,}000$ were trained on, nor what the $n = 2{,}000/n = 3{,}000$ crossover checks (filtered from the $n = 10{,}000$ file) or the RQ4 checkpoints (trained on the same files as RQ1–RQ3) were trained on. Checking the stored preference files directly against the training scripts that consumed them shows the following.

- **`slurm/build_preferences_scale.sh`**, used to build the $n = 5{,}000$ and $n = 10{,}000$ preference files, never passes `-alignscore-evaluation-mode` on its command line. The flag's default in `src/judging/build_reward_preferences.py` is `nli_sp`, not `nli`, so both files were built under `nli_sp`. The aggregation exponent is unaffected: both files correctly carry $\alpha = 0.0$, since `-alpha 0.0` is passed explicitly.
- The $n = 1{,}000$ file at `data/preferences_1000/` is dated to the original, pre-optimization run of `slurm/build_preferences_1000.sh` (Appendix D, Section D.2), which explicitly passes `-alpha 0.5` and, like the scale script, never sets `-alignscore-evaluation-mode`, so it too defaults to `nli_sp`. A separate file built correctly under `nli`, $\alpha = 0.0$ exists at `data/preferences_1000_alpha0_mode_nli/`. This correct file is what the six-run campaign (Appendix D, Section D.4) actually used: its four "seed validation" runs were each built fresh by `slurm/mode_nli_seed_confirm.sh`, which does pass `-alignscore-evaluation-mode nli` explicitly, and the run-metadata JSON files for those four runs (`outputs/mode_sweep_alpha0/mode_nli_seed\{7,100,314,2026\}/`) confirm both the correct configuration and an exact match to the six-run figures reported in Appendix D, Table D.3. The twenty-run extended campaign that supersedes it (Section 6.1) does not share this property: its own stored run-metadata, for every one of its twenty seeds, records `data/preferences_1000/` as the input, i.e. the stale $\alpha = 0.5$/`nli_sp` file, not `data/preferences_1000_alpha0_mode_nli/`. The RQ4 checkpoints at $n = 1{,}000$ were trained from the same stale file. In short: the six-run campaign at $n = 1{,}000$ is correctly configured; the twenty-run campaign that supersedes it, and every RQ4 result built on it, is not.
- The RQ4 held-out ground truth (Chapter 4, Section 4.8) is not affected in the same way: the build script `slurm/build_ground_truth.sh` does pass

`-alignscore-evaluation-mode nli` explicitly, and this is the only stage in the pipeline for which the intended and executed configurations agree.

The practical consequence is that every reward model whose accuracy is reported as a confirmatory result in this thesis, the RQ1/RQ3 twenty-run campaigns at all three scales, the $n = 2{,}000/n = 3{,}000$ crossover checks, and the RQ4 checkpoints, was trained on preference data built under `nli_sp`, not the `nli` configuration that Section 6.5.2 selected and that this thesis describes elsewhere as locked. At $n = 1{,}000$ specifically, the aggregation exponent is additionally the original $\alpha = 0.5$ penalty formula rather than the locked $\alpha = 0.0$ simple mean. The RQ4 ground-truth criterion itself was scored under the correctly-locked `nli`, $\alpha = 0.0$ configuration, so the training data and the ground-truth criterion in RQ4 were built under different judge configurations from one another, a further consequence addressed in Chapter 6, Section 6.4 and recorded as a limitation in Chapter 8, Section 8.4.

Two things are unaffected by this. First, every reported number is a genuine measurement of the data that was actually used: the run-metadata files trace every seed's result back to the exact preference file consumed, and the accuracies, gaps, and statistics reported throughout Chapter 6 are correctly computed from those runs. What is incorrect is the description, repeated at several points in this thesis prior to this discovery, of that data as the `nli`, $\alpha = 0.0$ configuration. Second, the comparison between holistic and GCA within each campaign remains fair: both conditions in a given campaign were built from the same candidate pool under the same (if mislabeled) judge configuration, so the discrepancy does not advantage one condition over the other within a run. What it does undermine is the specific mechanistic account in Chapter 7, Section 7.2, that the `nli` mode's absence of internal decomposition is why GCA shows an advantage: the confirmatory campaigns were not run under that mode, and that account is revisited in light of this finding at the same location.

Why the aggregation exponent differs between $n = 1{,}000$ and the larger scales, and what that means for the scale comparison. The two discrepancies above do not have the same origin. The $n = 5{,}000$ and $n = 10{,}000$ preference files were built after $\alpha = 0.0$ had already been locked in, by a script that passes `-alpha 0.0` explicitly and only omits the mode flag; their aggregation exponent is correct. The $n = 1{,}000$ campaign, however, read a preference file left over from before that decision was made, so it is not only the wrong mode but also the wrong exponent, $\alpha = 0.5$ instead of 0.0. This means the comparison across scale in Section 6.2 changes two things between $n = 1{,}000$ and the two larger sizes, dataset size and the aggregation exponent, not dataset size alone, and a genuine effect of the exponent could in principle imitate a genuine effect of scale.

Two pieces of evidence bear on this, though neither settles it. The six-run campaign (Appendix D, Section D.4), run on preference data correctly built under $\alpha = 0.0$, also finds GCA ahead at $n = 1{,}000$ (mean +3.08 percentage points, five of six runs), close in direction and magnitude to the twenty-run campaign's $+3.95$ points under $\alpha = 0.5$.

A single-run check under $\alpha = 0.0$ in the mode sweep (Chapter 6, Table 6.15, `nli` row) likewise favors GCA. Both configurations of α, in other words, produce a positive gap at $n = 1{,}000$ in the evidence collected so far, which weighs against the exponent alone being responsible for the advantage. What is missing is the comparison that would settle the question directly: a twenty-run campaign at $n = 1{,}000$ under the correctly locked $\alpha = 0.0$, matching the resolution of the $n = 5{,}000/n = 10{,}000$ campaigns exactly. Without it, the possibility that part of the decline across scale reflects the change in α rather than scale alone cannot be excluded, and this is recorded as a limitation (Chapter 8, Section 8.4).

This was found during a final review pass after the experimental phase of the project had concluded. Re-running the affected campaigns under the correctly-configured data would settle whether the reported effects persist under the intended configuration; it was not attempted here, both because of the time remaining before submission and because it constitutes new experimental work beyond what had already been agreed with the supervisors for this stage of the project.

5.5. Reward-Model Training

Reward-model training is implemented in `src/reward_model/train.py` (`BradleyTerryRewardModel`, `bradley_terry_loss`) and orchestrated by `src/reward_model/run_training.py`, which trains the holistic and GCA conditions from a single invocation. Two training modes are used in this thesis, selected by whether `run_training.py` is invoked with `-kfold`. Every RQ1–RQ3 result (Chapter 6, and the superseded campaigns in Appendix D) uses k-fold cross-validation (`_kfold_cv` in `run_training.py`): the preference pairs are shuffled using a seeded `random.Random` instance, split into $k = 5$ contiguous folds, and for each fold a freshly initialized reward model is trained on the other four folds and evaluated on the held-out one; only each fold's accuracy is kept, and the trained weights are discarded. The RQ4 ground-truth evaluation (Chapter 6, Section 6.4) instead omits `-kfold`, which routes through `train_reward_model` in `train.py`: a single model is trained on the full preference set for a given seed, and its checkpoint (encoder weights and reward head) is written to disk rather than discarded, since RQ4 needs the trained model itself, not only a cross-validation accuracy (Section 4.8). Both modes share the same architecture, loss, optimizer, and hyperparameters; they differ only in whether the data is split into folds and in whether the trained weights are kept. One implementation detail is worth stating precisely, since it affects how "training seed" should be interpreted. The `-seed` argument is passed to Python's `random` module, which governs the fold-assignment shuffle in the k-fold path (and the train/seed initialization in the non-k-fold path).

In the version of `_kfold_cv` used for the original six-run campaign (Appendix D, Section D.4), PyTorch's own generator was not seeded anywhere in the cross-validation path, so every other source of stochasticity was uncontrolled: the randomly initialized reward head, the shuffling of training batches by the `DataLoader`, and the dropout masks applied during training. Changing `–seed` therefore changed only the fold assignment, while these other sources varied between process executions regardless of the seed value, which is why two executions launched with the same seed value produced different results (Appendix D, Section D.4) and why the original campaign could not be extended past six runs without first addressing this.

This was fixed before the twenty-run campaign that produces every headline result in this thesis: `_kfold_cv` now seeds PyTorch and CUDA explicitly at the start of each fold, supplies an explicit generator to the training `DataLoader`, and seeds the `DataLoader` worker processes. Under this corrected procedure, a run is fully determined by its seed value on fixed hardware, and this is the training procedure that produced every twenty-run campaign reported in Chapter 6 (Sections 6.1 and 6.2 onward) and the reproduction instructions in Appendix B. The original, unseeded behavior is retained here as the explanation for why the six-run campaign's individual runs differ even at a repeated seed value, and is discussed further as a source of run-to-run variation in Chapter 7, Section 7.9.

The final training configuration, used for every reported reward-model result, is: `FacebookAI/roberta-base` [28] backbone, 5 epochs, batch size 8, learning rate 2×10^{-5} with a linear warmup over 10% of total steps, maximum sequence length 512 tokens (article truncated to 2,000 characters before tokenization, concatenated with the summary), and 5-fold cross-validation. This is confirmed by `slurm/train_rm_1000.sh`, the Slurm script used for the headline $n = 1{,}000$ result. One inconsistency in the codebase is worth flagging explicitly, since it could otherwise look like an error in the reported results: the in-code default for `–backbone` in both `train.py` and `run_training.py` is `microsoft/deberta-v3-base`, a leftover from an earlier version of the training script. Every Slurm script used to produce a result reported in this thesis overrides this default and passes `–backbone FacebookAI/roberta-base` explicitly; the `deberta-v3-base` default was never actually used to produce a reported number. The one exception is the very first Bradley–Terry result on the $n = 495$ set (Appendix D, Section D.2), which predates the standardization on `roberta-base` and is reported as such.

Source-document truncation. The reward model does not see the full source article. Two limits apply in sequence. The article string is first truncated to its leading 2,000 characters (`–max-article-chars`, applied in `PreferencePairDataset`), and the truncated article is then concatenated with the candidate summary and tokenized under a 512-token cap (`–max-length`), so the article prefix is shortened further to make room for the summary. Subset selection admits articles up to 8,000 characters (Section 5.2), so for the longer articles in the pool the reward model observes well under half of the source text.

This matters for interpreting the reward-model results. The evaluator that produced the preference labels scores summaries against the article under its own chunking scheme, whereas the reward model must reconstruct those labels from a prefix only. Where the evidence supporting or contradicting a claim lies beyond the retained prefix, that evidence is simply unavailable to the reward model, and the corresponding preference is not recoverable from its input in principle rather than merely in practice. How large this effect is was not measured here, and the near-chance accuracies reported in Chapter 6 should not be attributed to truncation without evidence; it is one plausible contributing factor among several discussed in Chapter 7, Section 7.4. Varying `-max-article-chars` while holding everything else fixed would isolate it, and this is recorded as a limitation (Chapter 8, Limitation 14).

5.6. Compute Infrastructure

All experiments run on MOGON NHR (`mogon-nhr-01.zdv.uni-mainz.de`), the University of Mainz's high-performance computing cluster, using Slurm partition `a100dl` under account `nhr-haloed`, on NVIDIA A100-SXM4-40GB GPUs. All Slurm scripts producing results reported in Chapter 6 exclude one cluster node found to have a faulty CUDA runtime, via `#SBATCH -exclude`; this affects scheduling only and not the computation performed.

5.7. Reproducibility

The pipeline is config-driven throughout: dataset, generation, and judging parameters live in `configs/*.yaml` and can be overridden per run from the command line, and every script accepts explicit flags for the parameters that matter for a given experiment (margin, α, AlignScore evaluation mode, reward-model hyperparameters, and so on) rather than hardcoding them.

Seeding is now complete for the procedure used to produce every headline result. Subset selection and cross-validation fold assignment are explicitly seeded through Python's `random` module, and, since the fix described in Section 5.5, PyTorch and CUDA are seeded per fold as well, with an explicit generator for the training `DataLoader` and seeded worker processes. Under this corrected procedure a run is fully determined by its seed value on fixed hardware. This was not always the case: the original six-run campaign at $n = 1{,}000$ (Appendix D, Section D.4) predates the fix, and two executions with identical arguments could differ under that earlier version of the code, which is why that campaign's per-run figures do not exactly reproduce from the seed value alone. Every pipeline script writes a JSON run-metadata file (`src/utils/logging.py`, `save_run_metadata`) recording its configuration, output artifact paths, and a timestamped run ID, so that any reported number can in principle be traced back to the exact invocation that produced it.

The configuration required to reproduce the $n = 1{,}000$ result is given in full in Table 4.1 (Chapter 4), together with the twenty training seeds (1–20) listed in Chapter 6, Table 6.1, and is implemented by `slurm/train_rm_1000.sh` and `slurm/seed_campaign.sh`. The six seeds in Appendix D, Table D.3 (42, 42, 7, 100, 314, 2026) reproduce only the earlier, superseded six-run campaign under the pre-fix code and are retained for that historical comparison, not as the current reproduction path. The codebase, including all scripts referenced in this chapter, is available at the project's GitHub repository: `https://github.com/hasnat23/master-thesis-rlaif-gca`.

6. Results

This chapter is organized by research question and leads with the confirmatory results: the RQ1 learnability comparison at $n = 1{,}000$ (Section 6.1), its behavior across dataset scale for RQ3 (Section 6.2), the effect sizes and equivalence tests that establish what those results do and do not bound (Section 6.3), and the independent ground-truth evaluation answering RQ4 (Section 6.4). Every confirmatory figure is a twenty-run campaign; the reasoning behind that choice is given in Chapter 4, Section 4.5. All four sections use the locked configuration stated in Chapter 4, Table 4.1; Section 6.5 then states the two decisions behind that configuration, aggregation formula and evaluator mode, reporting only as much exploratory evidence as is needed to justify them. The superseded experiments that motivated those decisions, including an early DPO prototype, the reward-model baselines, the single-run checks that failed to reproduce, and the six-run precursor campaign, are recorded in Appendix D rather than here.

6.1. RQ1: Preference Learnability at $n = 1{,}000$

The confirmatory campaign trains the locked configuration (Chapter 4, Table 4.1) across twenty independently seeded runs at $n = 1{,}000$, seeds 1 through 20, fixed in advance of running any of them so that no seed could be selected after seeing its result. It extends an earlier six-run pilot at the same configuration, detailed together with the configuration-selection evidence in Section 6.5 below. That pilot exposed a limitation in the cross-validation implementation that had to be fixed before further runs could be added: its seed argument controlled only the fold-assignment shuffle, while PyTorch's own random-number generator was never seeded, so the reward head's weight initialization, the training `DataLoader`'s batch order, and dropout were all left uncontrolled (Chapter 5, Section 5.5). This was corrected by seeding PyTorch and CUDA per fold, supplying an explicit generator to the training `DataLoader`, and seeding the `DataLoader` workers, so that a run is fully determined by its seed on fixed hardware. Table 6.1 lists every run of the resulting twenty-run campaign.

The preference data for this campaign, found during final review

The intention, consistent with the six-run campaign it extends, was to train this campaign on the `nli`-mode, $\alpha = 0.0$ preference file at `data/preferences_1000_alpha0_mode_nli/`. The stored run-metadata for all twenty seeds shows the training script instead read `data/preferences_1000/`, the earlier, pre-optimization file built under `nli_sp` and $\alpha = 0.5$ (Chapter 5, Section 5.4). Every number in this section and Table 6.1 is an accurate report of the runs as executed; the comparison between holistic and GCA remains fair, since both conditions were built from the same (mislabeled) file, but the campaign does not test the configuration Section 6.5.2 selected. Chapter 7, Section 7.2 discusses what this means for the mode-based interpretation offered there.

Table 6.1.: Per-run results for the extended campaign, seeds 1–20, $n = 1{,}000$, as executed under `nli_sp` mode, $\alpha = 0.5$, margin 0 (see caveat above; `nli`/$\alpha = 0.0$ was intended).

Seed	Holistic	GCA	Gap (GCA − Holistic, pp)
1	0.5280	0.5920	+6.40
2	0.5400	0.5450	+0.50
3	0.5200	0.5580	+3.80
4	0.5360	0.5700	+3.40
5	0.5470	0.5770	+3.00
6	0.5380	0.5610	+2.30
7	0.5050	0.5700	+6.50
8	0.5340	0.5430	+0.90
9	0.5300	0.5830	+5.30
10	0.5270	0.5550	+2.80
11	0.5090	0.5830	+7.40
12	0.5260	0.5730	+4.70
13	0.5320	0.5680	+3.60
14	0.5490	0.5690	+2.00
15	0.5220	0.5710	+4.90
16	0.5280	0.5690	+4.10
17	0.5290	0.5740	+4.50
18	0.5400	0.5740	+3.40
19	0.5220	0.5730	+5.10
20	0.5310	0.5750	+4.40

GCA led in all twenty runs. Figure 6.1 plots every per-run gap against the pooled mean. Table 6.2 reports the run-level statistics, which is the appropriate unit of independent replication (Chapter 4, Section 4.5; Chapter 7, Section 7.3).

Table 6.2.: Run-level statistics across the extended twenty-run campaign. The bootstrap interval resamples runs (not folds) 10,000 times with a fixed seed of 0. The two-sided Wilcoxon p-value is scipy's default computation; an independent exact permutation test (enumerating all 2^{20} sign assignments) gives $p \approx 1.9 \times 10^{-6}$, the theoretical floor for twenty same-signed observations. The two figures differ because one pair of runs (seeds 4 and 18) produced an identical gap of $+3.4$ percentage points, and scipy's default method falls back from the exact permutation calculation to a normal approximation whenever a tie is present; both figures are far below the conventional threshold, so the substantive conclusion does not depend on which is reported.

Runs	GCA ahead	Mean gap	SD	95% bootstrap CI	Wilcoxon p (two-sided)
20	20/20	**+3.95** pp	1.78 pp	$[+3.19, +4.70]$ pp	8.8×10^{-5}

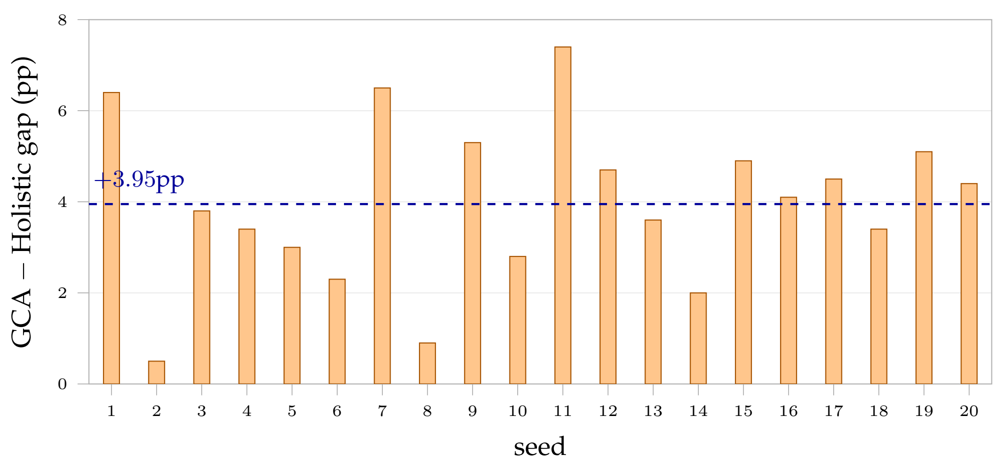


Figure 6.1.: Per-run GCA−Holistic accuracy gap across all twenty runs of the extended campaign, against the pooled mean. Every run favors GCA; the spread (1.78 percentage points, standard deviation) is narrower than in the six-run campaign, consistent with the earlier spread having reflected sampling noise from too few runs rather than genuine instability of the effect.

Headline learnability result (RQ1)

At $n = 1{,}000$, GCA preference construction produces a more learnable reward-model signal than holistic construction: mean gap $+3.95$ percentage points across twenty independently seeded runs, GCA ahead in all twenty, run-level Wilcoxon $p < 0.001$, Cohen's $d = +2.22$. The configuration this campaign was trained on is the one stated in the caveat above, not the intended one.

The mean gap of $+3.95$ percentage points is close to the six-run estimate of $+3.08$ points (Appendix D, Section D.4), so the additional runs refined the precision of the estimate rather than revealing a different effect. Both conditions nonetheless sit close to chance in absolute terms, 0.5297 and 0.5691 against a chance level of 0.5, so the finding concerns which of two weak preference signals is more consistently recoverable, not whether either yields a strong reward model (Chapter 7, Section 7.4).

6.2. RQ3: Robustness Across Dataset Scale

The same configuration was run at $n = 5{,}000$ and $n = 10{,}000$, each extended to twenty independently seeded runs under the same corrected, deterministic training procedure. These subsets are nested supersets of the $n = 1{,}000$ set rather than independent samples (Chapter 4, Section 4.7), so they measure how the effect behaves as data is added to the same pool rather than providing out-of-sample replication. Table 6.3 reports the run-level statistics at all three scales.

The $n = 5{,}000$ and $n = 10{,}000$ preference files carry the same discrepancy documented for $n = 1{,}000$ in the caveat above, though only in the evaluation mode: `slurm/build_preferences_scale.sh` never sets `-alignscore-evaluation-mode`, so both files were built under `nli_sp` rather than the intended `nli` (Chapter 5, Section 5.4). The aggregation exponent at these two scales is unaffected, since `-alpha 0.0` is passed explicitly. Every $n = 5{,}000/n = 10{,}000$ result in this thesis, including the $n = 2{,}000/n = 3{,}000$ crossover checks filtered from the $n = 10{,}000$ file, is therefore reported under `nli_sp`, not `nli`. The $n = 1{,}000$ campaign, by contrast, was built under $\alpha = 0.5$, not the $\alpha = 0.0$ used here, so the comparison across scale that follows changes the aggregation exponent alongside dataset size rather than varying size alone; what evidence bears on whether this confounds the comparison is discussed in Chapter 5, Section 5.4.

At $n = 5{,}000$ and $n = 10{,}000$ the mean gap is close to zero in both cases, GCA leads in a minority of runs at each scale (7 of 20 and 9 of 20) rather than consistently, and the two-sided Wilcoxon tests give $p = 0.3410$ and $p = 0.5882$, results indistinguishable from chance. With twenty runs at each scale, exactly matching the resolution of the $n = 1{,}000$ campaign, this is not an underpowered result in either direction: with twenty paired observations the test could have detected an effect as small as the $n = 1{,}000$ one had it been present, and it did not.

Table 6.3.: Repeated-run statistics at all three dataset sizes. All three rows are twenty-run campaigns using the same corrected training procedure; the $n = 1{,}000$ row is repeated from Table 6.2.

Dataset size	Runs	GCA ahead	Mean gap	95% bootstrap CI	Wilcoxon p
1,000	20	20/20	$+3.95$pp	$[+3.19, +4.70]$pp	8.8×10^{-5}
5,000	20	7/20	-0.22pp	$[-0.66, +0.21]$pp	0.3410
10,000	20	9/20	-0.11pp	$[-0.39, +0.18]$pp	0.5882

The confidence intervals settle the question directly. The $n = 1{,}000$ interval, $[+3.19, +4.70]$pp, does not overlap $[-0.66, +0.21]$pp at $n = 5{,}000$ or $[-0.39, +0.18]$pp at $n = 10{,}000$, and the gap between the nearest edges is more than two and a half percentage points in both cases. The GCA advantage observed at $n = 1{,}000$ is not present at $n = 5{,}000$ or $n = 10{,}000$, and this is supported by repeated, properly powered evidence at each scale rather than by runs of unequal weight.

Table 6.4 gives the underlying mean accuracies behind these gaps, and Figure 6.2 plots them.

Table 6.4.: Selected configuration at three dataset sizes, each averaged over twenty runs.

Dataset size	Holistic mean	GCA mean	Gap (GCA − Holistic)
1,000	0.5297	0.5691	$+0.0395$
5,000	0.5842	0.5820	-0.0022
10,000	0.5859	0.5848	-0.0011

Absolute accuracy continues to rise with dataset size for both conditions, from roughly 0.53–0.57 at $n = 1{,}000$ to roughly 0.582–0.586 at the larger scales, while the gap between conditions collapses from a significant 3.95 percentage points to a non-significant fraction of a point. Reporting only $n = 1{,}000$ would have substantially overstated how generally the GCA advantage holds.

One property of the training setup is worth naming explicitly here, since it bears on the first half of that claim. Epoch count, batch size, and learning rate schedule are held fixed at every scale (Table 4.1), but the number of optimizer steps is not: `src/reward_model/train.py` sets `total_steps = len(train_loader) * epochs`, so at a fixed batch size the number of gradient updates scales with the number of preference pairs. A model trained at $n = 10{,}000$ therefore receives roughly ten times as many optimizer steps as one trained at $n = 1{,}000$, not only ten times as much distinct data. The two are coupled by this design and nothing here separates them, so part of the rise in absolute accuracy with n may reflect more optimization rather than more information alone. This does not bear on the GCA-versus-holistic comparison itself, since both conditions receive

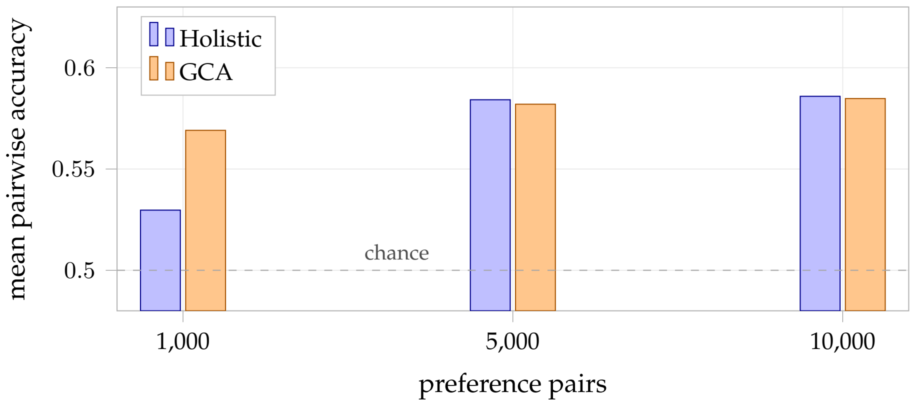


Figure 6.2.: Accuracy at the three dataset sizes, each averaged over twenty repeated runs. Absolute accuracy rises for both conditions as data is added, while the two conditions converge to statistically indistinguishable means at the larger scales.

the same step count at a given n, but it qualifies what "more data helps both conditions" is evidence of (Chapter 7, Section 7.5).

6.3. Effect Sizes, Equivalence, and Statistical Power

The tests reported so far establish whether each gap is distinguishable from zero. Two further questions bear directly on how the results should be read, and neither is answered by a p-value. How large is the $n = 1{,}000$ effect in standardized terms? And at $n = 5{,}000$ and $n = 10{,}000$, does the failure to reject the null reflect a genuinely negligible effect, or merely insufficient data to detect one? The second question matters because the claim made in Section 6.2, that the advantage is *absent* at the larger scales rather than merely unconfirmed, is an assertion about absence, and a non-significant result does not by itself support one. Table 6.5 reports the relevant quantities, computed over the run-level differences by `analysis/effect_sizes_and_equivalence.py`.

The $n = 1{,}000$ effect is large in standardized terms. Cohen's $d = +2.22$ places the mean gap more than two standard deviations of the run-to-run spread above zero, which is a large effect by any conventional reading of the scale [7]. The rank-biserial correlation reaches its maximum value of $+1.00$, reflecting that all twenty runs favor GCA without exception. These two figures should be read together with the absolute accuracies, however, and they measure different things: the effect is large *relative to the variability of the measurement*, while remaining small in absolute terms, since it moves a reward model from roughly 0.53 to roughly 0.57 pairwise accuracy against a chance level

Table 6.5.: Effect sizes, equivalence tests, and minimum detectable effects across the three twenty-run campaigns. Cohen's d and the rank-biserial correlation r_{rb} are computed over run-level gaps. The TOST column gives the larger of the two one-sided p-values at an equivalence bound of ± 1.0 percentage points; a value below 0.05 establishes equivalence at that bound. The tightest bound is the smallest δ at which equivalence still holds ($p < 0.05$). MDE is the smallest true effect detectable at 80% power given each campaign's observed standard deviation.

n	Mean gap	SD	Cohen's d	r_{rb}	TOST p ($\pm$1pp)	Tightest bound	MDE
1,000	$+3.95$pp	1.78pp	$+2.22$	$+1.00$	–	–	1.12pp
5,000	-0.22pp	1.03pp	-0.21	-0.24	0.0015	$\pm$0.61pp	0.64pp
10,000	-0.11pp	0.67pp	-0.16	-0.14	< 0.0001	$\pm$0.36pp	0.42pp

of 0.5 (Chapter 7, Section 7.4). A large d here indicates a highly reproducible difference, not a practically decisive one.

The larger-scale nulls are positive evidence of absence. At both $n = 5{,}000$ and $n = 10{,}000$, the two one-sided tests reject the hypothesis that the true effect is as large as ± 1 percentage point in either direction ($p = 0.0015$ and $p < 0.0001$ respectively). The effect at these scales is therefore not merely undetected; it is positively bounded. Tightening the bound until equivalence just fails gives the strongest statement the data support: at $n = 5{,}000$ any true effect larger than ± 0.61 percentage points can be ruled out, and at $n = 10{,}000$ any effect larger than ± 0.36 percentage points. Both bounds are far below the $+3.95$-point advantage measured at $n = 1{,}000$, and below the ± 1-point threshold at which an effect would begin to be practically interesting at all. This is what licenses the language of confirmed absence used in Section 6.2, which a non-significant Wilcoxon result alone would not.

The design had ample power to detect the effect it did not find. Given each campaign's observed run-to-run standard deviation, a paired test over twenty runs detects an effect of 1.12 percentage points at $n = 1{,}000$, 0.64 points at $n = 5{,}000$, and 0.42 points at $n = 10{,}000$, each at 80% power and $\alpha = 0.05$. Had an effect of the size observed at $n = 1{,}000$ been present at either larger scale, it would have exceeded the detection threshold there by a factor of more than six. The larger-scale campaigns are therefore not underpowered with respect to the effect they were testing for. Note that the smaller standard deviations at the larger scales, 1.03 and 0.67 points against 1.78 at $n = 1{,}000$, also mean those campaigns are intrinsically more sensitive, not less, so the failure to find an effect there cannot be attributed to noisier measurement.

6.4. RQ4: Precision Against an Independent Ground Truth

The experiments above establish that GCA preferences are more learnable at $n = 1{,}000$: a reward model recovers them more consistently under cross-validation. They do not establish that the recovered signal is more accurate, because the validation labels are supplied by the same evaluator that built the training preferences (Chapter 4, Section 4.4). This section reports RQ4, evaluated against the independent, held-out ground truth described in Chapter 4, Section 4.8: 500 articles disjoint from every training pool used in this thesis, scored by the same locked AlignScore-GCA configuration but never used to train any reward model. The reward models evaluated here are trained on the same preference data, seeds, and hyperparameters as RQ1–RQ3, but they are not the same trained weights: RQ1–RQ3 train five fold models per run under k-fold cross-validation and discard all five after recording their accuracy, so no single reusable model exists from that procedure (Chapter 5, Section 5.5). RQ4 therefore required one additional, non-k-fold training pass per seed and scale, using the same preference files, that trains a single model on the full preference set and retains its checkpoint (Chapter 4, Section 4.8). Every such checkpoint was trained once for this evaluation; every result that follows below reuses those same checkpoints without retraining them again, including the two hypothesis tests in Sections 6.4.4 and 6.4.6.

One consequence of this difference is worth flagging before the two sets of results are read together. The RQ1–RQ3 accuracies report each fold model trained on 80% of the preference set and evaluated on the held-out 20%, whereas the RQ4 models trained below see 100% of the same preference set during training. The comparison between RQ1–RQ3 learnability and RQ4 ground-truth precision (Chapter 7, Sections 7.5 and 7.6) is therefore a comparison across two things that differ in more than just the evaluation target: the training data fraction differs as well, and RQ4's numbers alone cannot separate what is due to scoring against an independent ground truth from what is due to training on the full set rather than four fifths of it. Nothing in this thesis isolates that second factor, and it is recorded as a limitation in Chapter 8, Section 8.4.

One further qualification applies to every number in this section, not only to the training-data-fraction point above. The held-out *articles* used below are disjoint from every training pool, but the *ranking* both reward models are scored against is built from GCA's own aggregation rule (sentence-level AlignScore, mean at $\alpha = 0.0$), the same rule the GCA reward model is trained to predict; the holistic reward model is scored against a criterion built from a rule it never saw (Chapter 4, Section 4.8). The results below are an evaluation on independent data, not an evaluation against a criterion neutral between the two conditions, and the two are not the same thing.

Table 6.6 states the four ground-truth constructions concretely, since the difference between them drives most of the pattern reported below. All four draw on the same pool of 1,000 scored held-out summaries and differ only in which pairs of that pool are compared, and therefore in how large a quality gap separates the two sides.

Table 6.6.: The four ground-truth constructions, all built from the same pool of 1,000 scored held-out candidate summaries (500 articles, two candidates each). "Top/bottom X%" sorts the whole pool by AlignScore-GCA score and compares the highest-scoring X% against the lowest-scoring X%, a global split across articles rather than a per-article comparison. Narrowing X gives fewer summaries per side but a larger and less ambiguous quality gap between them.

Construction	Summaries/side	Score gap	Comparison made
Same-article pairs	500 pairs	≈ 0.04	Each article's own two candidates, mirroring the training pairs
Top/bottom 25%	250	≥ 0.36	Best 250 of the pool against worst 250
Top/bottom 10%	100	≥ 0.66	Best 100 against worst 100
Top/bottom 5% (all-pairs)	50	≥ 0.79	Best 50 against worst 50, every high summary against every low one (2,500 comparisons) rather than one random pairing

6.4.1. Results at $n = 1{,}000$

Table 6.7 reports the success rate (the fraction of comparisons in which the reward model scores the higher-quality summary above the lower-quality one, Chapter 4, Section 4.8) for both reward models, averaged over the same twenty seeds used in Section 6.1, across all four ground-truth constructions.

Table 6.7.: Ground-truth precision at $n = 1{,}000$: success rate against four independent ground-truth constructions, averaged over the same twenty seeds used throughout Section 6.1.

Construction	Holistic	GCA	Gap (pp)	95% CI (pp)	Wilcoxon p
Same-article pairs	55.16%	55.30%	+0.14	[−0.76, +1.06]	0.760
Top/bottom 25%	71.94%	75.18%	+3.24	[−0.40, +7.36]	0.191
Top/bottom 10%	76.75%	86.00%	+9.25	[+5.70, +13.15]	< 0.001
Top/bottom 5% (all-pairs)	76.01%	83.39%	+7.38	[+3.22, +11.69]	0.004

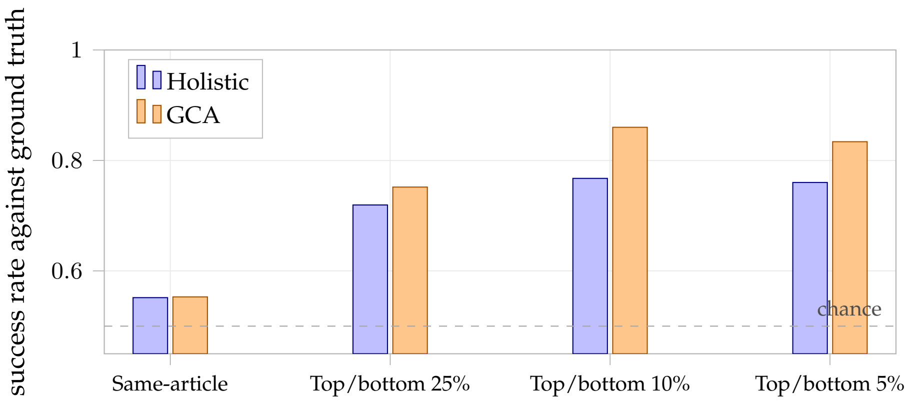


Figure 6.3.: Ground-truth precision at $n = 1{,}000$ (Table 6.7). GCA is indistinguishable from holistic on the closest comparison and ahead on all three constructions with a genuine quality gap, significantly so on two of the three.

The four constructions differ only in how large a quality gap separates the two summaries being compared (Chapter 4, Section 4.8): same-article pairs are typically close in quality (mean GCA score gap ≈ 0.04), while the percentile-split constructions compare progressively more clear-cut cases as X shrinks. The pattern across the four rows is consistent with that ordering, and Figure 6.3 shows it directly: the two conditions sit on top of one another on the closest comparison and separate progressively as the quality gap widens. On the closest comparison, same-article pairs, the two reward models are statistically indistinguishable ($p = 0.760$); a TOST equivalence test rules out any true difference larger than ± 0.96 percentage points at the 5% level. On the three constructions with a genuine, sizeable quality gap, GCA is ahead, and the advantage is significant on two of the three (top/bottom 10%, $d = +1.06$, a large effect by conventional standards [7]; and top/bottom 5% all-pairs, $d = +0.74$). The top/bottom 25% gap is directionally the same but does not reach significance at this seed count ($p = 0.191$). Neither reward model approaches 100% on any construction, plausibly because both are limited by the same 2,000-character article truncation (Chapter 5, Section 5.5), which bounds how much evidence either model can check a summary against regardless of how large the quality gap between the two candidates is.

The ranking used to compute success rate throughout this section is built from GCA's own aggregation rule, not a rule neutral to both conditions (Chapter 4, Section 4.8). Every GCA-versus-holistic gap reported here, including the $+9.25$ and $+7.38$ point advantages above, should be read with that in mind: they are established against a GCA-defined criterion, and the same evaluation under a holistic-defined criterion has not been run (Chapter 8, Limitation 4).

6.4.2. Scale Dependence: $n = 5{,}000$ and $n = 10{,}000$

The same four ground-truth constructions were then evaluated against reward models trained at $n = 5{,}000$ and $n = 10{,}000$, again over twenty seeds each, using the same checkpoints trained for Section 6.2. Only the training-set size changes; the held-out set, the four constructions, and the evaluation procedure are identical to Section 6.4.1.

Table 6.8 reports the underlying success rates for both conditions at all three sizes, and Table 6.9 reports the GCA-minus-holistic gaps derived from them. Both are given because the gaps alone are actively misleading about what happens as the training set grows, for reasons the absolute rates make immediately visible.

Table 6.8.: Ground-truth success rate (%) for each condition, by construction and training-set size. All values are twenty-seed means. Both conditions improve substantially as the training set grows; the gap between them is reported separately in Table 6.9.

	$n = 1{,}000$		$n = 5{,}000$		$n = 10{,}000$	
Construction	Hol.	GCA	Hol.	GCA	Hol.	GCA
Same-article pairs	55.16	55.30	55.04	54.61	58.10	56.65
Top/bottom 25%	71.94	75.18	75.06	74.64	81.16	79.04
Top/bottom 10%	76.75	86.00	83.25	83.35	91.95	90.00
Top/bottom 5% (all-pairs)	76.01	83.39	82.00	82.73	90.84	88.72

The advantage found at $n = 1{,}000$ does not merely fail to persist, as the learnability advantage did (Section 6.2); it reverses. At $n = 5{,}000$ every gap is within noise of zero, matching the learnability result at that scale.

Table 6.9.: GCA-minus-holistic ground-truth precision gap, by construction and training-set size, derived from Table 6.8. All three columns are twenty-seed means. p-values are for the $n = 10{,}000$ column.

Construction	$n = 1{,}000$ (pp)	$n = 5{,}000$ (pp)	$n = 10{,}000$ (pp)	Wilcoxon p (n=10,000)
Same-article pairs	$+0.14$	-0.43	-1.45	0.010
Top/bottom 25%	$+3.24$	-0.42	-2.12	0.036
Top/bottom 10%	$+9.25$	$+0.10$	-1.95	0.058
Top/bottom 5% (all-pairs)	$+7.38$	$+0.73$	-2.12	0.025

The ground-truth reversal (RQ4)

At $n = 10{,}000$, holistic is significantly ahead of GCA on three of the four ground-truth constructions ($p = 0.010$ to 0.036), with the fourth (top/bottom 10%) narrowly short of the conventional threshold ($p = 0.058$) while pointing the same way. This is the first result in this thesis, on any metric, in which holistic significantly outperforms GCA rather than the two being indistinguishable.

The reversal is holistic improving faster, not GCA degrading. Table 6.8 makes a point that Table 6.9 cannot, and that matters for how the reversal should be understood. GCA does not get worse as the training set grows. On the top/bottom 10% construction its success rate rises from 86.00% at $n = 1{,}000$ to 90.00% at $n = 10{,}000$, and the same upward movement holds on every construction except the same-article one. What produces the negative gap at $n = 10{,}000$ is that holistic improves *faster* over the same interval, from 76.75% to 91.95% on that construction, a gain of 15.2 percentage points against GCA's 4.0. Both conditions benefit from additional training data; they simply benefit at different rates, and holistic's steeper trajectory carries it past GCA somewhere between $n = 3{,}000$ and $n = 5{,}000$ (Section 6.4.3). Figure 6.4 plots this directly.

This reframing does not soften the finding for practical purposes: if the question is which construction procedure to use at $n = 10{,}000$, holistic is the better choice, and significantly so. But it does constrain what any explanation of the reversal has to account for. A mechanism that predicts GCA becoming actively worse with more data is inconsistent with Table 6.8. What requires explaining is why GCA's returns to additional training data are so much flatter than holistic's, which is the form the two candidate mechanisms tested in Sections 6.4.4 and 6.4.6 take.

6.4.3. Locating the Crossover: $n = 2{,}000$ and $n = 3{,}000$

Two further training-set sizes were run to locate where the crossover from a GCA advantage to a holistic advantage occurs. $n = 2{,}000$ and $n = 3{,}000$ were verified by

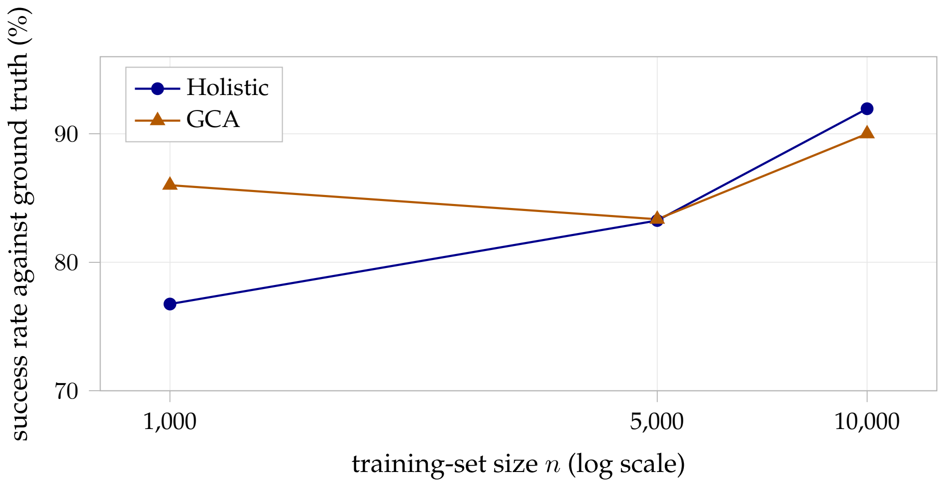


Figure 6.4.: Absolute ground-truth success rate on the top/bottom 10% construction, by training-set size (Table 6.8). Both conditions improve as training data grow. GCA starts far ahead and gains 4.0 percentage points overall; holistic starts lower and gains 15.2, overtaking GCA between $n = 5{,}000$ and $n = 10{,}000$. The reversal in Table 6.9 is this difference in slope, not a decline in GCA's own performance.

direct index comparison to be exact subsets of the $n = 10{,}000$ draw used throughout this thesis (seed 200), so their preference pairs were constructed by filtering the existing $n = 10{,}000$ preference files by article identifier rather than by regenerating candidates or re-running AlignScore. Given the number of scales already covered and the time remaining before submission, these two intermediate points were each run at five seeds rather than twenty; they are reported as a supplementary, lower-precision check on where the crossover falls, not as an equally-powered addition to Table 6.9. Table 6.10 places all five sizes side by side, and Figure 6.5 plots the same values as a curve per construction.

Table 6.10.: GCA-minus-holistic ground-truth precision gap at all five training-set sizes tested. The $n = 2{,}000$ and $n = 3{,}000$ columns are five-seed means; the remaining three are twenty-seed means, repeated from Table 6.9.

Construction	n=1,000	n=2,000	n=3,000	n=5,000	n=10,000
Same-article pairs	+0.14	−1.72	+0.48	−0.43	−1.45
Top/bottom 25%	+3.24	−0.16	−0.88	−0.42	−2.12
Top/bottom 10%	+9.25	+3.40	+3.80	+0.10	−1.95
Top/bottom 5% (all-pairs)	+7.38	+1.83	+1.42	+0.73	−2.12

For the three constructions with a genuine effect at $n = 1{,}000$ (top/bottom

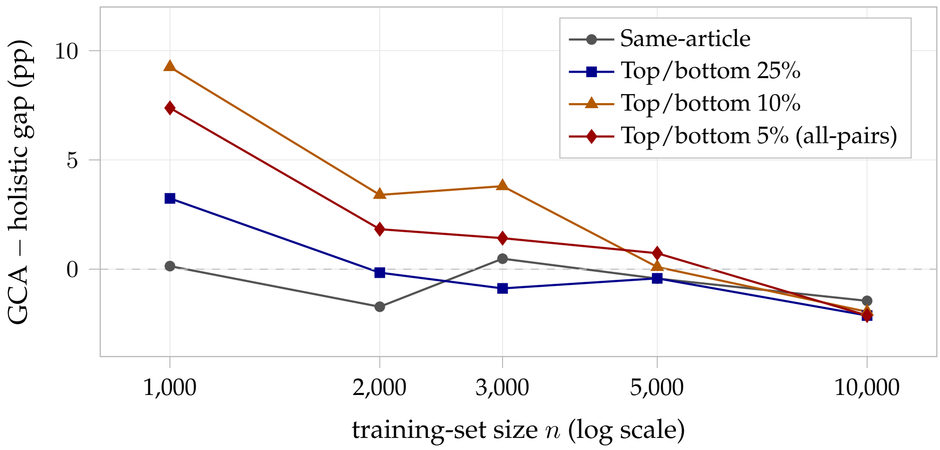


Figure 6.5.: GCA-minus-holistic ground-truth precision gap (Table 6.10) across all five training-set sizes tested. The n = 2,000 and n = 3,000 points are five-seed means; the rest are twenty-seed means. The three constructions with a genuine effect at n = 1,000 decline together and are all negative by n = 10,000, passing through a null region around n = 3,000–5,000 in which none is distinguishable from zero.

25%/10%/5%), the gap declines steadily from n = 1,000 through n = 2,000 and n = 3,000, sits within noise of zero at n = 5,000, and is significantly negative at n = 10,000. The same-article row, which never showed a reliable effect at any scale, stays close to zero throughout and does not display a comparable trend, consistent with there being nothing directional to decline in that condition.

The transition should be read as a region rather than a point, and the data do not support locating it more precisely than that. Statistical significance marks it out only at the two ends: the GCA advantage is significant at n = 1,000 and the holistic advantage is significant at n = 10,000, while no gap at n = 2,000, n = 3,000, or n = 5,000 is distinguishable from zero in either direction. The point estimates cross zero at different places depending on the construction, which is itself a reason not to over-read them: top/bottom 25% is already marginally negative by n = 2,000, whereas top/bottom 10% and top/bottom 5% remain marginally positive at n = 5,000 and turn over only between there and n = 10,000. With five seeds at the two intermediate points, differences of this size are well inside the run-to-run variation documented in Section 6.1. What the intermediate scales establish is therefore the *shape* of the decline and the fact that it passes through a null region around n = 3,000–5,000, not a single crossover value.

6.4.4. A First Candidate Tested and Rejected: Near-Tie Training Pairs

With the reversal established and its location narrowed to a region, the question becomes what produces it. Two candidate mechanisms were tested directly against data already on disk. Both were chosen because they make a specific, checkable prediction about *where* the effect should be largest, so that the data can contradict them rather than merely fail to confirm them, and because neither required any new reward-model training.

The first concerns the training data rather than the evaluation criterion. GCA's sentence-level aggregation might produce a larger share of near-zero-margin, effectively noise-level preference pairs than holistic scoring does, since every training preference file in this thesis is built with no margin filter, following prior feedback to use all candidate pairs rather than the smaller, margin-filtered set used in an earlier proposal-era configuration (`slurm/build_preferences_1000.sh`). If GCA's training pairs contain disproportionately more near-tie comparisons, a reward model trained on more of them might fit that noise rather than genuine signal. That would predict a GCA disadvantage that grows with the training set, which is the pattern Table 6.9 shows.

This was checked directly against the stored per-pair score gaps in the existing preference files, which required no additional model training or AlignScore scoring. Table 6.11 reports the fraction of training pairs with a score gap below 0.05, for both conditions at all three main training-set sizes, computed from the files each campaign actually trained on (Chapter 5, Section 5.4) rather than from the intended `nli`/$\alpha = 0.0$ files.

Table 6.11.: Fraction of training preference pairs with a score gap below 0.05, by condition and training-set size, computed from the preference files each campaign actually trained on.

n	Holistic (frac. < 0.05)	GCA (frac. < 0.05)	Holistic (count)	GCA (count)
1,000	24.6%	21.3%	246	213
5,000	24.6%	24.4%	1,232	1,218
10,000	24.2%	24.2%	2,416	2,425

The mechanism does not hold up against this data. At $n = 5{,}000$ and $n = 10{,}000$, the near-tie fraction is essentially identical between the two conditions, not larger for GCA, so there is no growing asymmetry at the scales where GCA actually loses. At $n = 1{,}000$, the scale at which GCA *leads*, GCA has *fewer* near-tie pairs than holistic (21.3% versus 24.6%), not more. The hypothesis requires GCA's near-tie share to be disproportionately larger, and to grow relative to holistic's as n grows, so that fitting that noise increasingly hurts GCA at scale; instead the largest gap between conditions occurs at the one scale where GCA wins, and it points the wrong way, while the two scales where GCA loses show next to no gap at all. Near-tie training-pair prevalence is therefore rejected as an explanation for the reversal.

6.4.5. How Often Holistic and GCA Agree

A related question, raised independently of the two hypotheses above, is how often the two conditions' labels agree in the first place. Holistic and GCA are scored from the same candidate pool, so on any pair where both conditions reach the same A/B decision, the two reward models are trained on literally the same (article, chosen, rejected) triple; the labels can only differ on pairs where the two decisions disagree. Table 6.12 reports this directly, computed from the same preference files as Table 6.11, at every scale for which the actually-trained files are available, and additionally for the intended `nli`/$\alpha = 0.0$ configuration at $n = 1{,}000$ (`data/preferences_1000_alpha0_mode_nli/`), where it can be measured without new AlignScore scoring.

Table 6.12.: Fraction of preference pairs on which holistic and GCA reach the same A/B decision, by training-set size and evaluator configuration.

n	Configuration	Pairs compared	Agreement
1,000	`nli`, $\alpha = 0.0$ (intended)	1,000	63.9%
1,000	`nli_sp`, $\alpha = 0.5$ (as trained)	1,000	82.2%
5,000	`nli_sp`, $\alpha = 0.0$ (as trained)	4,998	97.5%
10,000	`nli_sp`, $\alpha = 0.0$ (as trained)	9,991	97.9%

Two patterns stand out. First, agreement rises sharply with training-set size under the configuration actually used for the confirmatory campaigns: from 82.2% at $n = 1{,}000$ to 97.5–97.9% at $n = 5{,}000/n = 10{,}000$. This bears directly on the RQ1 result. Because agreeing pairs give both conditions identical training signal, the disagreeing pairs, roughly 18% of the training set at $n = 1{,}000$ but only 2–3% at $n = 5{,}000$ and $n = 10{,}000$, are the only pairs that can possibly produce a difference in what the two reward models learn, absent second-order effects from training jointly on a slightly different overall set. A shrinking pool of disagreeing pairs is consistent with, and offers a direct mechanistic account of, a shrinking gap between conditions: there is simply less room at large n for the two label sets to diverge, independent of any claim about which construction procedure is more informative per label. Second, evaluator mode affects agreement by more than dataset size does at $n = 1{,}000$: the intended `nli` configuration agrees only 63.9% of the time, well below `nli_sp`'s 82.2%, consistent with Chapter 7, Section 7.2's account of `nli_sp` performing an internal decomposition that already resembles what GCA does externally, which should make the two conditions' decisions converge more often under `nli_sp` than under `nli`. This pattern holds even though, as reported in Section 6.1 above, the resulting reward-model accuracy gap does not track it in the way the mode-based account of Section 7.2 predicts.

The corresponding figures for $n = 5{,}000$ and $n = 10{,}000$ under the intended `nli` configuration cannot be computed without rebuilding those preference files under the correct evaluator-mode flag and rescoring with AlignScore, which is new computation beyond what this thesis attempts (Chapter 8, Limitation 1).

6.4.6. A Second Candidate Tested and Rejected: A Length/Sentence-Count Shortcut

A second candidate explanation for the reversal was tested directly. If the GCA reward model learns to rely on summary length or sentence count as a shortcut for quality, and comes to rely on it more heavily as training data grow, that shortcut would be expected to transfer poorly to the ground truth's cross-article comparisons and could explain why GCA falls behind at large n. This requires no new training: every checkpoint already trained for RQ4 was used to score each of the 1,000 held-out ground-truth summaries, and the resulting scores were correlated against each summary's character length and sentence count (`analysis/check_length_shortcut.py`).

Table 6.13.: Correlation between predicted reward and summary character length, by condition and training-set size, averaged over five seeds at each scale.

n	Holistic r	GCA r	Gap (GCA − Holistic)
2,000	-0.037	-0.046	-0.009
3,000	-0.089	-0.132	-0.043
10,000	-0.198	-0.201	-0.003

The correlation is negative throughout for both conditions, meaning both reward models score shorter summaries higher, not longer ones, so this is not the length-inflation shortcut informally expected. More importantly for the mechanism under test, the hypothesis predicts that the GCA-minus-holistic gap in this correlation should be largest at $n = 10{,}000$, the scale at which GCA starts losing. Table 6.13 shows the opposite: the gap peaks at $n = 3{,}000$ (-0.043) and nearly vanishes at $n = 10{,}000$ (-0.003), where the two conditions converge to almost identical length-reliance. This mechanism is therefore rejected on the same logic as the near-tie-training-pair hypothesis (Section 6.4.4): the pattern needed to explain a GCA-specific effect is absent from the data.

One genuine pattern does emerge from this check, independent of the rejected hypothesis: both conditions become markedly more correlated with summary length as the training set grows, from $r \approx -0.04$ at $n = 2{,}000$ to $r \approx -0.20$ at $n = 10{,}000$, roughly a five-fold increase in magnitude for both. Because this growth affects holistic and GCA symmetrically, it cannot explain the reversal, but it is a candidate contributor to the low absolute accuracy ceiling neither condition breaks through (Chapter 7, Section 7.4).

6.4.7. What the RQ4 Investigation Establishes

The sequence of experiments in this section followed a single line of enquiry, and it is worth stating what it collectively supports before Chapter 7 interprets it.

Against a held-out set that no reward model was trained on, GCA is substantially and significantly more precise than holistic at $n = 1{,}000$, provided the two summaries being compared differ in quality by enough for the comparison to be meaningful: $+9.25$ percentage points on the top/bottom 10% construction ($p < 0.001$) and $+7.38$ on top/bottom 5% ($p = 0.004$), with no detectable difference on same-article pairs, where the candidates are close in quality by construction (Section 6.4.1). Extending the same evaluation to larger training sets shows the advantage does not survive: it is within noise of zero at $n = 5{,}000$ and significantly negative at $n = 10{,}000$ on three of four constructions (Section 6.4.2). Intermediate five-seed points at $n = 2{,}000$ and $n = 3{,}000$ show the decline passing through a null region around $n = 3{,}000$–$5{,}000$ (Section 6.4.3). The absolute rates locate the cause of the reversal in differing rates of improvement rather than in any degradation of GCA (Table 6.8).

Two mechanisms that would account for that difference in slope were then tested and both rejected, each because the data contradicted the hypothesis's own prediction about where its effect should concentrate: near-tie training pairs (Section 6.4.4) and a length or sentence-count shortcut (Section 6.4.6).

Two properties of how the ground truth is constructed limit how the significance pattern above should be read, and both follow from the same fact: every scale point and every construction scores the same fixed pool of 1,000 held-out summaries (500 articles), never an independently drawn one. First, the four constructions are nested rather than independent tests: the top/bottom 5% pairs are a subset of the top/bottom 10% pairs, which are in turn a subset of the top/bottom 25% pairs, all drawn from the same 1,000-summary ranking. No multiple-comparison correction is applied across them, and none of the "three of four constructions significant" statements in this section should be read as three independent confirmations; they are three overlapping views of the same underlying ranking at different degrees of overlap. Second, because the same 500 articles are reused at every training-set size, the five-point curve in Section 6.4.3 is not five independent samples of ground-truth precision: any idiosyncrasy of this particular held-out draw is shared by every point on it, so the curve should be read as showing how the trained models' behavior changes on one fixed evaluation instrument, not as an average over repeated independent draws of the ground truth itself. Chapter 8, Section 8.4 records both points.

The reversal is characterized, not explained

RQ4 establishes that the advantage is real at small scale, that it reverses at large scale, and roughly where the crossover falls. It does not establish *why*. Two candidate mechanisms have been eliminated, which narrows the space of remaining explanations and rules out the two most immediately available ones, but no mechanism has been confirmed. Chapter 7, Section 7.6 discusses what remains.

6.5. Selecting the Final Configuration

The four preceding sections report the confirmatory results under the locked configuration stated in Chapter 4, Table 4.1. This section states how that configuration was chosen. Two design decisions are not fixed by the study design and had to be settled empirically: how per-sentence scores are aggregated into a summary-level score (the exponent α of Chapter 4, Equation 4.1), and which AlignScore evaluation mode the judge is queried in (Chapter 3, Section 3.3). Both were settled on the $n = 1{,}000$ setting, and both are exploratory in the sense of Chapter 4, Section 4.6: the configuration was chosen after observing outcomes, so results on that setting cannot also serve as independent confirmation of the choice.

6.5.1. Aggregation Formula

The proposal-era weakest-link penalty ($\alpha = 0.5$) produced a GCA reward model that *underperformed* the holistic one, and GCA decisions agreed with the corresponding holistic decision on only 82.2% of pairs (Appendix D, Section D.2). Nine aggregation strategies were therefore compared by how well each agreed with the holistic decision on the same 1,000 pairs (`analysis/test_gca_formulas.py`, `analysis/test_advanced_aggregation.py`). Table 6.14 reports the results; the simple mean ($\alpha = 0.0$) had the highest agreement of any strategy tested, 15.4 percentage points above the original penalty formula.

On the strength of this result, α was changed from 0.5 to 0.0 throughout the codebase. One qualification governs how the table should be read: the criterion is agreement with the holistic decision, which measures internal consistency between two procedures driven by the same evaluator, not external factual correctness. A strategy could agree closely with holistic scoring while reproducing the same errors, and nothing here would distinguish those cases. The criterion is used as a diagnostic for whether the penalty term was amplifying disagreement, not as evidence that the simple mean produces better labels.

The formula change did not, on its own, produce a GCA advantage: holding the judge mode fixed, the gap widened slightly rather than closing (Appendix D, Section D.3). That is what motivated examining the judge mode next.

Table 6.14.: Agreement with the holistic decision across nine GCA aggregation strategies, $n = 1{,}000$ pairs. Plotted in rank order in Appendix D, Figure D.1.

Strategy	Agreement with holistic	vs. $\alpha = 0.5$ baseline
Simple mean ($\alpha = 0.0$)	97.6%	$+15.4$ pp
Confidence-weighted	96.0%	$+13.8$ pp
Ensemble	93.3%	$+11.1$ pp
Length-weighted	91.5%	$+9.3$ pp
Weighted by position	88.2%	$+6.0$ pp
Robust mean (10% trim)	85.4%	$+3.2$ pp
Harmonic mean	82.9%	$+0.7$ pp
Original penalty ($\alpha = 0.5$)	82.2%	baseline
Minimum	78.7%	-3.5 pp

6.5.2. Evaluator Mode

With $\alpha = 0.0$ fixed, the four AlignScore evaluation modes were compared under the same $n = 1{,}000$ training setup. Table 6.15 shows that the choice of mode, not only the aggregation formula, is associated with whether GCA leads at all. The `nli` mode produced the largest GCA advantage; `nli_sp`, the mode inherited from the earliest experiments, is one of the two modes that favor holistic.

Table 6.15.: AlignScore judge-mode sweep ($n = 1{,}000$, $\alpha = 0.0$). Each mode was run once; plotted in Appendix D, Figure D.2.

Mode	Holistic mean acc.	GCA mean acc.	Gap (GCA $-$ Holistic)
`nli`	0.523	0.583	$+0.060$
`bin`	0.510	0.564	$+0.054$
`bin_sp`	0.554	0.562	$+0.008$
`nli_sp`	0.578	0.557	-0.021

The ordering follows a systematic pattern rather than varying arbitrarily. The two modes in which GCA leads by a clear margin, `nli` and `bin`, are precisely the two that operate on the context and claim without splitting them. The two modes in which the advantage is negligible or negative, `bin_sp` and `nli_sp`, are the split variants, in which the evaluator itself segments the input and aggregates over the segments (Chapter 3, Section 3.3). Holistic accuracy also rises from 0.510–0.523 in the unsplit modes to 0.554–0.578 in the split modes, which is consistent with the split modes supplying a stronger holistic signal rather than with GCA becoming worse. The pattern suggests that external sentence-level decomposition may contribute most where the evaluator does not already decompose internally, and little where it does.

Two limits on this evidence must be carried forward. Each mode was run once, and single runs at this dataset size do not reproduce reliably (Appendix D, Section D.3), so the comparison across modes is suggestive rather than established. And, as Chapter 5, Section 5.4 reports, the confirmatory campaigns that follow did not in fact remain on the `nli` mode this sweep selected. Chapter 7, Section 7.2 sets out what the confirmatory evidence now permits to be concluded from Table 6.15, which is substantially less than was originally supposed.

6.5.3. From Single Runs to Repeated Campaigns

Before the selected configuration could be evaluated, two single-run results had already failed to reproduce: a promising hyperparameter configuration reversed direction on a same-seed rerun, and two nominally identical `nli_sp` runs differed by 0.8 and 0.4 percentage points (Appendix D, Section D.3). A six-run campaign on the selected `nli`, $\alpha = 0.0$ configuration followed, giving a mean gap of $+3.08$ percentage points with GCA ahead in five of six runs (Appendix D, Section D.4). That campaign could not establish run-level significance at any effect size, because with six paired observations and one discordant run the smallest attainable two-sided Wilcoxon p-value is 0.0625. Extending to twenty runs, reported next, was the direct response.

6.6. Summary of Results

Table 6.16 summarizes how the results in this chapter bear on the research questions from Chapter 4, Section 4.1. Full interpretation is left to Chapter 7.

Table 6.16.: Summary of findings against the research questions.

	Finding
RQ1	At $n = 1{,}000$, GCA leads in all 20 runs of the extended campaign (Section 6.1) with a mean difference of $+3.95$ percentage points, a run-level two-sided Wilcoxon $p < 0.001$, a bootstrap 95% CI entirely above zero, and a large standardized effect ($d = +2.22$, $r_{\text{rb}} = +1.00$). Twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$ (Section 6.2), at the same resolution, confirm the advantage is genuinely absent at those scales, not merely unreproduced: mean gaps of -0.22pp ($p = 0.3410$) and -0.11pp ($p = 0.5882$), with confidence intervals that do not overlap the $n = 1{,}000$ result, and equivalence tests excluding effects larger than ± 0.61pp and ± 0.36pp (Section 6.3).
RQ2	Aggregation strategy and evaluator mode are associated with substantial outcome differences (Section 6.5). The aggregation result is the firmer of the two: the weakest-link penalty measurably hurt GCA relative to the simple mean. The evaluator-mode comparison is exploratory, each mode having been run once, and the confirmatory campaigns do not support the mode-based account originally drawn from it (Chapter 7, Section 7.2).
RQ3	Stable at fixed scale, not stable across scale. All 20 runs at $n = 1{,}000$ favor GCA (SD 1.78pp); twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$ are similarly consistent around a mean of approximately zero (SD 1.03pp and 0.67pp). The advantage is confirmed present at $n = 1{,}000$ and confirmed absent at the two larger, nested dataset sizes, with non-overlapping confidence intervals at every comparison and equivalence established at both larger scales (Section 6.3).
RQ4	No, and the pattern is stronger than for RQ1: against an independent, held-out ground truth, GCA is significantly more precise than holistic at $n = 1{,}000$ on clear-cut comparisons (up to $+9.25$pp, $p < 0.001$; Section 6.4.1), indistinguishable from holistic at $n = 5{,}000$, and significantly *less* precise than holistic at $n = 10{,}000$ on three of four constructions (-1.45 to -2.12pp, $p = 0.010$–0.036; Section 6.4.2). Five-seed checks at $n = 2{,}000/n = 3{,}000$ show the decline passing through a null region around $n = 3{,}000$–$5{,}000$ (Section 6.4.3). Two candidate explanations were tested directly and rejected: that GCA's training pairs contain disproportionately more near-zero-margin comparisons (Section 6.4.4), and that GCA increasingly relies on a length/sentence-count shortcut relative to holistic (Section 6.4.6). The mechanism behind the reversal remains open.

7. Discussion

7.1. What the Experiments Establish, and What They Do Not

The experiments support a narrow and conditional claim. Under one specific configuration, at one dataset size, preferences constructed by sentence-level aggregation are more consistently recoverable by a Bradley–Terry reward model than preferences constructed holistically from the same evaluator and the same candidate summaries. The difference averages $+3.95$ percentage points over twenty independently seeded runs and favors GCA in every one of them, with a run-level Wilcoxon $p < 0.001$ (Chapter 6, Section 6.1).

Four claims that might appear to follow do not. The results do not establish that GCA labels are factually more accurate, since no independent annotation was collected and AlignScore supplies both sets of labels (Section 7.8). They do not establish that a summarization model trained on GCA preferences produces more faithful summaries, since the final study contains no policy-optimization stage. They do not establish that GCA is generally preferable, since Chapter 6 reports configurations and scales at which it is not. And they do not establish a strong effect in absolute terms, since both reward models remain near chance (Section 7.4). What the study contributes is evidence about *when* feedback granularity changes the structure of a preference-learning signal, and the remainder of this chapter examines the conditions under which it does.

7.2. Evaluator-Internal Decomposition: A Hypothesis the Data Do Not Support

One interpretation of the mode sweep in Chapter 6, Section 6.5.2 shaped this thesis's design and is worth stating, and then retracting, explicitly: that GCA's advantage requires the `nli` evaluation mode, because `nli` lacks the internal sentence decomposition that `nli_sp` performs. On this account, external and evaluator-internal decomposition act as partial substitutes, so applying GCA on top of a judge that already decomposes internally adds little.

The mechanism is plausible and one component of it is measurable. The `_sp` suffix enables a preprocessing step internal to AlignScore [41]: the context is split into roughly 350-token chunks and the claim into sentences, and the score is the maximum entailment across chunks per claim sentence, averaged over claim sentences. Plain `nli` skips this and scores the full context against the full claim in one pass. A holistic `nli_sp` call should therefore closely track the mean of separately computed per-sentence scores of the same summary, and on the stored $n = 200$ data it does: across all 400 summaries the two quantities correlate at $r = 0.983$, agree to within 0.001 for 234 of 400, and have a median absolute difference of 0.000 (Figure 7.1). Under `nli_sp`, holistic scoring and mean-aggregated sentence scoring are close to the same computation.

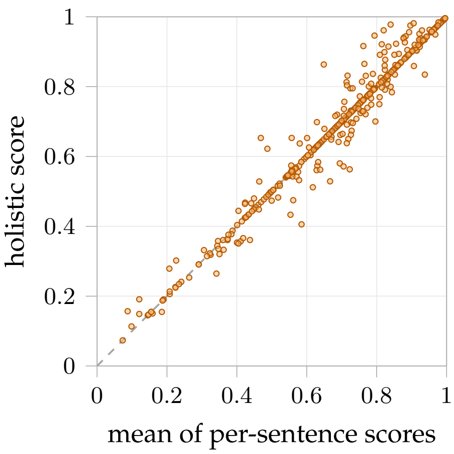


Figure 7.1.: Holistic AlignScore against the mean of the separately computed per-sentence scores, for all 400 candidate summaries in the $n = 200$ preference set under evaluator mode `nli_sp`. The dashed line is $y = x$. The two quantities are near-identical ($r = 0.983$), which is what the split mode's internal sentence-averaging predicts. The corresponding files for the final `nli` configuration are not retained in the repository, so the same check could not be run for the unsplit mode.

What the account cannot survive is this thesis's own confirmatory evidence. It predicts a GCA advantage under `nli` and none under `nli_sp`. Both halves of that prediction fail. The twenty-run campaign was itself trained under `nli_sp` (Chapter 5, Section 5.4) and produced a GCA advantage of $+3.95$ points with GCA ahead in all twenty runs; the six-run campaign, genuinely trained under `nli` (Appendix D, Section D.4), produced $+3.08$ points with GCA ahead in five of six. Once each mode is tested with enough runs to distinguish an effect from noise, GCA leads under both. The single-run `nli_sp` result that appeared to confirm the account (-2.1 points, Table 6.15) is best read as another instance of the single-run instability documented in Section 7.9, not as a stable property of the mode.

The proposition that evaluator-internal decomposition reduces the marginal benefit of external GCA is therefore not a finding of this thesis, and on the evidence now available it is not a well-supported hypothesis either. Two things survive. The $r = 0.983$ correlation is a genuine measurement of what `nli_sp` computes, independent of what it implies for GCA. And the proposition remains a candidate worth testing under a correctly-configured repetition (Chapter 8, Limitation 1), since what refutes it here is a pair of campaigns neither of which was designed to isolate the mode. The broader methodological point in Section 7.7, that a judge's internal behavior should be checked before an effect is attributed to one's own pipeline, stands on the measurement rather than on the refuted account.

7.3. How Much Weight the Statistical Evidence Can Carry

Two properties of the reported statistics govern how much they can be asked to support: the unit at which independence holds, and the difference between statistical significance and practical size.

The unit of independent replication is the run, not the cross-validation fold. Within a single run the five folds partition one dataset: any two folds share four fifths of their training data, and all five evaluate models trained from the same pool of preference pairs. Their errors are correlated, so a test assuming independence treats the evidence as stronger than it is, and the effective sample size lies between the number of runs and the number of folds, closer to the former. Every fold-level interval and p-value in this thesis is therefore reported as an exploratory summary of spread, never as a hypothesis test, and every confirmatory statistic is computed over run-level differences (Chapter 4, Section 4.5). This distinction is not cosmetic: on the six-run precursor campaign the fold-pooled p-value was 0.0034 while the correct run-level test could not fall below 0.0625, a difference of more than an order of magnitude arising purely from the unit of analysis (Appendix D, Section D.4).

At the run level, the twenty-run campaign gives a mean gap of $+3.95$ percentage points, a bootstrap 95% confidence interval of $[+3.19, +4.70]$ points that does not cross zero, and a two-sided Wilcoxon $p < 0.001$ (Chapter 6, Section 6.1). Significance alone understates how consistent the difference is: measured against the run-to-run spread it must be distinguished from, Cohen's $d = +2.22$ and the rank-biserial correlation attains its maximum of $+1.00$, because no run favors holistic. The p-value and the effect size answer different questions and both are needed. What is large is the difference relative to the noise in the measurement; what remains small, as the next section discusses, is the difference in absolute terms.

Reaching significance changes what can be claimed, but not by as much as it may appear. It establishes that GCA preferences are *consistently more learnable* than holistic preferences by this reward-model architecture, under this evaluator configuration, at this dataset size, to a degree unlikely to arise from run-to-run noise alone. It does not by itself establish a large effect (Section 7.4), an effect that generalizes to other scales (Section 7.5), or a mechanism (Section 7.2). What the evidence supports is a directionally consistent, statistically significant advantage of approximately three to four percentage points under this configuration, scoped to the conditions under which it was measured.

7.4. Both Reward Models Are Weak in Absolute Terms

A comparison can be sound while both conditions are weak

At $n = 1{,}000$, the holistic reward model reaches 0.5297 and the GCA reward model 0.5691 pairwise accuracy, against a chance level of 0.5. In absolute terms these models recover very little of the preference signal: the better of the two is correct on roughly 57 of every 100 held-out pairs, where guessing would be correct on 50. The comparison between conditions remains valid; what it is a comparison *of* is modest in absolute terms.

This materially qualifies what the thesis shows. The finding is that GCA preferences are *more learnable* than holistic preferences under the selected configuration, not that either produces a reward model one would want to deploy. A reward model at 0.56 accuracy would provide a very noisy training signal to any downstream policy, and the significant $+3.95$ percentage-point advantage established in the extended campaign (Chapter 6, Section 6.1) is an improvement to a weak signal rather than the difference between a weak and a strong one.

Several factors plausibly contribute to the low absolute numbers, and this thesis can separate them only partially. The task itself is genuinely hard: both candidate summaries in a pair are generated by the same model from the same article and differ only in sampling temperature, so many pairs are near-ties in factual quality, and a judge's preference between them may encode little learnable structure. The RoBERTa-base reward model [28] is also small, and articles are truncated to 2,000 characters before tokenization (Chapter 5, Section 5.5), so the model may simply not see enough of the source to verify many claims. The accuracies at $n = 5{,}000$ and $n = 10{,}000$ are higher for both conditions (0.58–0.59), which is consistent with more training data helping the reward model in general, even as the gap between conditions closes.

A preliminary ablation tested the truncation explanation directly by varying the retained article length under the RoBERTa-base backbone, three seeds per setting (Table 7.1). The result runs against the truncation hypothesis as originally stated: accuracy was *higher* with less retained context (500 and 1,000 characters) than at the thesis's own 2,000-character setting, for both conditions. A plausible reading is that 2,000 characters is not actually more informative in practice, because the concatenation of article prefix and summary already exceeds the 512-token cap at that length (Chapter 5, Section 5.5), so how much of the 2,000-character budget survives tokenization varies unpredictably with summary length across examples. A shorter, more consistently-applied budget may therefore produce a more uniform input distribution even though it discards more raw text. This ablation does not yet isolate that explanation from a simple backbone effect, since it was run only under RoBERTa-base; the two prepared but unexecuted arms of the same ablation, under `deberta-v3-base` at 512 and 1,024 tokens, would hold the backbone fixed while varying context length and are recorded alongside the truncation limitation (Chapter 8, Limitation 14). Taken as it stands, this result reframes rather than resolves the truncation question: raw article length is not the limiting factor it was hypothesized to be, at least not in the straightforward way of "more characters, more evidence, higher accuracy," and the low absolute ceiling remains only partially explained.

Table 7.1.: Truncation ablation, RoBERTa-base arms (three seeds each). All three rows use the 512-token cap; `max_article_chars` is the character budget before tokenization. The `deberta-v3-base` arms (512 and 1,024 tokens) were prepared but not completed; see Chapter 8.

Max article chars	Holistic	GCA	Gap (GCA − Holistic, pp)
500	0.5553	0.5850	+2.97
1,000	0.5663	0.5903	+2.40
2,000	0.5290	0.5480	+1.90[1]

7.5. Interpreting the Larger-Scale Results

The $n = 1{,}000$ advantage was initially not reproduced with comparable magnitude in single runs at $n = 5{,}000$ and $n = 10{,}000$ (Chapter 6, Section 6.2), and this section originally posed a question it could not answer: does that movement reflect a genuine dataset-size effect, or is it the same run-to-run variation already documented at $n = 1{,}000$, simply unmeasured at the larger sizes because only one run was performed at each? The distinguishing test was stated explicitly at the time: if the scale effect is statistical noise, repeated runs at $n = 5{,}000$ should produce a confidence interval overlapping the $n = 1{,}000$ estimate; if it reflects a real, scale-dependent change, they should not. Twenty-run campaigns at both $n = 5{,}000$ and $n = 10{,}000$, using the same corrected, deterministic training procedure as the $n = 1{,}000$ extension and matching its resolution exactly, now answer that question.

The result is unambiguous. At $n = 5{,}000$, the mean gap across twenty runs is -0.22

[1]The three seeds used here (1, 2, 3) are a subset of the twenty used in the headline campaign (Chapter 6, Section 6.1), not an independent draw. The 2,000-character row's holistic figure, 0.5290, matches the mean of seeds 1–3 in that campaign (0.5293) closely, as expected under the same configuration and seeds. Its GCA figure, 0.5480, is 1.7 percentage points below the corresponding three-seed mean there (0.5650); this size of gap is within the run-to-run spread already documented for GCA at $n = 1{,}000$ (individual seeds range from 0.543 to 0.592, Table 6.1), and is expected when averaging three seeds rather than twenty. It should therefore be read as sampling noise from a small seed count, not as a discrepancy between this ablation and the headline result it is compared against.

percentage points (95% CI $[-0.66, +0.21]$pp, Wilcoxon $p = 0.3410$); at $n = 10{,}000$, -0.11 percentage points (95% CI $[-0.39, +0.18]$pp, $p = 0.5882$). Both intervals are centred almost exactly on zero and are indistinguishable from a null result. Neither overlaps the $n = 1{,}000$ interval, $[+3.19, +4.70]$pp, and the margin is not close: the nearest edges are separated by more than two and a half percentage points at both scales. This is the pattern the genuine-scale-effect explanation predicted and the statistical-noise explanation did not.

One further step is required before the language of absence is justified, because a non-significant test result is compatible both with a negligible effect and with an effect the design was too small to see. The distinction is settled by the equivalence tests reported in Chapter 6, Section 6.3, which reverse the burden of proof and ask whether effects above a given size can be *excluded*. They can: at $n = 5{,}000$ any true effect larger than ± 0.61 percentage points is ruled out at the 5% level, and at $n = 10{,}000$ any effect larger than ± 0.36 points. The design was also amply powered for the effect it was testing for, detecting 0.64 and 0.42 points respectively at 80% power, against the 3.95-point effect measured at $n = 1{,}000$. The GCA advantage observed at $n = 1{,}000$ is therefore not an artifact that would have appeared at any scale if only enough runs had been collected, nor an effect that persists below the resolution of these experiments; it is specific to the smaller dataset size, and it disappears, not merely shrinks, once training data becomes more plentiful.

This resolves what could previously only be raised as a hypothesis. Because this thesis holds candidate pool, evaluator, and reward-model architecture fixed while varying only preference-construction granularity and dataset size, the mechanism proposed earlier remains the most natural reading: at $n = 1{,}000$ the reward model is fitting a small dataset, and how cleanly the preference relation is structured matters a great deal, since a labeling that is internally more consistent is easier to recover from few examples. As the dataset grows, the model sees more examples of the same underlying relation, and additional data can compensate for a noisier labeling, so the benefit of a cleaner preference construction shrinks toward the point of being undetectable. Consistent with this account, absolute accuracy rises for *both* conditions from $n = 1{,}000$ to the larger scales (from 0.5297 and 0.5691 to roughly 0.582–0.586 at both $n = 5{,}000$ and $n = 10{,}000$), exactly as the two conditions converge. What the repeated-run evidence newly establishes is the *shape* of this pattern, a real effect at small scale that is absent at larger scale rather than merely a shrinking one, not the mechanism behind it: nothing in this design isolates article length, sentence count, or segmentation quality as an independent variable, so the account above remains the most plausible explanation rather than a demonstrated one.

One piece of direct, rather than inferred, support for it comes from how often holistic and GCA actually disagree (Chapter 6, Section 6.4.5). On any pair where the two conditions reach the same decision, both reward models train on an identical (article, chosen, rejected) triple, so the disagreeing pairs are what any gap between conditions has to come from. That fraction falls from roughly 18% of pairs at $n = 1{,}000$ to 2–3%

at $n = 5{,}000$ and $n = 10{,}000$, under the configuration these campaigns were actually trained on. This does not by itself prove the convergence account above, since it says only how much room the two conditions have to differ, not why GCA uses that room better at small n, but it is independent, measurable evidence that the room genuinely shrinks in step with the vanishing gap, rather than an assumption required to make the account work.

Read this way, the larger-scale results are a substantive finding rather than an unresolved caveat. GCA provides a confirmed learnability advantage when preference data are scarce, and no detectable advantage once enough data are available for holistic preferences to be recoverable on their own terms. This is also the regime in which automatic preference construction is most attractive in practice, since data scarcity is precisely where collecting human labels would be most burdensome; the result therefore bounds GCA's practical usefulness to a specific and identifiable regime rather than leaving that boundary unmeasured, as the original single-run design had.

7.6. The Ground-Truth Reversal: A Stronger and Stranger Pattern

The account just given, that additional training data let holistic preferences "catch up" to a cleaner but data-hungry GCA signal, predicts *convergence*: as n grows, the gap should shrink toward zero and stay there, because both conditions are approaching the same ceiling from below. That prediction holds for learnability (Section 7.5). It does not hold for RQ4. Against the independent, held-out ground truth (Chapter 4, Section 4.8), GCA is significantly more precise than holistic at $n = 1{,}000$ on comparisons with a genuine quality gap, statistically indistinguishable at $n = 5{,}000$ (consistent with convergence so far), and then significantly *less* precise than holistic at $n = 10{,}000$ on three of four ground-truth constructions (Chapter 6, Section 6.4.2). A convergence account has nothing to say about why the line should keep moving past the point where the two conditions meet. Something in addition to "holistic catches up" is needed to explain why GCA falls *behind*, and this section considers what that might be and what can currently be ruled out.

One clarification narrows the target before any mechanism is proposed. The reversal is not GCA deteriorating. The absolute success rates (Chapter 6, Table 6.8) show GCA improving with additional training data on every construction except the same-article one, from 86.00% to 90.00% on top/bottom 10% between $n = 1{,}000$ and $n = 10{,}000$. Holistic improves over the same interval from 76.75% to 91.95%. Both conditions therefore benefit from more data, as the convergence account expects; what the account does not predict is that they benefit at such different rates, holistic gaining roughly four times as much as GCA. Any adequate explanation has to account for GCA's flatter returns to scale rather than for a decline that does not occur, and this is the form in which the two candidate mechanisms below were posed.

One candidate was tested directly rather than left as speculation. If GCA's uniformly margin-0 training pairs contain a growing share of near-zero-margin, effectively noise-level comparisons relative to holistic's, a reward model trained on more of them at larger n might be fitting that noise rather than genuine signal, which would predict exactly this pattern: fine at small n, where there is not enough data to fit the noise precisely, and increasingly harmful as n grows. Checked against the stored per-pair score gaps in the actual training data, this does not hold: at $n = 5{,}000$ and $n = 10{,}000$, the scales at which GCA actually loses, the near-tie fraction is essentially identical between the two conditions rather than larger for GCA, and at $n = 1{,}000$, where GCA leads, GCA has *fewer* near-tie pairs than holistic (21.3% against 24.6%). The one scale showing an appreciable gap between the conditions is thus the scale at which GCA wins, and the gap points the wrong way for the hypothesis (Chapter 6, Section 6.4.4). Ruling this out is informative on its own terms: it means the reversal is not simply an artifact of an unfiltered margin setting that happened to interact badly with GCA's aggregation, but it leaves the actual mechanism open.

A second candidate follows the same logic as the temperature-confound check in Section 7.10, but asks a different question of it. That section shows that summary length and sentence count do not carry the *training label*: a classifier predicting the evaluator's decision from length or sentence count alone performs at chance (Chapter 4, Table 4.4). This does not establish that a *trained reward model* cannot come to rely on length or sentence count as a shortcut regardless of whether the label technically depends on it: a model can learn a spurious but locally predictive correlate of its training labels even when that correlate does not determine the labels outright, and it can lean on that correlate more heavily the more training examples reinforce it [12]. This is a different empirical question from the one Section 7.10 already answers, because it is about what the trained model does at inference time on unseen ground-truth summaries, not about what the label construction encodes. Unlike the mode-interaction hypothesis of Section 7.2, this one required no new training to test: every RQ4 checkpoint was already on disk, so it could be scored directly against the held-out ground-truth summaries and its predicted reward correlated with each summary's length and sentence count (Chapter 6, Section 6.4.6). The hypothesis predicts the GCA-minus-holistic gap in this correlation should be largest at $n = 10{,}000$, where GCA loses; instead the gap peaks at $n = 3{,}000$ and nearly vanishes at $n = 10{,}000$, where the two conditions converge to almost identical length-reliance. This candidate is therefore also rejected. What does emerge is a symmetric pattern that affects both conditions rather than distinguishing them: both grow markedly more correlated with summary length as n increases, a five-fold increase in magnitude from $n = 2{,}000$ to $n = 10{,}000$ for each. That pattern cannot explain the reversal, since it does not differentiate GCA from holistic, but it is a plausible contributor to the low absolute accuracy ceiling neither condition breaks (Section 7.4). With both mechanical candidates ruled out, the mechanism behind the reversal remains open and is recorded as a limitation (Chapter 8, Limitation 21).

Whatever the mechanism, the practical implication is stronger than the learnability result alone would suggest. Section 7.5's closing statement, that GCA helps when data are scarce and does no harm once data are plentiful, has to be revised. Against the ground truth, GCA does not merely stop helping at scale; it becomes the worse choice. The regime in which GCA is advisable is narrower than the learnability result on its own implies: small training sets *and* comparisons with a genuine quality gap between the two options being ranked, not large training sets, and not the closest, most ambiguous comparisons even at small n (Chapter 6, Table 6.7, same-article row).

7.7. Implications for RLAIF Preference Construction

Three implications follow for anyone constructing AI-generated preference data from a fixed factuality judge, independent of whether the specific GCA method studied here is adopted.

First, the aggregation formula used to combine sentence-level scores into a preference label is not a minor implementation detail. The original weakest-link-penalty formula ($\alpha = 0.5$) actively hurt GCA relative to a plain mean (Chapter 6, Table 6.14), which runs counter to the intuition that motivated it: a formula designed to be more sensitive to localized errors was, in practice, more sensitive to noise than to signal. Aggregation-formula choices of this kind should be validated empirically rather than assumed correct from first principles. This is a qualification worth attaching to the broader case for fine-grained supervision [40]: locating errors more precisely helps only if the rule that converts those locations into a training signal is itself sound.

Second, judge configuration can matter more than the granularity decision that is nominally the point of comparison. Section 7.2's reading of AlignScore's mode behavior suggests that part of what looked like a question about holistic-versus-sentence-level supervision was really a question about how much sentence-level decomposition the judge was already doing internally. Anyone comparing supervision granularities on top of a judge with its own internal aggregation behavior should check what that judge already does before attributing an effect to the granularity of their own pipeline. Surveys of automatic judging emphasize sensitivity to prompt and configuration choices [13, 43]; the present result adds that a judge's internal decomposition can also absorb the very manipulation an experiment intends to study.

Third, a result validated at one dataset scale should not be treated as general without a scale check. This is a caution this thesis can make credibly only because Section 7.5's robustness check was actually run rather than assumed unnecessary; had the study stopped at the $n = 1{,}000$ result, it would have reported a materially less qualified conclusion than the evidence supports.

7.8. Judge Reliability Without Human Auditing

No independent human validation of the factuality labels was collected in this study, and it is worth stating precisely what this does and does not leave open. The twenty repeated runs in Chapter 6, Section 6.1 establish that the difference between the two preference sets is repeatedly observed: it survives changes in fold assignment and appears in every run of the campaign, so it is not confined to a single favorable run.

Reproducibility and correctness are separate properties, however. That a preference relation is consistently recoverable says nothing about whether the underlying AlignScore judgments agree with what a human reader would consider faithful to the source. A judge can produce labels that are highly reproducible and systematically wrong, and nothing in this study would detect that case. The labels function here as the reference standard, not as verified ground truth, and every result is conditional on them in that sense. Establishing agreement with human factuality judgments requires annotation this study did not collect, quantified against a chance-corrected agreement measure such as Cohen's κ [6], and the question is recorded as an open limitation in Chapter 8. The concern is not peculiar to this study: reviews of reinforcement learning from human feedback identify the quality of the preference data itself, and the difficulty of auditing it, as a central open problem [3], and substituting an automatic judge for human annotators relocates that problem rather than removing it.

7.9. Stochastic Variation Between Runs

Run-to-run variation is large enough in this setting to determine what conclusions can be drawn, and it deserves treatment as a finding rather than as an inconvenience. In the six-run precursor campaign, two runs launched at the same seed differed by 4.7 percentage points ($+0.060$ versus $+0.013$), and the gap across those six runs ranged from -0.010 to $+0.060$ (Appendix D, Section D.4). A study reporting only one of them could honestly have concluded either that GCA helps substantially or that it does not help at all.

This is a known hazard rather than a peculiarity of the present setup. Fine-tuning pretrained language models is sensitive to random seed alone, with weight initialization and data order contributing comparably to the variance of out-of-sample performance [8], and reporting score distributions across several runs rather than a single figure has been urged for precisely this reason [34]. Here the variation had a specific and locatable cause: PyTorch's generator was not seeded in the cross-validation path, so weight initialization, batch order and dropout varied between executions regardless of the seed value (Chapter 5, Section 5.5). That property was fixed before the twenty-run campaign, each of whose seeds is consequently an independent and exactly reproducible sample, so the concern no longer applies to the results reported as headline findings.

The practical implication generalizes beyond this study regardless of the fix. At the accuracy levels and dataset sizes involved here, differences of one to two percentage points between conditions are within the range that a single under-seeded run can produce by chance. Comparisons of preference-construction methods reported from single training runs at this scale, without controlling every source of training randomness, should be treated with corresponding caution.

7.10. Confounds That Remain

Two confounds could not be removed within this design, and both plausibly affect the reported numbers.

The candidate pairs differ in decoding temperature, and Chapter 4, Table 4.3 shows the consequence is not small: the lower-temperature candidate is preferred in 68.5% of holistic pairs and 63.0% of GCA pairs on the data where this could be measured. The asymmetry also differs between conditions by roughly five percentage points, so the two preference sets are not equivalent with respect to this confound.

How much this threatens the reported comparison depends on whether the asymmetry is *learnable* from the summary text, and that question has now been examined rather than left open. The reward model never observes decoding temperature or candidate position; it sees only (article, summary) pairs (Chapter 5, Section 5.5). For the temperature skew to inflate its accuracy, some textual correlate of temperature would have to be recoverable from the summary. The surface-feature baselines in Chapter 4, Table 4.4 show that the two most obvious such correlates do not qualify: classifiers deciding purely from summary length or sentence count reach 0.465–0.535 accuracy against the judge's own decisions, which is chance, and the mean length difference between chosen and rejected summaries is 2 characters under holistic scoring and -6 under GCA. Whatever the reward models are fitting, it is not summary length.

This weakens the confound but does not dispose of it. Lexical and stylistic correlates of temperature beyond length, such as word-frequency profiles, hedging, or named-entity density, were not tested, and a reward model with a pretrained encoder is well placed to pick up exactly that kind of regularity. What can be said is that the simple and most frequently raised version of the objection, that the models are covertly learning to prefer longer or shorter summaries, is empirically excluded. Sampling both candidates at the same temperature would remove the confound at source and remains the cleanest correction available.

The reward model also sees only a truncated prefix of each source article (Chapter 5, Section 5.5), while subset selection admits articles several times longer than the retained prefix. Where the evidence bearing on a claim lies beyond that prefix, the corresponding preference is not recoverable from the model's input even in principle. This is a plausible contributor to the low absolute accuracies discussed in Section 7.4, though its magnitude was not measured, and it should not be assumed to be the dominant cause without the ablation described there.

7.11. Generalizability

Every result reported here is obtained under one combination of choices: CNN/DailyMail as the corpus, AlignScore as the evaluator, Mistral-7B-Instruct-v0.3 as the candidate generator, and a RoBERTa-base Bradley–Terry model as the reward model. Each of these bounds the conclusions in a distinct way.

The evaluator is the most consequential. Section 7.2 raises the possibility that whether external decomposition helps is related to whether the evaluator already decomposes internally, which would make the finding partly *about* the evaluator's architecture rather than simply conditional on AlignScore as a fixed instrument. If that reading is correct, a different factuality model with different internal aggregation would shift the result. The corpus matters because CNN/DailyMail summaries are short, extractive-leaning, and drawn from a single register; domains with longer or more abstractive summaries would change both the number of sentences per summary and the difficulty of the underlying factuality judgments. The reward-model backbone matters because a stronger encoder, or one with a longer context window, might recover more of the preference relation and thereby narrow or widen the difference between conditions.

Taken together, these dependencies mean the study is best read as characterizing behavior under one specific configuration, and as motivating a hypothesis about the interaction between external and evaluator-internal decomposition, rather than as establishing a quantity that transfers directly to other setups.

8. Conclusion

8.1. Summary

This thesis asked a controlled question: does converting sentence-level factuality scores into a summary-level preference through Granular Credit Assignment produce a more learnable reward-model signal than scoring each candidate summary holistically. The two conditions were compared under an identical candidate pool, an identical factuality judge, and an identical Bradley–Terry training recipe, so that the only manipulated factor was the preference-construction procedure itself.

Averaged over an extended campaign of twenty independently seeded runs at $n = 1{,}000$ (Chapter 6, Section 6.1), GCA preferences yield an advantage of approximately $+3.95$ percentage points in reward-model pairwise accuracy, favoring GCA in all twenty runs, with a bootstrap 95% confidence interval of $[+3.19, +4.70]$ points and a run-level Wilcoxon $p < 0.001$. An earlier six-run campaign had given a consistent but structurally underpowered estimate of $+3.08$ points that could not, with only six paired observations, reach the conventional significance threshold; the extension confirms that direction rather than revealing a different effect. The advantage does not persist as the dataset grows: twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$, using the same corrected procedure and matching the $n = 1{,}000$ campaign's resolution exactly, give mean gaps of -0.22 and -0.11 percentage points respectively, both statistically indistinguishable from zero (95% CIs $[-0.66, +0.21]$ and $[-0.39, +0.18]$ points), with neither interval overlapping the significant $n = 1{,}000$ result. Absolute accuracy improves for both conditions as the dataset grows even as their difference vanishes. Uncertainty estimates computed over cross-validation folds remain anti-conservative, because folds within a run are not independent, but the run-level estimates at all three scales are now each supported by a properly powered test.

The experiments therefore do not support a universal claim that GCA is preferable to holistic factuality supervision. They show that sentence-level aggregation produces a statistically significant, repeatedly confirmed improvement in the learnability of automatically generated pairwise preferences specifically when training data are scarce, and no detectable improvement once training data become more plentiful. An exploratory single-run sweep suggested that the advantage appears only under evaluator modes that do not themselves decompose the input, motivating the hypothesis that external and evaluator-internal decomposition act partly as substitutes; the confirmatory campaigns do not support that hypothesis, which is discussed in Chapter 7, Section 7.2 and revisited under RQ2 below. What the aggregation comparison does establish is that the rule used to combine sentence scores matters: a weakest-link penalty performed measurably worse than a simple mean. Taken together, the results indicate that the usefulness of granular credit assignment is bounded to a specific, data-scarce regime, and depends within that regime on the aggregation rule chosen, rather than following from finer granularity as such.

A fourth question (RQ4, Chapter 4, Section 4.1) asked whether this learnability advantage also translates into more accurate reward-model judgments against an independent, held-out ground truth of 500 articles disjoint from every training pool used in this thesis, scored by the same locked evaluator but never used to train any reward model (Section 4.8). It does, but only in the same narrow, data-scarce regime, and the pattern is more pronounced than for learnability: at $n = 1{,}000$, GCA is significantly more precise than holistic on comparisons with a genuine quality gap (up to $+9.25$ percentage points, $p < 0.001$), at $n = 5{,}000$ the two are indistinguishable, and at $n = 10{,}000$ holistic is significantly *more* precise than GCA on three of four ground-truth constructions (Chapter 6, Section 6.4). The advantage does not merely fail to persist as training data grow, as the learnability result also showed; it reverses. Five-seed checks at $n = 2{,}000$ and $n = 3{,}000$ show the decline passing through a null region around $n = 3{,}000$–$5{,}000$, in which no gap is distinguishable from zero in either direction. Two candidate mechanisms were tested directly and both rejected. That GCA's uniformly unfiltered training pairs contain disproportionately more near-zero-margin comparisons than holistic's does not hold: at $n = 5{,}000$ and $n = 10{,}000$, where GCA loses, the near-tie fraction is essentially identical between the two conditions, and at $n = 1{,}000$, where GCA leads, GCA has *fewer* near-tie pairs than holistic (21.3% against 24.6%), so the only appreciable gap runs opposite to what the hypothesis requires (Section 6.4.4). That GCA increasingly relies on a summary-length or sentence-count shortcut relative to holistic as training data grow also does not hold: scoring the held-out ground truth with the already-trained checkpoints shows the GCA-minus-holistic gap in this correlation peaks at $n = 3{,}000$ and nearly vanishes at $n = 10{,}000$, the opposite of what the hypothesis predicts (Section 6.4.6). The mechanism behind the reversal remains open (Limitation 21).

8.2. Answers to the Research Questions

RQ1 (does GCA produce a more learnable preference signal than holistic scoring?) is answered *yes at small scale, no at larger scale, both with statistical support*. An average advantage of $+3.95$ percentage points is observed across twenty independent runs at $n = 1{,}000$, favoring GCA in all twenty, with a run-level Wilcoxon $p < 0.001$, a bootstrap confidence interval that does not cross zero, and a large standardized effect ($d = +2.22$). Twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$, the same resolution as $n = 1{,}000$, give mean gaps statistically indistinguishable from zero ($p = 0.3410$ and $p = 0.5882$), with confidence intervals that do not overlap the $n = 1{,}000$ result; equivalence tests at those scales additionally exclude any effect larger than ± 0.61 and ± 0.36 percentage points (Section 6.3), so the absence is established positively rather than inferred from a failure to reject. The answer is therefore not merely configuration-dependent in an unresolved sense: it is a confirmed advantage at one dataset scale and a confirmed absence of advantage at two larger ones, with both conclusions resting on repeated, adequately powered evidence rather than single runs.

RQ2 (how do evaluator configuration and aggregation strategy affect relative learnability?) is answered *partially, and less conclusively than this thesis originally believed*. The aggregation formula clearly matters: the original weakest-link penalty measurably hurt GCA relative to the simple mean (Chapter 6, Section 6.5.1). Evaluator mode was believed to matter in the same way, based on an exploratory single-run sweep in which the two modes performing internal decomposition showed no GCA advantage. That belief does not survive the discovery, during final review, that the confirmatory twenty-run campaign was itself trained under one of those two modes (`nli_sp`, Chapter 5, Section 5.4) and nonetheless reproduced a GCA advantage as large as the one originally attributed to the other mode. Taken together with the six-run campaign, which genuinely used the other mode and also favored GCA, the confirmatory evidence now available points against evaluator mode being the deciding factor, which is the opposite of what the exploratory sweep suggested (Chapter 7, Section 7.2).

RQ3 (how stable are the differences across runs and dataset sizes?) is answered *differently, and now conclusively, across the two axes it asks about*. Across repeated runs at a fixed dataset size, the difference is highly stable: all twenty runs of the extended campaign at $n = 1{,}000$ favor GCA, with a standard deviation of 1.78 percentage points around a mean of $+3.95$, and the twenty-run campaigns at $n = 5{,}000$ and $n = 10{,}000$ are similarly consistent around a mean of approximately zero (SD 1.03 and 0.67 points respectively). Across dataset sizes, stability does not hold, and this is no longer an open question left by single or few runs: twenty repeated runs at each of $n = 5{,}000$ and $n = 10{,}000$, the same resolution used at $n = 1{,}000$, confirm a null result whose confidence intervals do not overlap the significant $n = 1{,}000$ estimate (Section 7.5). Run-to-run stability at fixed scale is well established at all three sizes tested; stability across scale does not hold, and this absence is now itself a properly powered finding rather than an artifact of insufficient replication.

RQ4 (does the learnability advantage also hold against an independent, held-out ground truth, and as training data grow?) is answered *yes at $n = 1{,}000$, and no at $n = 10{,}000$, with the direction reversing rather than merely vanishing*. Against a held-out set disjoint from every training pool used in this thesis (Section 4.8), GCA is significantly more precise than holistic at $n = 1{,}000$ on comparisons with a genuine quality gap ($+9.25$ percentage points on the sharpest construction, $p < 0.001$; Chapter 6, Section 6.4.1), statistically indistinguishable from holistic at $n = 5{,}000$, and significantly *less* precise than holistic at $n = 10{,}000$ on three of four constructions ($p = 0.010$–0.036; Section 6.4.2). Five-seed checks at $n = 2{,}000$ and $n = 3{,}000$ show the decline passing through a null region around $n = 3{,}000$–$5{,}000$ (Section 6.4.3). This answer is stronger, in one sense, than RQ1's: it was obtained against data the trained reward models never saw and never influenced their selection, so it is not subject to the nesting caveat that qualifies RQ1–RQ3 (Section 4.7). It does not, however, remove the evaluator-dependence caveat that qualifies every result in this thesis: the ground-truth labels are still constructed by AlignScore, the same evaluator used throughout (Section 8.4). Nor is the comparison as neutral as "held out" suggests: the ranking used to score both conditions is built from GCA's own aggregation rule, so it is in-distribution for GCA and out-of-distribution for holistic in a way that disjoint articles alone do not fix, and the $n = 1{,}000$ advantage in particular has not been checked against a holistic-defined criterion (Section 8.4).

8.3. Status of the Working Hypotheses

Chapter 4, Section 4.1 stated three working hypotheses that guided the design. Their status against the evidence collected is as follows.

H1 (sentence-level aggregated preferences are more learnable than holistic preferences, because they localize the factuality signal within a long output) is *supported at $n = 1{,}000$ and refuted at larger scale*. The predicted direction holds decisively where training data are scarce, in all twenty runs, with a large standardized effect ($d = +2.22$). It does not hold at $n = 5{,}000$ or $n = 10{,}000$, where equivalence tests exclude effects larger than ± 0.61 and ± 0.36 percentage points respectively (Chapter 6, Section 6.3). The hypothesis as originally stated was unconditional and is therefore not supported in that form; what survives is a scale-bounded version of it.

H2 (the evaluator's configuration affects whether any such difference is observable) is *not supported by the confirmatory evidence, contrary to what this thesis concluded before a final-review discovery*. The exploratory single-run sweep suggested exactly this, with GCA leading only under modes that skip AlignScore's internal decomposition (Chapter 6, Section 6.5.2). But the twenty-run campaign that was meant to confirm this, and that this thesis believed had, was actually trained under the mode predicted *not* to show an advantage, and showed one anyway, as large as the single-run result it superseded (Chapter 5, Section 5.4; Chapter 7, Section 7.2). The six-run campaign, run under the other mode, also favored GCA. H2 is retained here, marked as refuted rather than removed, because the reasoning behind it is not itself wrong and remains worth testing under a correctly configured repetition (Limitation 1); what is refuted is the specific claim that this thesis's own confirmatory campaigns support it.

H3 (effects observed at one dataset scale need not persist at another) is *supported, and is the hypothesis the evidence bears on most directly*. It was included as a methodological precaution rather than a substantive prediction, and it proved to be the one that determined the shape of the final result: an effect that is significant, large, and perfectly consistent at $n = 1{,}000$ is confirmed absent at both larger scales. Had the study stopped at $n = 1{,}000$, as the original design would have permitted, it would have reported an unqualified positive finding that the fuller evidence does not support.

8.4. Limitations

1. **The confirmatory campaigns were not trained on the intended configuration.** Found during final review, after the experimental phase had concluded: the twenty-run campaigns at all three scales, the $n = 2{,}000/n = 3{,}000$ crossover checks, and the RQ4 checkpoints were trained on preference data built under AlignScore's `nli_sp` mode, not the `nli` mode selected in Chapter 6, Section 6.5.2 and described elsewhere in this thesis as locked; at $n = 1{,}000$ specifically, the aggregation exponent is additionally $\alpha = 0.5$ rather than the locked $\alpha = 0.0$. The comparison between holistic and GCA within each campaign remains fair, since both conditions in a given run were built under the same configuration, and every reported number is a genuine measurement of the run that produced it. What is not sound is this thesis's account, prior to this discovery, of *why* GCA leads at $n = 1{,}000$: that account attributes the advantage to AlignScore's `nli` mode specifically, and the confirmatory campaign that was believed to test this was not run under that mode. A second consequence follows from the α discrepancy specifically: because $n = 1{,}000$ was built under $\alpha = 0.5$ while $n = 5{,}000$ and $n = 10{,}000$ were correctly built under $\alpha = 0.0$, the RQ3 scale comparison (Chapter 6, Section 6.2) varies the aggregation exponent alongside dataset size rather than dataset size alone, so an effect of the exponent could in principle imitate an effect of scale. Existing evidence, a six-run campaign correctly built under $\alpha = 0.0$ that also favors GCA at $n = 1{,}000$, weighs against this, but no properly powered campaign exists at $n = 1{,}000$ under $\alpha = 0.0$ to compare against the larger scales on equal terms, so the possibility cannot be excluded (Chapter 5, Section 5.4). Both issues are reported in full there, and the evaluator-mode consequence is addressed in Chapter 7, Section 7.2. Neither was corrected by re-running the affected campaigns, both for lack of time before submission and because doing so is new experimental work beyond what had been agreed with the supervisors at this stage of the project.
2. **No human factuality audit.** No independent human annotation was collected in the final study, so the evaluator's judgments were never checked against human assessments of faithfulness (Chapter 7, Section 7.8).
3. **The labels are not independent ground truth.** AlignScore supplies both the holistic and the sentence-level scores, so it functions as the reference standard rather than as a verified one. Every result is conditional on that evaluator. RQ4 (Chapter 6, Section 6.4) evaluates trained reward models against summaries and articles they never saw during training, which removes the nesting concern discussed below, but it still scores those summaries with AlignScore-GCA, so it does not remove

evaluator-dependence: a systematic bias in AlignScore itself would still be present in both the training preferences and the RQ4 ground truth alike.

4. **The RQ4 ground-truth criterion is GCA's own rule, not a neutral one.** The ranking RQ4 scores both reward models against is built from GCA's aggregation formula (sentence-level AlignScore, mean at $\alpha = 0.0$), the same rule the GCA reward model is trained to predict; the holistic reward model is scored against a criterion built from a rule it never saw (Chapter 4, Section 4.8). The held-out *articles* are genuinely disjoint from every training pool, but the *criterion* is not neutral between the two conditions in the way this thesis originally claimed. Some part of GCA's RQ4 advantage, including the $+9.25$ percentage-point result at $n = 1{,}000$ (Chapter 6, Section 6.4.1), could in principle reflect this asymmetry rather than a general precision advantage, and the data collected here cannot rule that out.

5. **The locked aggregation rule does not solve the problem that motivated it.** The thesis is motivated by a summary that is mostly accurate but contains one fabricated sentence, which a holistic judge may still prefer over a uniformly mediocre but non-fabricating alternative (Chapter 1, Section 1.1). The locked configuration, however, is the simple mean ($\alpha = 0.0$), and a mean dilutes a single low-scoring sentence in exactly the way a holistic judgment does; Figure 4.2 (Chapter 4, Section 4.2) shows this directly, with the weakest-link penalty formula, not the locked mean, producing the sentence-level sensitivity the motivating example calls for. That penalty formula was tested and rejected for performing worse overall (Chapter 6, Section 6.5.1). What this thesis's confirmed result actually establishes, then, is that GCA's benefit at $n = 1{,}000$ comes from scoring sentences independently and averaging the result, not from penalizing the weakest one; the fabrication-masking scenario that motivated the project in the first place remains unresolved by the configuration that was found to work best.

6. **Preference learnability is not factual correctness.** The dependent variable measures how consistently a preference relation can be recovered by the chosen architecture, not whether the preferences are factually right (Chapter 4, Section 4.4).

7. **RQ4 models are trained on more data than RQ1–RQ3 models.** The RQ1–RQ3 accuracies are computed from fold models trained on 80% of the preference set; the RQ4 checkpoints are trained on the full 100%, since k-fold cross-validation retains no reusable single model (Chapter 5, Section 5.5; Chapter 6, Section 6.4). Comparing the two sets of results, as Chapter 7, Sections 7.5 and 7.6 do, is therefore a comparison across both a different evaluation target and a different training-data fraction at once. Nothing in this thesis separates the two, so part of the difference between the learnability and ground-truth patterns could in principle be attributable to the training-data fraction rather than to the evaluation target alone.

8. **No downstream policy evaluation.** The final study contains no DPO or other policy-optimization stage, so it does not establish whether models trained on either preference set generate more factually consistent summaries.

9. **Both reward models are weak in absolute terms.** At $n = 1{,}000$ the conditions reach 0.5297 and 0.5691 pairwise accuracy against a chance level of 0.5, so the comparison is between two weak signals (Chapter 7, Section 7.4).

10. **Cross-validation folds are not independent.** Uncertainty estimates computed by pooling folds across runs treat correlated observations as independent and are therefore anti-conservative regardless of campaign size; the run-level test is the appropriate unit of replication and is what the reported significance result is based on (Section 7.3).

11. **Incomplete determinism in the original six-run campaign.** In the campaign that first produced the $+3.08$-point estimate, subset selection and cross-validation fold assignment were explicitly seeded, but PyTorch's generator was not seeded in the cross-validation path, leaving weight initialization, batch ordering and dropout uncontrolled (Chapter 5, Section 5.5). This was fixed before the extended twenty-run campaign was carried out, so the headline significance result in this thesis is based on fully deterministic, exactly reproducible runs; the fix does not retroactively apply to the original six runs, which remain a documented instance of the underlying problem (Chapter 7, Section 7.9).

12. **Configuration selected on observed results.** The aggregation strategy and evaluator mode were chosen after inspecting $n = 1{,}000$ outcomes (Chapter 4, Section 4.6), and the $n = 5{,}000/n = 10{,}000$/RQ1–RQ3 subsets are nested supersets of that same data (Section 4.7). RQ4 is the one result in this thesis that evaluates the selected configuration on data disjoint from the set it was chosen on (Section 4.8); it was formulated specifically to close this gap, and does so for the training-data axis, though not for the evaluator-choice axis noted above.

13. **Candidate generation confounds temperature with condition.** The two candidates differ in decoding temperature, and the lower-temperature candidate is preferred substantially more often under both conditions and to differing degrees (Chapter 4, Section 4.10). Summary length and sentence count are excluded as carriers of this information, both performing at chance against the evaluator's decisions (Table 4.4), but subtler temperature-correlated textual features were not tested and could still carry part of the label information.

14. **Source-document truncation.** The reward model receives only a truncated prefix of each article, so evidence located later in the source is unavailable to it (Chapter 5, Section 5.5).

15. **Evaluator and dataset specificity.** All findings are obtained on CNN/DailyMail with AlignScore, Mistral-7B-Instruct-v0.3 candidates, and a RoBERTa-base reward model, and need not transfer to other domains, evaluators, or architectures.

16. **Sentence segmentation was not evaluated.** Summaries are split with a regex-based splitter whose accuracy was never measured (Chapter 5, Section 5.4).

17. **Nested dataset sizes.** The $n = 1{,}000$, $n = 2{,}000$, $n = 3{,}000$, $n = 5{,}000$ and $n = 10{,}000$ subsets overlap almost completely with one another (Chapter 4, Section 4.7), so the RQ1–RQ3 runs at each scale are not independent samples of one another. The RQ4 held-out ground truth (Section 4.8) was constructed to be disjoint from all five, so it is not affected by this limitation, but it is itself a single fixed 500-article set reused across every scale, so the five RQ4 scale points share the same evaluation sample rather than being independently sampled test sets.

18. **RQ4's four constructions are nested, and uncorrected.** The top/bottom 5%, 10%, and 25% constructions are subsets of one another, all drawn from the same ranked pool of 1,000 held-out summaries, and no multiple-comparison correction is applied across the four constructions tested at each scale (Chapter 6, Section 6.4.7). Statements of the form "significant on three of four constructions" should accordingly be read as three overlapping views of one ranking, not as three independent replications.

19. **No error-category analysis.** No analysis links the observed effects to specific categories of factual error; this was not part of the final research questions.

20. **Unequal replication at the two intermediate scales.** The $n = 2{,}000$ and $n = 3{,}000$ points used to locate the RQ4 crossover (Section 6.4.3) were each run at five seeds rather than the twenty used at $n = 1{,}000$, $n = 5{,}000$, and $n = 10{,}000$, to keep the total experiment volume tractable in the time remaining before submission. They locate the crossover to a region rather than a precise value.

21. **The reversal's mechanism is unexplained.** Two candidate explanations for why GCA's ground-truth precision reverses at large training-set sizes were tested directly and both rejected: that its training pairs contain more near-zero-margin comparisons than holistic's (Section 6.4.4), and that it increasingly relies on a summary-length or sentence-count shortcut relative to holistic as training data grow (Section 6.4.6). No mechanism has been confirmed.

8.5. Future Work

The items below concern what someone could build on top of this work, given that it works, rather than the open methodological questions of whether it does; those are cataloged as limitations above and are prerequisites for the confirmatory result to be trusted at face value, not directions for extending it.

- **Adaptive and learned aggregation rules.** This thesis compares a small, fixed family of closed-form aggregation formulas parameterized by a single global exponent α (Chapter 4, Section 4.3). A natural extension is an aggregation rule that is not fixed in advance: a lightweight model that learns how much weight to give the weakest sentence conditioned on the rest of the summary, or a penalty strength calibrated per instance rather than set once for the whole dataset. Limitation 5 above notes that the locked mean does not, in fact, penalize the fabricated-sentence case that originally motivated this line of work; an adaptive rule is one way to pursue that original goal without reintroducing the noise that made the fixed weakest-link penalty perform worse overall (Section 6.5.1).

- **Testing whether the advantage generalizes.** The confirmed $n = 1{,}000$ result is obtained with one evaluator (AlignScore), one dataset (CNN/DailyMail), one candidate generator (Mistral-7B-Instruct), and one reward-model backbone (RoBERTa-base). Establishing that granular feedback's advantage is a property of the construction method, rather than of this particular combination, requires repeating the same controlled comparison with at least one factor substituted at a time: a different factuality judge, a different summarization domain, or a different reward-model architecture. This is the comparison needed to move from "GCA outperforms holistic scoring in this setting" toward a claim about granular feedback construction in general.

- **Downstream policy evaluation, run as an isolated follow-up.** Training a summarization policy on each preference set and evaluating the factual consistency of its generations would test whether the preference-learnability advantage measured here propagates to better generation quality, which is the practical outcome the learnability metric is a proxy for. This thesis deliberately excludes a policy-optimization stage because combining it with the preference-construction comparison made it impossible to attribute an observed difference to either stage individually (Chapter 4, Section 4.9). A future study could recover this comparison without reintroducing that problem by running policy optimization as a separate stage downstream of, and independent from, the preference-construction comparison established here, rather than folding both into one pipeline.

- **Cheaper human validation.** Rather than a full independent human audit of every judgment, an active-learning-style protocol that routes only the cases where the holistic and GCA judges disagree to a human annotator would give an evaluator-independent signal on exactly the comparisons that matter most, at a fraction of the annotation cost a full audit would require.

- **Extending granular feedback beyond summarization.** The decompose-then-aggregate idea behind GCA is not specific to single-document summarization; it applies wherever an output is long enough that a single holistic judgment can hide localized errors, such as long-form question answering or multi-turn dialogue. Testing it in those settings would indicate whether the narrow, conditional advantage found here is a general property of granular feedback or particular to summarization.

Bibliography

[1] Yuntao Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[2] Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3–4):324–345, 1952. doi: 10.1093/biomet/39.3-4.324.

[3] Stephen Casper et al. Open problems and fundamental limitations of reinforcement learning from human feedback. *Transactions on Machine Learning Research (TMLR)*, 2023. arXiv:2307.15217. TMLR Survey Certification, Featured Certification, and Outstanding Paper Finalist.

[4] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 4299–4307, 2017. arXiv:1706.03741.

[5] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2022. doi: 10.48550/arXiv.2210.11416.

[6] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

[7] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ, 2nd edition, 1988. doi: 10.4324/9780203771587.

[8] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah A. Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*, 2020. doi: 10.48550/arXiv.2002.06305.

[9] Esin Durmus, He He, and Mona Diab. FEQA: A question answering evaluation framework for faithfulness assessment in abstractive summarization. In *Proceedings of ACL*, 2020.

[10] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

[11] Alexander R. Fabbri et al. QAFactEval: Improved QA-based factual consistency evaluation for summarization. In *Proceedings of NAACL*, 2022.

[12] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. doi: 10.1038/s42256-020-00257-z.

[13] Jiawei Gu et al. A survey on LLM-as-a-judge. *arXiv preprint arXiv:2411.15594*, 2024.

[14] Yuzhe Gu, Wenwei Zhang, Chengqi Lyu, Dahua Lin, and Kai Chen. Mask-DPO: Generalizable fine-grained factuality alignment of LLMs. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2503.02846.

[15] Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations (ICLR)*, 2021. doi: 10.48550/arXiv.2006.03654.

[16] Karl Moritz Hermann, Tomáš Kočiský, Edward Grefenstette, Lasse Espeholt, Will Kay, Mustafa Suleyman, and Phil Blunsom. Teaching machines to read and comprehend. In *Advances in Neural Information Processing Systems 28 (NIPS)*, pages 1693–1701, 2015. arXiv:1506.03340.

[17] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.09685.

[18] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023. doi: 10.48550/arXiv.2310.06825.

[19] Ron Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1137–1143, 1995.

[20] Wojciech Kryściński et al. Evaluating the factual consistency of abstractive text summarization. In *Proceedings of EMNLP*, 2020.

[21] Philippe Laban et al. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 2022.

[22] Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.

[23] Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. RewardBench: Evaluating reward models for language modeling. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1755–1797, Albuquerque, New Mexico, 2025. arXiv:2403.13787.

[24] Harrison Lee et al. RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback. *arXiv preprint arXiv:2309.00267*, 2023.

[25] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.20050.

[26] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, 2004.

[27] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2511–2522, 2023. doi: 10.18653/v1/2023.emnlp-main.153.

[28] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019. doi: 10.48550/arXiv.1907.11692.

[29] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 12076–12100, 2023. doi: 10.18653/v1/2023.emnlp-main.741.

[30] Ramesh Nallapati, Bowen Zhou, Cícero dos Santos, Çağlar Gulçehre, and Bing Xiang. Abstractive text summarization using sequence-to-sequence RNNs and beyond. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning (CoNLL)*, pages 280–290, 2016. doi: 10.18653/v1/K16-1028.

[31] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 2022. arXiv:2203.02155.

[32] Wenjie Qiu, Yi-Chen Li, Xuqin Zhang, Tianyi Zhang, Yihang Zhang, Zongzhang Zhang, and Yang Yu. Sentence-level reward model can generalize better for aligning LLM from human preference. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Industry Track*, 2025. arXiv:2503.04793.

[33] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023. arXiv:2305.18290.

[34] Nils Reimers and Iryna Gurevych. Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 338–348, Copenhagen, Denmark, 2017. doi: 10.18653/v1/D17-1035.

[35] John Schulman et al. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[36] Abigail See, Peter J. Liu, and Christopher D. Manning. Get to the point: Summarization with pointer-generator networks. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1073–1083, 2017. doi: 10.18653/v1/P17-1099.

[37] Hwanjun Song, Hang Su, Igor Shalyminov, Jason Cai, and Saab Mansour. FineSurE: Fine-grained summarization evaluation using LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 906–922, 2024. doi: 10.18653/v1/2024.acl-long.51.

[38] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020. arXiv:2009.01325.

[39] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1 (6):80–83, 1945. doi: 10.2307/3001968.

[40] Zeqiu Wu, Yushi Hu, Weijia Shi, Nouha Dziri, Alane Suhr, Prithviraj Ammanabrolu, Noah A. Smith, Mari Ostendorf, and Hannaneh Hajishirzi. Fine-grained human feedback gives better rewards for language model training. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023. arXiv:2306.01693.

[41] Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11328–11348, Toronto, Canada, 2023. doi: 10.18653/v1/2023.acl-long.634.

[42] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*, 2020. doi: 10.48550/arXiv.1904.09675.

[43] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, et al. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*, 2023. arXiv:2306.05685.

[44] Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10586–10613, 2024.

A. Data and Code Availability

All source code, configuration files, Slurm job scripts, and analysis notebooks used to produce the results reported in this thesis are publicly available at:

`https://github.com/hasnat23/master-thesis-rlaif-gca`

The repository is organized as follows. `src/` contains the pipeline implementation, split into the modules referenced throughout Chapter 5: `src/data/` (subset selection and record schemas), `src/generation/` (candidate summary generation), `src/judging/` (the AlignScore judge wrapper and both preference-construction regimes), `src/reward_model/` (Bradley–Terry training and cross-validation), and `src/eval/` and `src/analysis/` (evaluation and statistics). `configs/` holds the YAML configuration files for subset selection, generation, and judging. `slurm/` holds the batch scripts submitted on MOGON NHR, including `slurm/train_rm_1000.sh`, which produced the headline $n = 1{,}000$ result. `analysis/` holds the standalone scripts used for the formula sweep, the judge-mode sweep, and the holistic-versus-GCA comparison, together with `analysis/aggregate_campaigns.py`, which computes the run-level means, bootstrap intervals and Wilcoxon tests for each seed campaign, and `analysis/effect_sizes_and_equivalence.py`, which computes the effect sizes, equivalence tests and minimum detectable effects reported in Chapter 6, Section 6.3. Both read the per-run JSON summaries under `outputs/` directly, so every reported figure can be regenerated from the stored run artifacts without retraining. `progress-updates/` contains the dated working notes kept over the course of the project.

The underlying corpus, CNN/DailyMail version `3.0.0` [16, 30], is publicly available through the Hugging Face `datasets` library and is not redistributed in the repository. The candidate summaries, preference files, and trained reward-model checkpoints generated during this project are likewise not committed, as they run to several gigabytes; they can be regenerated from the corpus using the configuration recorded in Appendix B. The factuality evaluator, `yzha/AlignScore` [41], and the candidate generator, Mistral-7B-Instruct-v0.3, are both publicly released model checkpoints obtained from the Hugging Face Hub.

B. Reproduction Configuration

This appendix records the exact configuration required to reproduce the headline $n = 1{,}000$ result. It restates in executable form what Chapter 4, Table 4.1 summarizes and Chapter 5 describes in prose. Under the corrected, fully-seeded training procedure described in Chapter 5, Section 5.5, a single run of these commands at a given seed value is deterministic on fixed hardware; running the full twenty-run headline campaign means repeating the reward-model training command below once per seed in $\{1, \ldots, 20\}$, as `slurm/seed_campaign.sh` does.

B.1. Summarization Prompt

Both candidate summaries for a given article are generated from the same fixed prompt template (`SUMMARIZATION_PROMPT` in `src/generation/candidates.py`); the two candidates differ only in the sampling parameters applied at decoding time.

```
Summarize the following news article in a concise paragraph.

Article:
{article}

Summary:
```

B.2. Pipeline Parameters

Table B.1 lists the parameters of each pipeline stage as used for the final comparison. Where a value differs from the default recorded in the corresponding YAML file, the command-line override in Section B.3 is authoritative; this is the case for the subset size, the tie margin, and the GCA exponent α, each of which changed over the course of the project as described in Chapter 4, Section 4.9. The evaluation mode and α below are the *intended* configuration; Chapter 5, Section 5.4 reports that the twenty-run campaign actually reproduced by the commands in Section B.3 was trained on preference data built under `nli_sp` and $\alpha = 0.5$, not the `nli` and $\alpha = 0.0$ shown here, an implementation oversight found during final review. Running the commands below as written reproduces the *intended* configuration, which is `nli`/$\alpha = 0.0$, not the one the headline campaign was actually trained on.

Table B.1.: Pipeline parameters for the final $n = 1{,}000$ comparison.

Stage	Parameter	Value
Subset selection	Corpus	CNN/DailyMail 3.0.0, `test` split
	Samples	1,000
	Subset seed	200
	Max article length	8,000 characters
	Max summary length	2,000 characters
Candidate generation	Model	Mistral-7B-Instruct-v0.3, `bfloat16`
	Candidate A	$T = 0.7$, top-$p = 0.9$
	Candidate B	$T = 1.0$, top-$p = 0.95$
	Max input tokens	1,024
	Max new tokens	256
	Batch size	4
Preference construction	Evaluator	`yzha/AlignScore`, `FacebookAI/roberta-base`
	Evaluation mode	`nli`
	GCA exponent	$\alpha = 0.0$ (simple mean)
	Tie margin	0
Reward-model training	Backbone	`FacebookAI/roberta-base`
	Epochs	5
	Batch size	8
	Learning rate	2×10^{-5}, linear warmup over 10% of steps
	Max sequence length	512 tokens
	Max article chars	2,000
	Cross-validation	5-fold; training seeds 1–20 (headline campaign, Chapter 6, Table 6.1)

B.3. Command-Line Invocations

The three stages are run in sequence. Preference construction produces both conditions from a single invocation, and reward-model training consumes both preference files so that the holistic and GCA models are trained under an identical recipe within one process.

```
# 1. Subset selection
python scripts/01_prepare_subset.py --n-samples 1000 --seed 200

# 2. Candidate generation
python scripts/02_generate_candidates.py \
    --config configs/generation_1000.yaml

# 3. Preference construction (both conditions)
python src/judging/build_reward_preferences.py \
    --mode both --alpha 0.0 --margin 0 \
    --alignscore-evaluation-mode nli \
    --alignscore-backbone FacebookAI/roberta-base

# 4. Reward-model training (5-fold CV, both conditions)
python src/reward_model/run_training.py \
    --holistic data/preferences_1000/\
holistic_reward_preferences_1000.jsonl \
    --gca      data/preferences_1000/\
gca_reward_preferences_1000.jsonl \
    --output-dir outputs/reward_models_1000 \
    --backbone FacebookAI/roberta-base \
    --epochs 5 --batch-size 8 --lr 2e-5 \
    --max-length 512 --max-article-chars 2000 \
    --kfold 5 --seed 1
```

Step 3 above, run exactly as written, produces the intended `nli`/$\alpha = 0.0$ preference data. The actual headline campaign was produced by the Slurm scripts `slurm/build_preferences_1000.sh` and `slurm/build_preferences_scale.sh` (below) instead of this exact command, and those scripts omit `–alignscore-evaluation-mode` (defaulting to `nli_sp`) and, for $n = 1{,}000$, pass `–alpha 0.5`; this is the discrepancy reported in Chapter 5, Section 5.4. A reader following the four commands above exactly as shown will reproduce the configuration this thesis intended, not the one the headline numbers in Chapter 6 were actually measured from.

This reproduces one run of the headline campaign, seed 1 of 20. Substituting each of `–seed 2` through `–seed 20` in turn reproduces the remaining nineteen.

In step 4 the two preference-file paths are shown split across lines for typesetting only; the trailing backslash inside a path is a line continuation in this listing, not part of the filename.

On the MOGON NHR cluster these are submitted as Slurm batch jobs (`slurm/generate_candidates_1000.sh`, `slurm/build_preferences_1000.sh`, `slurm/trai `n_rm_1000.sh`) on partition `a100dl` with one A100-SXM4-40GB GPU, 32 GB of host memory, and four CPU cores per task.

C. Example Preference Record

Every record written by the GCA preference-construction stage retains the full sentence-level detail behind the summary-level decision, which is what makes the qualitative disagreement analysis in Chapter 6 possible. The listing below shows one such record, abridged: the article, reference summary, and the `chosen`/`rejected` fields are truncated, and the sentence list for summary B is cut after the first three sentences. All numeric values are reproduced verbatim.

The record shown was produced by the earlier $n = 200$ run, which used the exploratory settings $\alpha = 0.5$ and margin $= 0.05$ rather than the final $\alpha = 0.0$ and margin $= 0$ (Chapter 4, Section 4.9). It is included to illustrate the record *schema* and the relationship between the sentence scores and the aggregate, not as an instance of the final configuration. Note in particular that `gca_score_a` $= 0.382$ is well below `mean_score_a` $= 0.629$: this is the penalty term at $\alpha = 0.5$ pulling the aggregate down towards the summary's weakest sentence, the behavior that Chapter 6, Section 6.5.1 found to perform worse than the simple mean.


```
{
  "sample_id": "cnn_00000_b0b34b2d843e",
  "article": "LONDON, England (Reuters) -- Harry Potter star Daniel ...",
  "reference_summary": "Harry Potter star Daniel Radcliffe gets ...",
  "summary_a": "Daniel Radcliffe, the 18-year-old actor who plays ...",
  "summary_b": "On Monday, July 23, 2007, Harry Potter actor Daniel ...",
  "gca_score_a": 0.382102,
  "gca_score_b": 0.169184,
  "gca_score_diff": 0.212918,
  "mean_score_a": 0.629039,
  "mean_score_b": 0.578186,
  "min_score_a": 0.232103,
  "min_score_b": 0.049505,
  "n_sentences_a": 3,
  "n_sentences_b": 6,
  "n_low_faithfulness_a": 1,
  "n_low_faithfulness_b": 3,
  "decision": "A",
  "judge_method": "gca",
  "judge_model": "yzha/AlignScore [FacebookAI/roberta-base]",
  "margin": 0.05,
  "alpha": 0.5,
  "sentence_details_a": [
    { "index": 0,
      "text": "Daniel Radcliffe, the 18-year-old actor who plays Harry
```

```
                Potter, has gained access to a 20 million pound fortune
                on his 18th birthday.",
        "score": 0.748840, "label": "faithful" },
      { "index": 1,
        "text": "Despite the newfound wealth, he has stated that he does
                not plan to be extravagant, and will instead use his
                money to buy books, CDs, and DVDs.",
        "score": 0.232103, "label": "hallucinated" },
      { "index": 2,
        "text": "Radcliffe has filmed a TV movie and an Australian film,
                and has made his stage debut in \"Equus.\" He is expected
                to face closer media scrutiny now that he is legally an
                adult.",
        "score": 0.906173, "label": "faithful" }
    ],
    "sentence_details_b": [
      { "index": 0, "text": "On Monday, July 23, 2007, ...",
        "score": 0.475803, "label": "hallucinated" },
      { "index": 1, "text": "Radcliffe has stated he has no ...",
        "score": 0.997014, "label": "faithful" },
      { "index": 2, "text": "Instead, he plans to use the money ...",
        "score": 0.049505, "label": "hallucinated" }
    ]
  }
```

Two features of this record are worth drawing out, because they illustrate mechanisms discussed in the main text. First, the `label` field is a threshold applied at a score of 0.5 for reporting convenience only; it plays no part in the aggregation, which operates on the continuous scores. Second, summary B contains both the highest-scoring sentence in the pair (0.997) and the lowest (0.050). Under the holistic condition this internal variation is not observable at all: the summary receives a single score. This is precisely the information that sentence-level decomposition exposes and holistic scoring discards, and whether exposing it yields a more learnable preference signal is the question the thesis sets out to answer.

D. Exploratory and Superseded Experiments

The main text reports the final design and the confirmed results. This appendix retains the record of the experiments that preceded them: an early prototype under a different design, the reward-model baselines that motivated the aggregation-formula change, the single-run checks that failed to reproduce, and the six-run campaign that the twenty-run campaign of Chapter 6, Section 6.1 superseded. None of these is a finding of this thesis. They are kept because two of them are load-bearing for decisions the main text makes: the observation that single runs at this dataset scale do not reproduce is what motivated the repeated-run protocol (Chapter 4, Section 4.5), and the six-run campaign is the one $n = 1{,}000$ campaign that was built on the intended evaluator configuration (Chapter 5, Section 5.4).

D.1. Early Prototype Results Under the DPO Design

An earlier version of the pipeline produced results under a DPO-based design before the study narrowed to reward-model comparison (Chapter 4, Section 4.9). They are reported here as context for that change, not as findings of the present study.

A FLAN-T5 [5] baseline on a 20-sample pilot gave ROUGE-1 0.3278, ROUGE-2 0.1201, ROUGE-L 0.2341 [26], and BERTScore F1 0.8612 [42]. On a 200-sample subset (subset seed 42), the original AlignScore-based preference construction, using the proposal-era margin of 0.05 and $\alpha = 0.5$, produced 154 usable holistic pairs out of 200 (77.0%, mean scores 0.7324 for the low-temperature candidate and 0.6530 for the high-temperature one) and 157 usable GCA pairs (78.5%, mean GCA scores 0.4066 and 0.3337). The two methods agreed on 121 of 200 decisions (60.5%).

Mistral-7B-Instruct-v0.3 was then fine-tuned with DPO on both preference sets (Chapter 3, Section 3.2), using LoRA adapters [17] on the attention and feed-forward projections rather than full fine-tuning: training loss reached 0.6902 for the holistic condition and 0.6913 for the GCA condition, each in roughly 134.5 seconds on an A100. On 200 held-out articles, baseline AlignScore was 0.7965, DPO-Holistic reached 0.8017, and DPO-GCA reached 0.7996. Only DPO-GCA's ROUGE-L differed from baseline at the conventional threshold (Wilcoxon $p = 0.015$); the AlignScore differences between conditions were small and not distinguishable from noise. These downstream results illustrate the attribution problem described in Chapter 4, Section 4.9: differences of this magnitude cannot be traced to the preference-construction procedure rather than to the optimization stage, which is why the final study measures the preference data directly.

D.2. Bradley–Terry Reward-Model Baselines

The first Bradley–Terry reward-model result used the $n = 495$ set (subset seed 100) with a `microsoft/deberta-v3-base` backbone [15], predating the standardization on `FacebookAI/roberta-base` (Chapter 5, Section 5.5). Under 5-fold cross-validation, the holistic reward model reached 58.1% mean pairwise accuracy and the GCA reward model reached 54.6%, both above the 50% chance level.

The first result on the $n = 1{,}000$ set, with the `roberta-base` backbone and the original configuration ($\alpha = 0.5$, margin 0.05), is summarized in Table D.1. This is the baseline the configuration-selection experiments of Chapter 6, Section 6.5 work from: holistic outperforms GCA by 1.2 percentage points.

Table D.1.: Bradley–Terry reward-model baseline at $n = 1{,}000$ ($\alpha = 0.5$, margin 0.05, `roberta-base`).

Condition	Mean pairwise accuracy	Gap (GCA − Holistic)
Holistic RM	57.20%	-1.20 pp
GCA RM	56.00%	

To understand why GCA underperformed, disagreement between the holistic and GCA decisions at $\alpha = 0.5$ was analyzed directly on the $n = 1{,}000$ pairs (`analysis/gca_vs_holistic_analysis.py`). The two methods disagreed on 17.8% of pairs. GCA scores showed higher discrimination between candidates than holistic scores (mean absolute score difference 0.2392 versus 0.1527), but only 82.2% of GCA decisions agreed with the corresponding holistic decision, suggesting that the extra discrimination was at least partly noise rather than signal. This motivated the aggregation-formula comparison reported in Chapter 6, Section 6.5.

Figure D.1 plots the nine aggregation strategies of Chapter 6, Table 6.14 in rank order.

D.3. Single-Run Checks That Did Not Reproduce

Two single-run results from this period failed to reproduce, and together they are why every confirmatory result in this thesis is a repeated-run campaign.

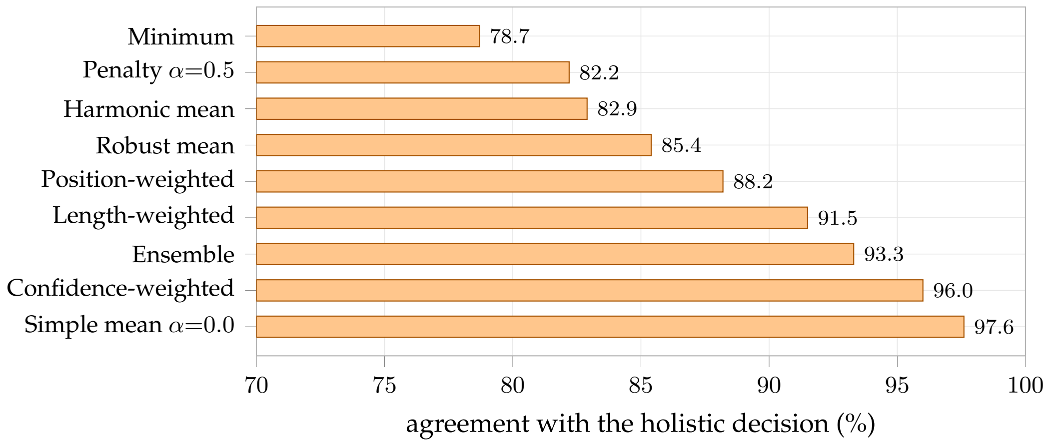


Figure D.1.: Agreement with the holistic decision across the nine aggregation strategies of Chapter 6, Table 6.14 ($n = 1{,}000$ pairs). Agreement measures consistency between two procedures driven by the same evaluator, not external factual correctness.

First, the aggregation-formula change was validated the same way the baseline was measured, by training reward models on the resulting preferences and checking pairwise accuracy, holding the judge mode fixed at the original `nli_sp` setting. Table D.2 shows that $\alpha = 0.0$ did not, on its own, close the gap: holistic accuracy improved more than GCA accuracy did, and the gap between them widened slightly rather than closing.

Table D.2.: Effect of the formula change alone ($n = 1{,}000$, judge mode `nli_sp` unchanged).

Metric	$\alpha = 0.5$ (baseline)	$\alpha = 0.0$	Change
Holistic mean accuracy	0.572	0.586	+0.014
GCA mean accuracy	0.560	0.561	+0.001
Gap (Holistic − GCA)	0.012	0.025	+0.013 (wider)

Second, a nine-configuration hyperparameter search was run, sweeping learning rate over $\{1 \times 10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}\}$ and epoch count over $\{3, 5, 7\}$, still at $\alpha = 0.0$. One configuration, learning rate 1×10^{-5} with 7 epochs, produced a GCA advantage of +0.014 (GCA 0.586 versus holistic 0.572). A same-seed confirmation rerun of that exact configuration produced the opposite result: holistic 0.577 versus GCA 0.560, a gap of -0.017. The apparent advantage did not reproduce. At this dataset size, a single training run, even with 5-fold cross-validation, can produce a result that looks like a real effect but falls within the range of ordinary fold-to-fold and initialization noise. This directly motivated moving away from hyperparameter tuning toward the judge-mode comparison, and toward the multi-seed validation protocol used for every confirmatory result.

Figure D.2 plots the judge-mode sweep of Chapter 6, Table 6.15, grouped by whether the evaluator performs its own internal splitting.

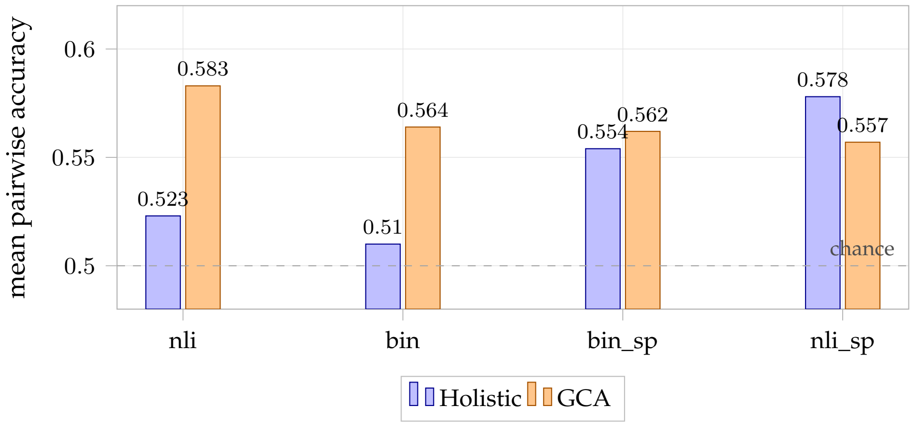


Figure D.2.: Evaluator-mode sweep ($n = 1{,}000$, $\alpha = 0.0$). The two left-hand modes do not split the input internally; the two right-hand modes do. Each bar is a single run, so differences of this size fall within the run-to-run variation documented in Section D.4, and the comparison is exploratory.

The `nli_sp` row of Table 6.15 (0.578 holistic, 0.557 GCA) is a separate single run from the $\alpha = 0.0$/`nli_sp` row in Table D.2 (0.586 holistic, 0.561 GCA), not a repetition of it: the two experiments were run independently, at different points in the exploratory process, before either PyTorch or CUDA was seeded. The 0.8 and 0.4 percentage-point differences between them, on what is nominally the same configuration, illustrate the same run-to-run instability.

D.4. The Six-Run Campaign at $n = 1{,}000$

The `nli` mode, $\alpha = 0.0$, margin 0 configuration was fixed and run across four further seed values, in addition to the original sweep and its confirmation rerun, for six runs and 30 cross-validation folds in total. Table D.3 lists every run. The six runs span five distinct seed values, not six: seed 42 appears twice, as the original sweep and its rerun. As discussed in Chapter 4, Section 4.5, those two runs are not duplicates, because the seed argument controlled only fold assignment and not the reward model's weight initialization under the code version in use at the time.

This campaign matters beyond its historical role. It is the one $n = 1{,}000$ campaign whose preference data was built under the intended `nli`, $\alpha = 0.0$ configuration; the twenty-run campaign that supersedes it was not (Chapter 5, Section 5.4).

Table D.3.: Per-run results for the selected `nli`-mode configuration (n = 1,000).

Run	Seed	Holistic	GCA	Gap (GCA − Holistic)
Original sweep	42	0.523	0.583	+0.060
Confirmation	42	0.543	0.556	+0.013
Seed validation 1	7	0.556	0.546	−0.010
Seed validation 2	100	0.510	0.561	+0.051
Seed validation 3	314	0.520	0.552	+0.032
Seed validation 4	2026	0.525	0.564	+0.039

GCA leads in five of the six runs; seed 7 is the only run in which holistic is ahead. Figure D.3 plots the per-run gaps against the pooled mean: the individual runs range from −0.010 to +0.060, a spread wide enough that any single run, taken on its own, would have been weak evidence in either direction. Averaging over all 30 fold-level readings gives the estimate this campaign supported, shown in Table D.4: a mean gap of +3.08 percentage points.

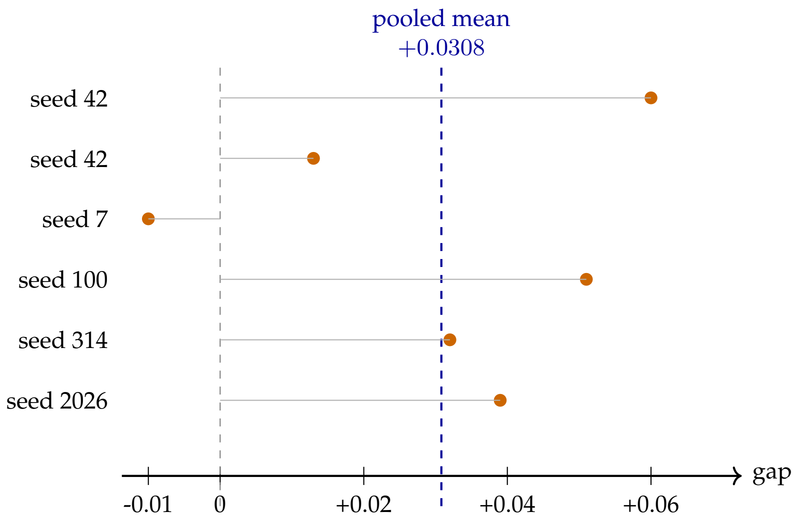


Figure D.3.: Per-run GCA−Holistic accuracy gap at n = 1,000, against the pooled mean. Five of six runs favor GCA, but the spread across runs is wide relative to the pooled effect, which is why no single run was treated as conclusive.

At the fold level, GCA leads in 22 of the 30 folds (73%), with no ties. Both conditions sit close to chance in absolute terms: 0.5295 and 0.5603 are 2.95 and 6.03 percentage points above the 0.5 chance level.

Table D.5 tracks how the fold-level estimate changed as runs were added. The interval narrows at each stage while the point estimate stays near three percentage points, though the independence caveat of Chapter 4, Section 4.5 applies to every row.

Table D.4.: Averaged result across 6 runs / 30 cross-validation folds at $n = 1{,}000$. The interval and p-value are computed over folds, which are not independent observations, and are therefore exploratory and anti-conservative (Chapter 4, Section 4.5).

Holistic mean	GCA mean	Gap	95% fold-level CI	Fold-level Wilcoxon p
0.5295	0.5603	**+0.0308**	$[+0.013, +0.047]$	0.0034

Table D.5.: Progression of the pooled statistical estimate as runs were added. The p-values are subject to the independence caveat discussed in Chapter 7, Section 7.3.

Stage	Runs	Folds	Pooled gap	95% CI	Wilcoxon p
Two-run	2	10	$+0.034$	$[+0.003, +0.064]$	0.084
Five-run	5	25	$+0.029$	$[+0.009, +0.048]$	0.0123
Six-run	6	30	$+0.031$	$[+0.013, +0.047]$	0.0034

At six runs, the fold-pooled p-value had already fallen to 0.0034, yet the run-level test could not fall below 0.0312 no matter how consistent the six runs were, because that is the smallest two-sided value the Wilcoxon signed-rank test can return with six paired observations; with five wins and one loss the attainable minimum is 0.0625. The six-run campaign was therefore not merely inconclusive by chance; it was structurally unable to establish run-level significance at the conventional threshold, independent of the true size of the effect. Resolving this required more runs, which in turn required the reward-model training procedure to be fully deterministic under a given seed, a property it did not previously have (Chapter 5, Section 5.5).